

Natural Sciences

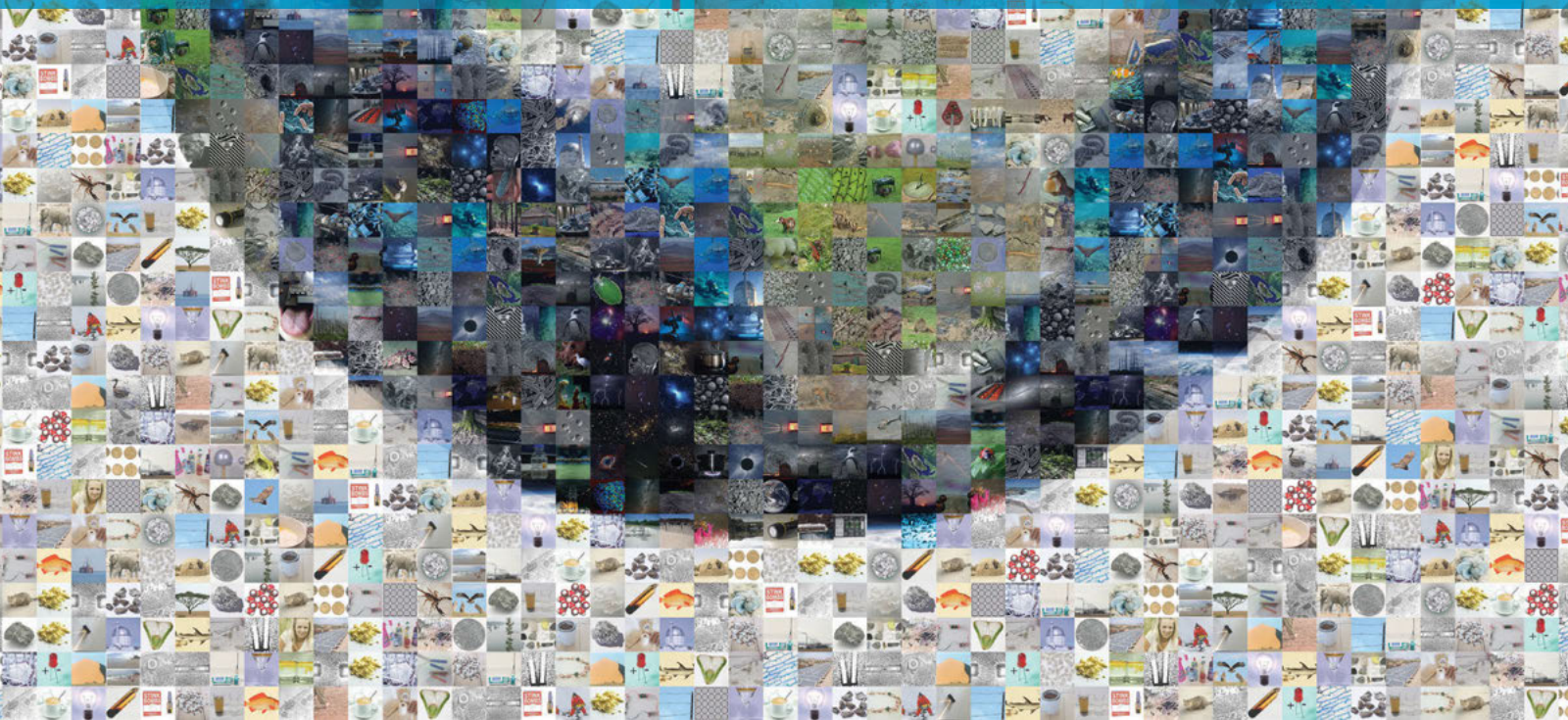
Grade 8-B (CAPS)

sasol
reaching new frontiers



EXPLORE

A World Without Boundaries



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

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No
Element

- Transition Metal
- Metal
- Metalloid
- Non-metal
- Noble Gas
- Lanthanide
- Actinide

57	58	59	60	61	62	63	64	65	66	67	68	69	70	71
La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu
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Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr

Natural Sciences

Grade 8-B

CAPS

developed by



funded by



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This book was written by Siyavula with the help, insight and collaboration of volunteer educators, academics, students and a diverse group of contributors. Siyavula believes in the power of community and collaboration by working with volunteers and networking across the country, enabled through our use of technology and online tools. The vision is to create and use open educational resources to transform the way we teach and learn, especially in South Africa.

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To learn more about the project and the Sasol Inzalo Foundation, visit the website at:

www.sasolinzalofoundation.org.za

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ENERGY AND CHANGE



KEY QUESTIONS:

- What is static electricity?
- What is friction?
- Why does my hair stand on end and crackle when I pull a jersey off?
- What is lightning?
- What does it mean to 'earth' an object?
- What does it mean when we say 'opposites attract'?

Have you ever pushed a trolley through the shops and suddenly felt a shock? Or pulled your school jersey over your head and heard it crackling? What causes those shocks and noises? Let's investigate.

1.1 Friction and static electricity

The effects of **static electricity** are all around us, but we do not always recognise it when we see or feel them. Or perhaps you have, but you never realised what was causing it. For example, have you ever felt a slight shock when you put a jersey over your head on a cold day, or perhaps you have observed your hair stand on end when you touch certain objects? Let's do a quick activity to demonstrate static electricity.



ACTIVITY: Sticky balloons

MATERIALS:

- balloons (or a plastic comb)
- small pieces of paper

INSTRUCTIONS:

1. Work in pairs.
2. Blow up a balloon and tie it closed so that the air does not escape.
3. Hold the balloon a short distance away from your hair or pieces of paper. What do you notice?

4. Rub your hair with the balloon.

5. Now hold the balloon a short distance away from your hair or pieces of paper. What do you see?



Did you see your hair 'rise' like this?!

VISIT

Watch this video about static electricity to understand why your hair stands on ends when you brush it or rub it against a balloon

bit.ly/1dZkVJr



QUESTION:

1. What did you do to make your hair or the pieces of paper stick to the balloon?

Let's look at an everyday example of static electricity. Sometimes when you comb your hair with a plastic comb your hair stands on end and makes crackling sounds. How does this happen?

You have dragged the surface of the plastic comb against the surfaces of your hair. When two surfaces are rubbed together there is **friction** between them. Friction is a resistance against the movement of an object as a result of its contact with another object. This means that when you rubbed the plastic comb along your hair, your hair resisted the movement of the comb and slowed it down.

The friction between two surfaces can cause electrons to be transferred from one surface to the other.

In order to understand how electrons can be transferred, we need to remember what we learnt about the structure of an atom last term in Matter and Materials.

VISIT

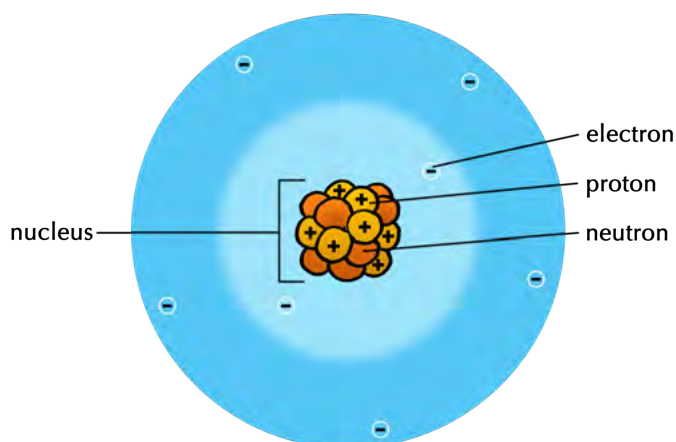
A simulation on friction between a carpet and John Travolta's foot.

bit.ly/GzGCEa



NEW WORDS

- friction
- static electricity
- electrostatic charge
- attract: to pull something closer
- repel: to push something away
- neutral
- discharge
- earth
- earthing



All atoms have a nucleus which contains protons and neutrons. The nucleus is held together by a very strong force, which means that the protons within a nucleus can be considered to be fixed there. The atom also contains electrons. Where are the electrons arranged in the atom?

What is the charge on a proton?

What is the charge on an electron?

What is the charge on a neutron?

TAKE NOTE

Remember, **like** charges **repel** and **opposite** charges **attract**.

The atom is held together by the **electrostatic attraction** between the positively charged nucleus and the negatively charged electrons. Within an atom, the electrons closest to the nucleus are the most strongly held, whilst those further away experience a weaker attraction.

Normally, atoms contain the same number of protons and electrons. This means that atoms are normally **neutral** because they have the same number of positive charges as negative charges, so the charges balance each other out. All objects are made up of atoms and since atoms are normally neutral, objects are also usually neutral.

However, when we rub two surfaces together, like when you comb your hair or rub a balloon against your hair, the friction can cause electrons to be transferred from one object to another. Remember, the protons are fixed in place in the nucleus and so they cannot be transferred between atoms, it is only electrons that are able to be transferred to another surface. Some objects give up electrons more easily than other objects. Look at the following diagram which explains how this happens.



Which object gave up some of its electrons in the diagram?

Does this object now have more positive or more negative charges?

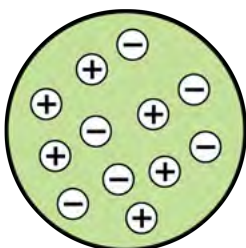
Which object gained electrons in the diagram?

Does this object now have more positive or more negative charges?

- When an object has more electrons than protons overall, then we say that the object is **negatively charged**.
- When an object has fewer electrons than protons overall, then we say that the object is **positively charged**.

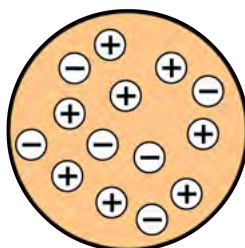
Have a look at the following diagram which illustrates this.

6 positive charges and
6 negative charges
 $6 + (-6) = 0$



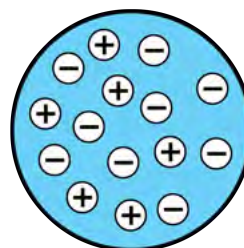
There is zero overall charge.
The object is neutral.

8 positive charges and
6 negative charges
 $8 + (-6) = 2$



The overall charge is +2.
The object is positively charged.

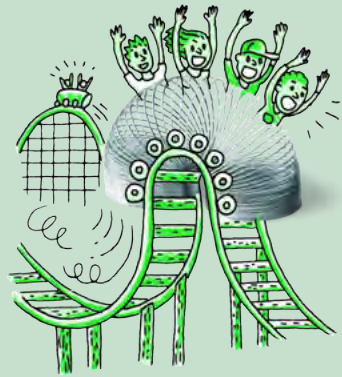
6 positive charges and
9 negative charges
 $6 + (-9) = -3$



The overall charge is -3.
The object is negatively charged.

So, we now understand the transfer of electrons that takes place as a result of

friction between objects. But, how did that result in your hair rising when you brought the charged balloon close to your hair in the last activity? Let's look at what happens when oppositely charged objects are brought together.



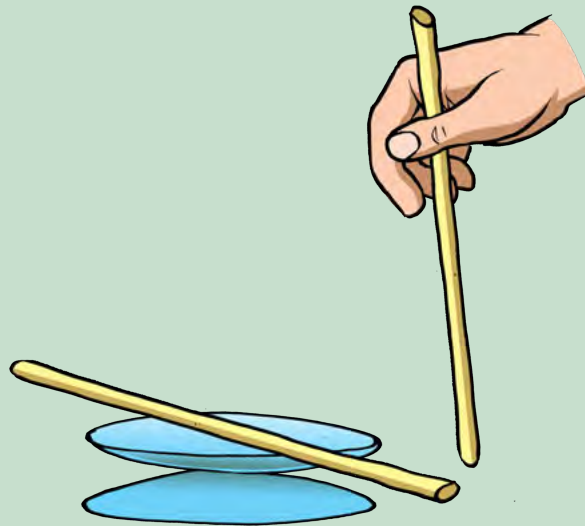
ACTIVITY: Turning the wheel

MATERIALS:

- 2 curved watch glasses
- 2 perspex rods
- cloth: wool or nylon
- plastic rod
- small pieces of torn paper

INSTRUCTIONS:

1. Place a watch glass upside down on the table.
2. Balance the second watch glass upright on the first watch glass.
3. Rub one of the perspex rods vigorously with the cloth.
4. Balance the perspex rod across the top of the watch glass.
5. Rub the second perspex rod vigorously with the same cloth.
6. Bring the second perspex rod close to the first perspex rod. What do you see happening?



7. Repeat the activity but instead of the second perspex rod, use the plastic rod. What do you see happening?
8. Next, bring a rod that you have rubbed close to small pieces of torn paper lying on the table. What do you observe?

QUESTIONS:

1. What happened when you brought the second perspex rod close to the first perspex rod?

2. What happened when you brought the plastic rod close to the first perspex rod?

3. What happened when you brought the plastic rod close to the pieces of paper?



When we rubbed the perspex rods with the cloth, electrons were transferred from the perspex to the cloth. What charge do the perspex rods now have?

Both the perspex rods now have the **same** charge. Did you notice that objects with the same charge tend to push each other away? We say that they are **repelling** each other.

When we rubbed the plastic rod with the cloth, electrons were transferred from the cloth to the plastic rod. What charge does the plastic rod now have?

The perspex rod and the plastic rod now have **opposite** charges. Did you notice that objects with different charge tend to pull each other together? We say that they are **attracting** each other.

In the example of the pieces of paper being attracted to the ruler, the paper starts off neutral. However, as the negatively charged plastic rod is brought closer, the electrons in the paper that are nearest to the rod will begin to move away, leaving behind a positive charge on the surfaces of the paper that are nearest to the rod. The paper is therefore attracted to the rod because opposite charges attract. Another example is dust that is attracted to newly polished glasses.

We have now observed the fundamental behaviour of charges.

In summary, we can say:

- If two negatively charged objects are brought close together, then they will repel each other.
- If two positively charged objects are brought close together, then they will repel each other.
- If a positively charged object is brought near to a negatively charged object, they will attract each other.

VISIT

Discover more with a simulation on rubbing balloons and a jersey.

bit.ly/GzwlIt



VISIT

Opposites attract and like repel (video)

bit.ly/16ThzBL



NEW WORDS

- flammable
- ignite



VISIT

A video showing the dangers of sparks of static electricity at a petrol station.

bit.ly/17mYLIC

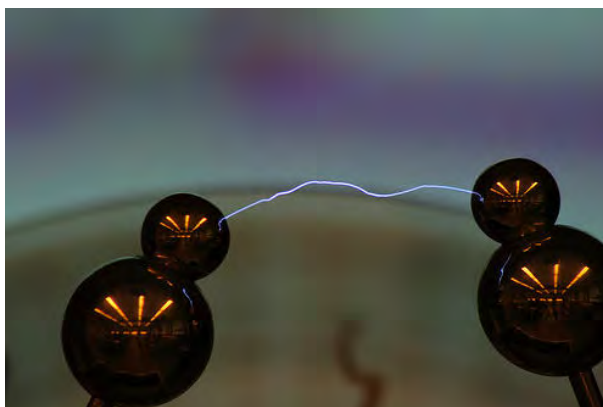


Do you now understand why your hair rises and is attracted to the balloon after you rub the balloon on your hair? Write a short description to explain what is happening using the words: electrons, transfer, negative charge, positive charge, opposite, attract, repel.

Sparks, shocks and earthing

A large build-up of charge on an object can be dangerous. When electrons transfer from a charged object to a neutral object we say that the charged object has discharged.

Discharging can take place when the objects touch each other. But the electrons can also transfer from one object to another when they are brought close, but not touching. When electrons move across an air gap they can heat the air enough to make it glow. The glow is called a **spark**.



An electrostatic spark between two objects.

Sparks can be harmless, but they can also be very dangerous. Sparks can cause **flammable** materials to **ignite**. You will probably have noticed that you may not smoke cigarettes or have open flames near petrol tanks at petrol stations. This is because petrol fumes are very explosive and only need a small amount of heat to start them burning. A small electrostatic spark is enough to ignite flammable petrol fumes.

Electrostatic discharge can also cause **electric shocks**. Have you ever been shocked by a shopping trolley while you are pushing it around a shop? Or have you walked across a carpeted room and then shocked yourself when you touch the door handle to leave the room? You have experienced an electric discharge. Electrons move from the door handle onto your skin and the movement of the electrons causes a small electric shock. Small electric shocks can be uncomfortable but mostly harmless. Large electric shocks are extremely dangerous and can cause injury and death.

Do you know where else we can see sparks due to static electricity? Look at the photo for a clue!



Lightning is a huge electrostatic discharge.

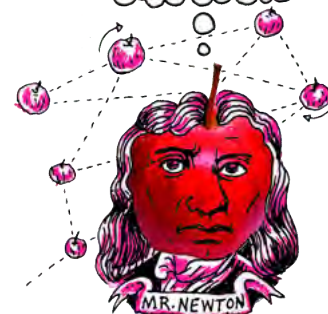
During a thunderstorm, there is friction in the atmosphere between the particles that make up clouds, causing the build-up of regions of charge. Once the difference in charge between two regions becomes great enough, electrostatic discharge becomes possible. A lightning flash is a massive discharge between charged regions within clouds, or between clouds and the Earth.

In order to discharge extra electrons safely from an object we must earth it. **Earthing** means that we connect the charged object to the ground (the Earth) with an electrical conductor. The extra electrons travel along the conductor and enter the ground without causing any harm. The Earth is so large that the extra charge does not have any overall effect.

For example, think of the metal trolleys in shopping centres. Have you ever noticed that they normally have a metal chain hanging at the bottom which drags along the floor? This is to earth the trolley if it gets a charge so that charge cannot build up on the trolley. This protects the person pushing the trolley from getting a shock.

DID YOU KNOW?

Lightning bolts can travel at about 210 000 km/h and get as hot as 30 000 °C.



VISIT

How to survive a lightning strike.

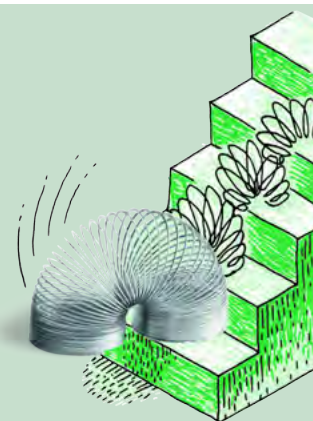
bit.ly/18nTOps



ACTIVITY: Research the practical applications of static electricity

INSTRUCTIONS:

1. Use the internet or your school or community library to find information about the practical applications of static electricity.
2. Research one useful effect of static electricity and one problem caused by static electricity.
3. Write a short paragraph explaining your research.



VISIT

Should a person touch 20 000 Volts? Visit this link to find out!

bit.ly/19mUtun



DID YOU KNOW?

The fundamental idea of using friction in a machine to generate a charge dates back to the 17th century, but the generator was only invented by Robert Van de Graaff in 1929 at Princeton University.



VISIT

Watch this video so see how a Van de Graaff generator works

bit.ly/1a5YNKE



We are now going to look at two instruments which demonstrate static electricity.

Van de Graaff generator

The Van de Graaff generator is a machine which uses friction to generate a large build-up of electric charge on a metal dome.

The Van de Graaff generator can be used to demonstrate the effects of an electrostatic charge. The big metal dome at the top becomes positively charged when the generator is turned on. When the dome is charged it can be discharged by bringing another insulated metal sphere close to the dome. The electrons will jump to the dome from the metal sphere and cause a spark.



These girls are touching the large dome of a Van de Graaff generator.

You can also touch the dome and your hair will rise. Why do you think this happens?

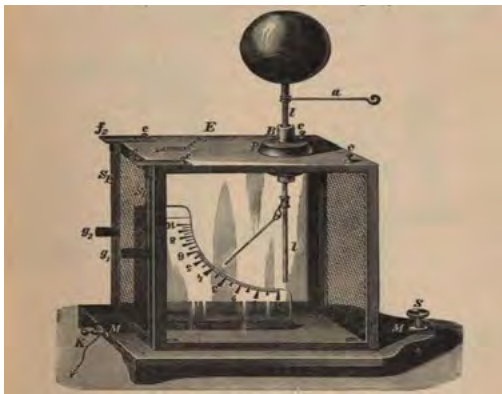
Electroscope

An electroscope is an early scientific instrument used to identify the presence of a charged object or it can be used to identify the type of charge on a charged object.

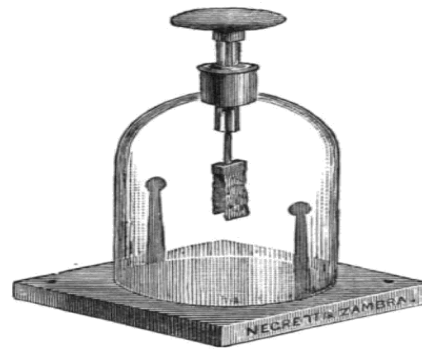


An electrostatic demonstrator used in a laboratory.

The following images show some drawings of different types of electroscopes.



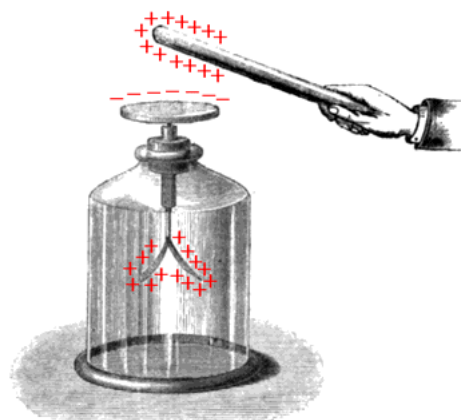
An early example of an electrostatic demonstrator with one gold strip at the bottom and a ball at the top.



Another example of an electrostatic demonstrator with a disc at the top and two gold foil strips at the bottom.

The electrostatic demonstrator is made up of an earthed metal box with glass windows. There is a metal rod hanging down and at the end are two strips of thin gold foil attached to it. A disc or ball is attached to the top of the metal rod, as seen in the illustrations above. When the metal ball or disc at the top is touched with a charged object, or a charged object is brought near to it, the gold foil strips spread apart, indicating that the object has a charge.

Look at the next illustration which shows how this works.



The positively charged rod attracts electrons to the disc from the gold foil strips. The disc at the top becomes negatively charged and the gold foil strips at the bottom become positively charged. Why do the gold foil strips move apart?

You can make a simple electroscope with everyday items. Let's try.



ACTIVITY: Making a simple electroscope

MATERIALS:

- glass jar, with lid
- 14 gauge copper wire, about 12 cm in length
- plastic straw or plastic tubing
- 2 small pieces of aluminium foil
- piece of wool cloth
- plastic ruler
- glass rod

INSTRUCTIONS:

1. Twist one end of the copper wire into a spiral shape. This will increase its surface area.
2. Make a hole in the jar lid and push a small piece of the plastic tubing through the hole.
3. Put the other end of the copper wire through the straw so that the spiral end is on the outside of the lid.
4. Make a hook out of the pointed end of the copper wire.
5. Cut two rectangular strips of aluminium foil.
6. Put each piece of aluminium foil onto the hook. Make a small hole in the aluminium foil to allow it to hang from the hook.
7. Carefully put the hook end of the copper wire into the glass jar and close the jar.
8. Rub the ruler with the wool cloth for a minute.
9. Bring the ruler close to the spiral end of the copper wire.

QUESTIONS:

1. What did you observe when you brought the ruler close to the copper wire?
-

2. What happens if you move the ruler away from the copper wire?
-

Why do the pieces of aluminium foil move apart? When you rubbed the plastic ruler with the wool cloth, the ruler became negatively charged. When the negatively charged ruler is brought close to the copper wire, the electrons on



the wire are repelled downwards towards the aluminium foil. The pieces of aluminium foil then have extra electrons on them and they both become negatively charged. Two objects which are negatively charged will repel each other and so the pieces of aluminium foil move away from each other.

3. Write a short paragraph to explain what would happen if you brought a positively charged object close to your electroscope.

SUMMARY:

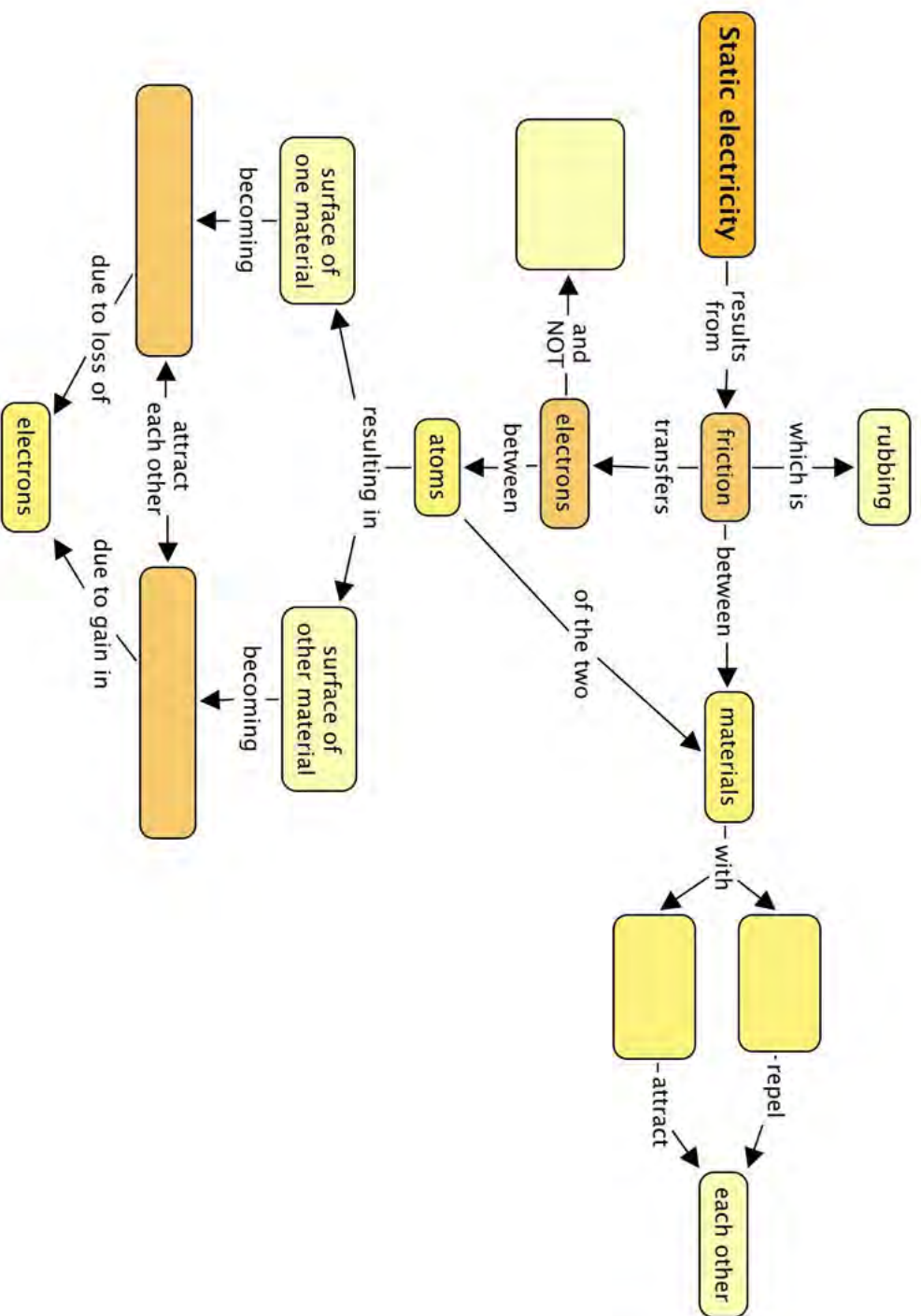
Key Concepts

- Objects are usually neutral because they have the same number of positive and negative charges.
- Objects can become negatively or positively charged when friction (rubbing) results in the transfer of electrons between objects.
- Protons and neutrons cannot be transferred, only electrons can be transferred by friction.
- If an object has more electrons than protons, then it is negatively charged.
- If an object has fewer electrons than protons, then it is positively charged.
- Like charges repel each other, i.e. negative repels negative; positive repels positive.
- Opposite charges attract each other, i.e. negative attracts positive; positive attracts negative.
- A discharge of the electrons from a charged object can cause sparks or shocks of static electricity, especially when the air is dry.

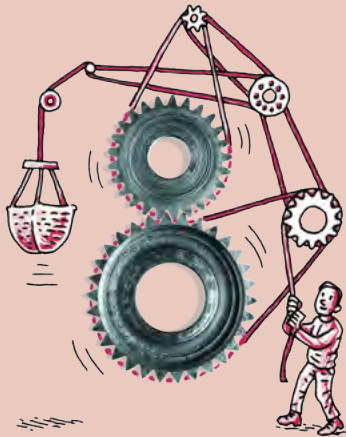
Concept Map

Complete the following concept map to summarise what you have learnt in this chapter about charge and static electricity.





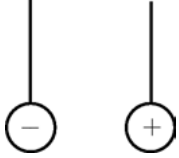
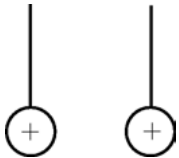
REVISION:



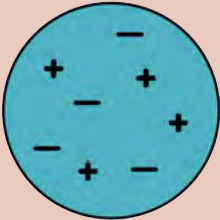
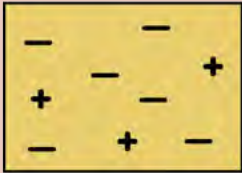
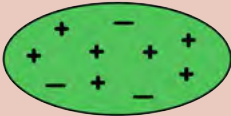
1. Complete the following sentences. Just write the missing word on the line below.
a) An object which has a **negative** charge is said to have _____ electrons than protons. [1 mark]

2. An object which has a **positive** charge is said to have _____ electrons than protons. [1 mark]

3. Sarah uses a plastic comb to comb her hair. The comb becomes negatively charged. The comb is negatively charged because the comb has: [1 mark]
a) gained electrons
b) gained protons
c) lost electrons
d) lost protons
4. A perspex strip was rubbed with a cloth and became positively charged. The correct explanation for why the perspex rod becomes positively charged is that: [1 mark]
a) the perspex rod got extra protons from the cloth.
b) the perspex rod got extra protons due to friction.
c) protons were created as the result of friction.
d) the perspex rod lost electrons to the cloth due to friction.
5. Look at the following images in the table. Redraw the images in the second column to show how the spheres will move because of the nature of the charges. Write an explanation in the last column. [6 marks]

Charged spheres	Draw how they will move	Explanation
		
		

6. Complete the table by working out the overall charge on each object. Show your calculations. State whether the object is positively charged, negatively charged or neutral and why. [9 marks]

Object	Overall charge	Why is it positive, negative or neutral?
		
		
		

7. The ruler in this photo has been rubbed with a cloth. Describe what is happening in this photo and why. [4 marks]



What is happening?

8. Sometimes, when you are pushing a trolley, you can get a small shock.
Explain why this would happen. [2 mark]

9. Why does your jersey make a crackling sound when you pull it over your head? [2 mark]

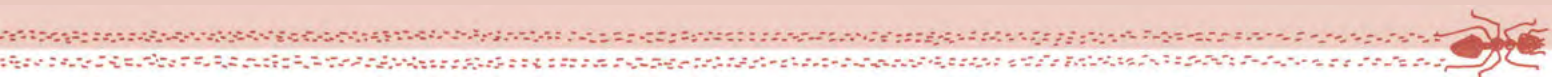
10. Why do trucks transporting petrol drag a short length of metal chain on the road as they drive? [2 mark]

11. What do you think these two girls are touching on the left of the photo?
Explain your answer and what is happening to them. [3 marks]



What is happening in this photo?

Total [32 marks]





KEY QUESTIONS:

- What is an electric current?
- What is an electric circuit?
- Where does the energy come from in a circuit?
- What are components?
- How do we draw electric circuits?
- What effects can an electric current produce?
- Why does the element in a light bulb glow and the element in a kettle become hot?
- What is an electromagnet and are they useful to us?
- How do you plate metal rings and earrings in gold to produce jewellery?

In the last chapter we looked at static electricity. We are now going to focus on current electricity. You will already be familiar with some of the concepts and terminology about electricity from previous grades. This year we are going to revise some of these concepts and also extend our knowledge about electricity.

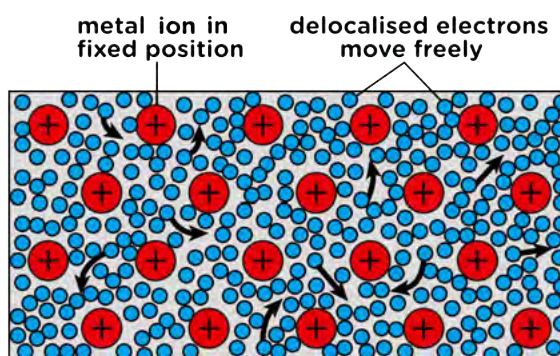
2.1 Circuits and current electricity

What is an electric current?

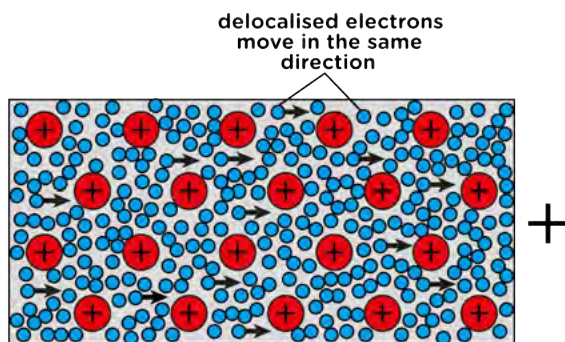
An electric current is the movement of charge in a closed, conducting circuit. As we know from Chapter 1, and also from Matter and Materials, the electrons in an atom are arranged in the outer space around the central nucleus. We saw in the last chapter how electrons can be transferred between objects resulting in a charge on the object. In metals, the electrons are able to move freely within the metal. The electrons are not associated with a particular atom in the metal. We say electrons in a metal are **delocalised**. Have a look at the following diagram which shows this.

NEW WORDS

- delocalised
- component
- conductor
- electric circuit
- electric current
- qualitative
- resistor
- switch



Conducting wire in an electric circuit is made of metal. If we supply it with a source of energy and a complete circuit, then the electrons will all move in the same general direction through the wire. This movement of electrons through a conductor is **electric current**.



TAKE NOTE

An ion is an atom that has a charge due to the loss or gain of electrons. Here the metal ions are positive as the electrons are delocalised.



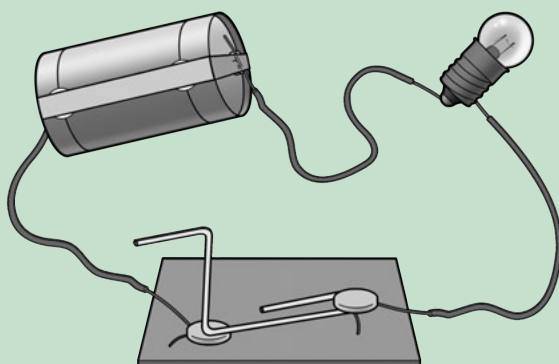
Do you remember what you learnt in Grade 6 and 7 about circuits? Let's revise briefly:

- An electric circuit needs a **source of energy** (a cell or battery).
- Cells have positive and negative terminals.
- A circuit is a **complete pathway** for electricity.
- The circuit must be **closed** in order for a device to work, such as a bulb which lights up.
- We can say that an electric circuit is a **closed system** which transfers electrical energy.
- A circuit is made up of various **components**, which we will look at in more detail.

ACTIVITY: A simple circuit

INSTRUCTIONS:

1. Look at the example of a simple circuit.
2. Answer the questions which follow.



QUESTIONS:

1. What are the parts that make up this system for transferring electrical energy?



2. Do you think this is an open or closed circuit? Explain your answer.

3. Which part is providing the source of energy?

4. What is the conducting material?

5. What type of energy does the battery have?

6. What is this energy transferred to when the circuit is closed and the electrons **move** through the wires?

7. What is the output of this system?

8. In most systems, the input energy is more than the useful output energy as some of the input energy is transferred to wasted output energy. In this simple circuit with a light bulb, what is the wasted output energy?

VISIT

Electricity and circuits
bit.ly/17ni2R4 and
Revise a simple circuit.
[video]
bit.ly/1eWpN5k



NEW WORDS

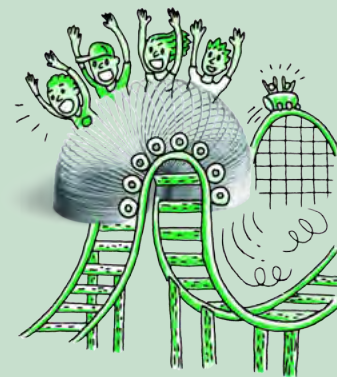
- ammeter
- cell



A complete circuit is a complete conducting pathway for electricity. It goes from one terminal of a cell along conducting material, through a device and back to the other terminal of the cell. Let's look at the components of a circuit.

2.2 Components of a circuit

You are probably already familiar with the components of an electric circuit from previous grades. Do you remember that we have a specific way of drawing the components in a circuit in an electric circuit diagram? Each component has a different symbol.



ACTIVITY: Components in an electric circuit

Complete the following table. List the function of the component and draw the circuit symbol. The last two rows have been filled in for you as you may not yet know these symbols, but we will be using them in this chapter.

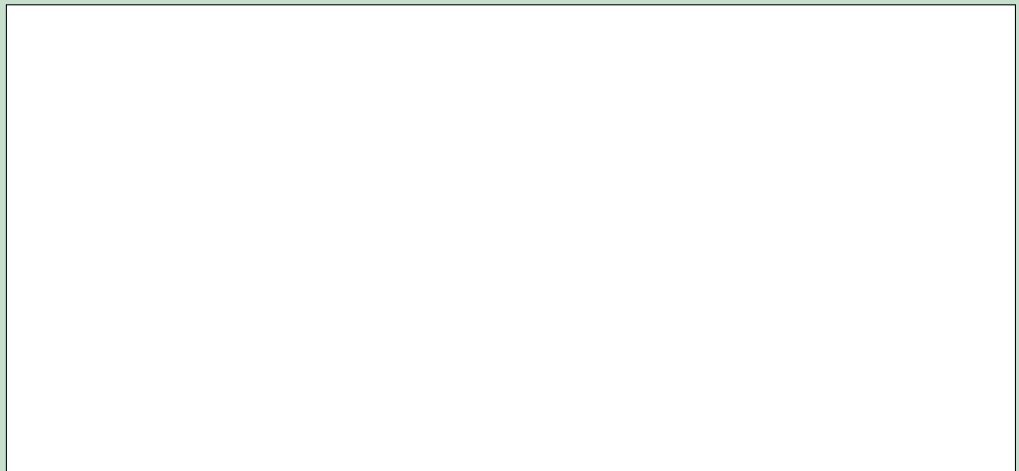
Component	Function	Symbol
Cell		
Torch bulb		
Open switch		
Closed switch		
Electrical wire		
Resistor	A component that opposes or inhibits electrical current in a circuit. It can also convert electrical energy to heat or light.	 or 
Variable resistor	A resistor whose resistance can be adjusted higher or lower.	

Let's now practice drawing some simple circuit diagrams. Draw the following circuit diagrams.

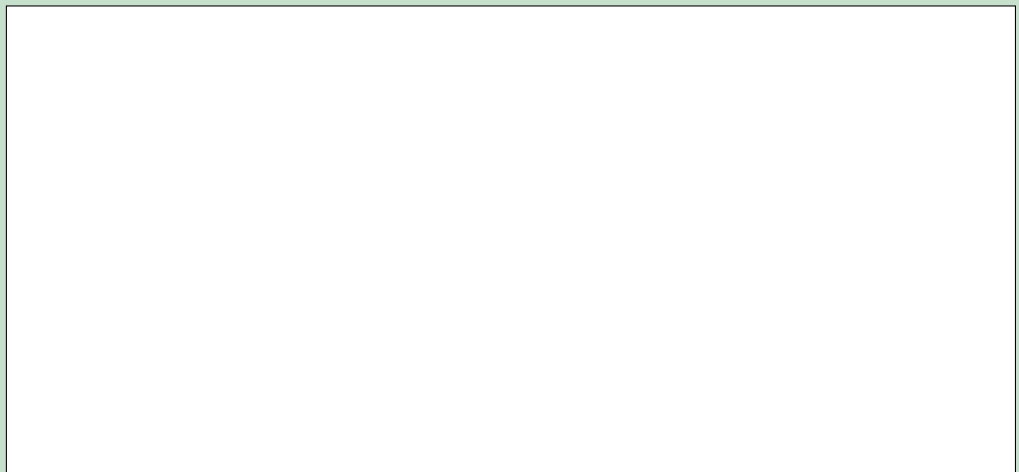
1. A closed circuit with one cell, two light bulbs and a switch.



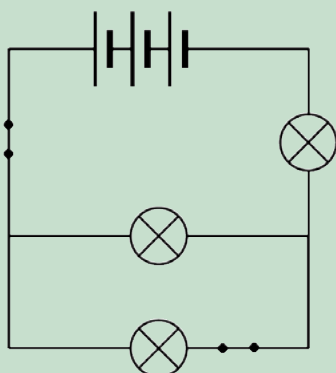
2. An open circuit with two cells, two light bulbs and a switch.



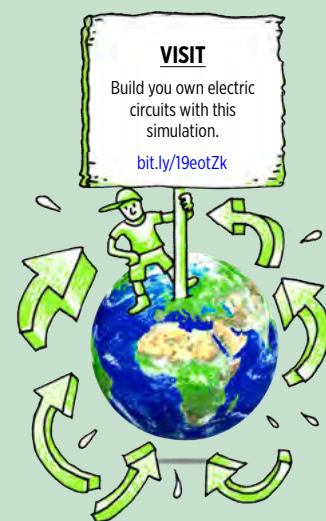
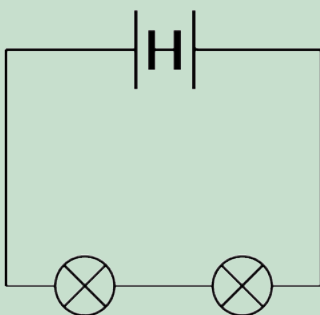
3. A closed circuit with 4 cells and one light bulb.



4. Look at the following circuit diagram. Identify the number of bulbs, switches and cells in this circuit.



5. What is wrong with the following circuit diagram? Does it represent a closed circuit? Explain your answer.



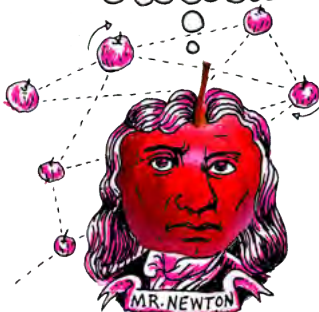
6. Why do you think it is useful to have a switch in a circuit?

7. Why are conducting wires made out of metal?

Let's take a closer look at the source of energy in electric circuits.

DID YOU KNOW?

All muscles in our bodies move in response to electrical impulses generated naturally in our bodies.



Cells

Electrical cells are the source of energy for the electric circuit. Where does that energy come from?

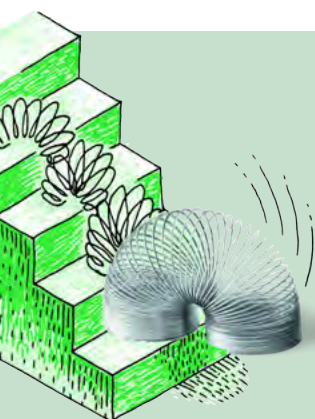
Inside the cell are a number of chemicals. These chemicals store **potential energy**. When a cell is in a complete circuit, the chemicals react with each other. As a result, electrons are given the potential energy they need to start moving through the circuit. When the electrons move they have both potential and kinetic energy. The **electric current** is the movement of electrons through the conducting wires.

Cells come in many different sizes. Different sized cells provide different amounts of energy to the electrical circuit. The types of cells you would use in toys, torches and other small appliances range in size from AAA, AA, C, D, and 9-volt sizes. AAA, AA, C and D cells usually have a rating of 1.5V, but the larger cells have a larger capacity. This means that the larger cells will last longer before going 'flat'. A cell goes flat when it is no longer able to supply energy through its chemical reactions.

When we buy cells in the shop they are usually referred to as batteries. This can be a bit confusing because a battery is really two or more cells connected together. So when we refer to a battery in circuit diagrams we need to draw two or more cells connected together.



Different sized batteries.



ACTIVITY: Recycling of batteries

Batteries which no longer work must not be thrown away in dustbins. They need to be recycled.

INSTRUCTIONS:

1. Work in small groups.
2. Find out why batteries should not be thrown away in normal dustbins. Write a paragraph to explain why.

3. Find out where you can recycle batteries in your community. Write down the details of the centre(s) closest to where you live.

Resistors

What are resistors? In order to work out what they are, let's first remind ourselves about conductors and insulators.

We are specifically looking at electricity so we can now talk about **electrical conductors and insulators**. An electrical conductor is a substance which allows electric charge to move through it. An insulator is a substance which does not allow electric charge to move through it.

Think back to our model of a metal wire and how the electrons are able to move through the wire. The metal wire is a conductor of electricity. Write down some materials which do not conduct electricity.



Why do you think most conducting wires are surrounded with plastic?

Resistors are a bit of both. They allow electrons to move through them, but do not make it easy. They are said to **resist** the movement of electrons. Resistors therefore influence the electric current in a circuit.

But, why would we want to resist the movement of electrons? Resistors can be extremely useful. Think about a kettle. If you look inside you will see a large metal coil.



Looking inside a kettle.

This metal coil is the heating element. If you plug in and switch on the kettle, the element heats up and heats the water. The element is a large resistor. When the electrons move through the resistor they expend a lot of energy in overcoming the resistance. This energy is transferred to the surroundings in the form of heat. This heat is useful to us as it heats our water.

A good example of where resistors are used is in light bulbs. Let's take a closer look at the different parts of a light bulb to see how it works.

DID YOU KNOW?

The first electric light was made by Humphry Davy in 1800. He invented an electric battery, and when he connected wires to it and a piece of carbon, the carbon glowed as the carbon is a resistor, producing light.





ACTIVITY: Resistance in a light bulb



An incandescent light bulb.

MATERIALS:

- light bulb
- lamp

INSTRUCTIONS:

1. If you have light bulbs available, have a close look at the different parts, otherwise have a look at the photos provided here.
2. Read the information about how a light bulb works and identify the parts that have been numbered.
3. Answer the questions that follow.

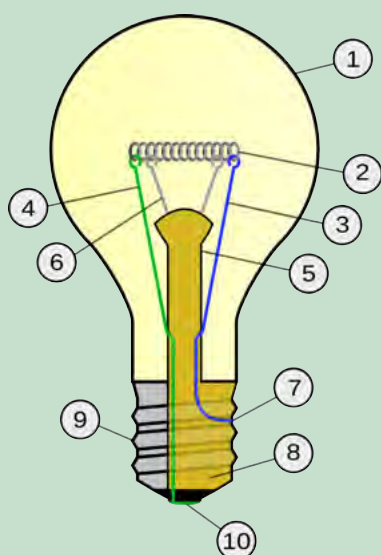


Diagram of the parts of a light bulb.

A light bulb consists of an air-tight enclosed glass case (number 1). At the base of the bulb are two metal contacts (numbers 7 and 10), which connect to the ends of an electrical circuit. The metal contacts are attached to two stiff wires, (numbers 3 and 4).

These wires are attached to a thin metal filament. Have a look at a light bulb. Can you identify the filament? This is number 2 in the diagram. The filament is made from tungsten wire. This is an element with high resistance.

QUESTIONS:

1. When the electrons move through the filament they experience high resistance. This means that they transfer a lot of their energy to the filament when they pass through. The energy is transferred to the surroundings in the form of heat and bright light. Describe the transfer of energy in this light bulb.

2. What is the useful energy output and what is the wasted energy output in this light bulb?

3. Can you see the filament is coiled? Why do you think this is so? Discuss this with your class and teacher.

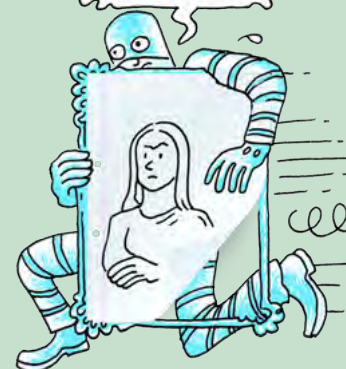
4. The filament is mounted on a glass stem (number 5). There are two small support wires to hold the filament up (number 6). Why do you think the stem is made of glass?

5. The inside of the base of the bulb is made from an insulating material. This is the yellow part labeled number 8. On the outside of this is a metal conducting cap to which the wire is attached at number 7. Why is the wire attached at 7 making contact with the metal conducting cap?

6. If you have a lamp in the classroom, screw the bulb into the lamp and turn it on to observe the filament glow and also getting hot.

TAKE NOTE

Incandescent means to emit light as a result of being heated.



VISIT

A fun game about electric circuits.
bit.ly/15lcr49

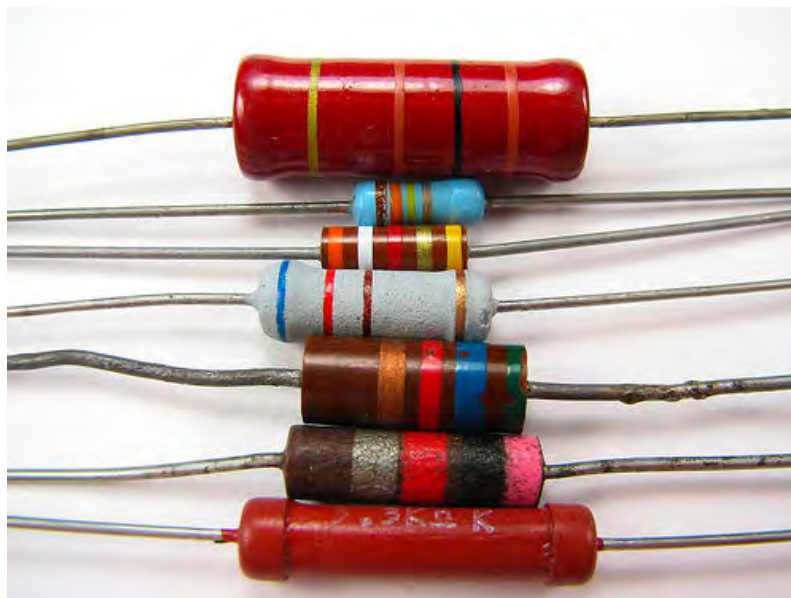


The amount of resistance a substance offers to the circuit is measured in ohms (Ω). If we want to use resistors to control the current flow, then we need to know the amount of resistance. There are some common resistors shown in the

photo.

DID YOU KNOW?

The inventor, Thomas Edison, experimented with thousands of different resistor materials until he eventually found the right material so that the bulb would glow for over 1500 hours.



Some common resistors.

Can you see that there are different coloured bands on the resistors? This isn't just to make them look pleasing to the eye. The coloured bands are actually a code that tells us the resistance of the resistor. We also get resistors where we can adjust the resistance ourselves. This is called a variable resistor. You have already seen the symbol for drawing a resistor in a circuit diagram. Draw a circuit diagram in the space below with two bulbs, two cells, an open switch and a resistor.

NEW WORDS

- variable
- fuse
- electromagnet
- electric current



An electric current can have various effects. Let's find out more about what these are.

2.3 Effects of an electric current

We are going to look at the effects of an electric current, and specifically how we use these effects. An electric current can:

- generate heat in a resistor;
- generate a magnetic field; and

- cause a chemical reaction in a solution.

Heating effect

As electrons move through a resistor they encounter resistance and they transfer some of their energy to the resistor itself. We saw this in the last section where we looked at the filament in a light bulb and the element in a kettle.

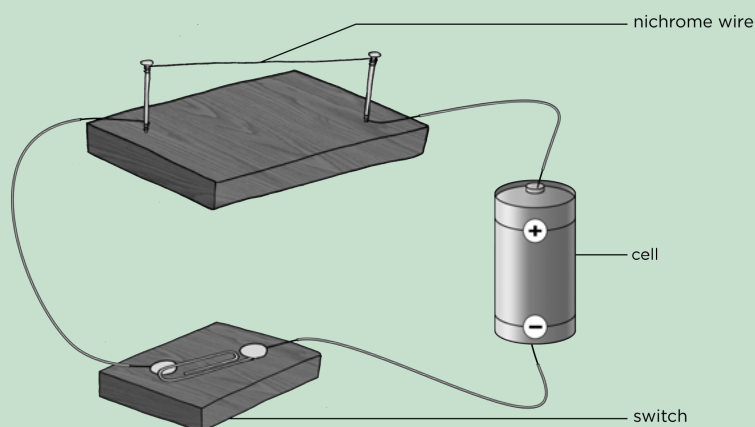
ACTIVITY: Heating a wire in a circuit

MATERIALS:

- 1,5 V cell
- conducting wires
- switch
- block of wood
- 2 nails
- hammer
- 10 cm of nichrome wire

INSTRUCTIONS:

1. Hammer the two nails into the block of wood and attach the nichrome wire between the nails.
2. Build the following circuit and keep the switch open.



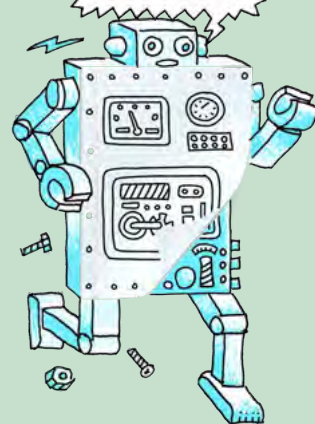
3. Feel the nichrome wire. Is it hot or cold?

4. Close the switch. Leave it on for a minute.
5. Open the switch again.
6. Feel the wire, briefly. Is it hot or cold?



TAKE NOTE

You can easily make your own switch by sticking two metal drawing pins into a piece of wood with a metal paper clip in between, as shown in the diagram.



QUESTIONS:

1. When you felt the nichrome wire after the circuit had been on for a while, you felt an increase in **temperature** in your skin as **thermal energy**, which was transferred from the wire to your skin. Explain the heating effect of the electric current in the resistance wire.

2. List 2 useful applications of the heating effect of an electric current.

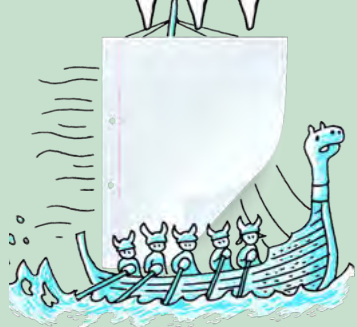
3. Choose one of the applications you listed in question 2 and explain how the heating effect of the electric current is used.

4. Look at the following photo of a toaster.



An electric toaster.

Can you see the glowing filament inside? Why does the element glow?



TAKE NOTE

Remember that heat and temperature are not the same thing. Temperature is a measure of how hot or cold something is (measured in $^{\circ}\text{C}$) whereas heat is the transfer of thermal energy from a hotter object to a colder object (measured in J).



So now we know that an electric current can cause objects to heat up. Let's look at a useful application of the heating effect.

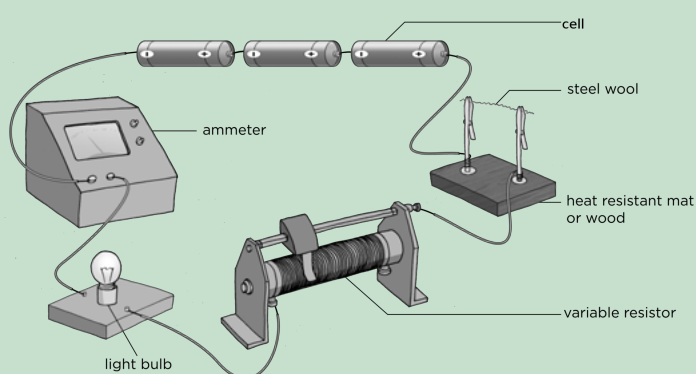
ACTIVITY: Melting metal

MATERIALS:

- three 1,5 V cells
- copper conducting wires with crocodile clips
- steel wool
- heat resistant mat or piece of wood
- torch light bulb
- variable resistor
- ammeter

INSTRUCTIONS

1. Set up a circuit according to the following picture.



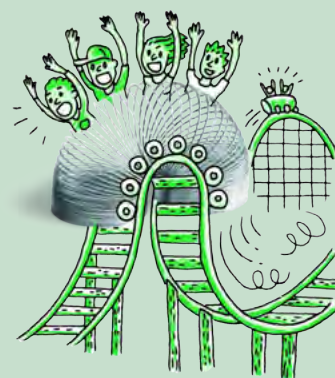
2. Twist a few strands of steel wool into a wire.
 3. Use the steel wool to complete the circuit.
 4. Set the variable resistor to its highest resistance.
 5. Close the switch. What do you observe?
-
6. Take note of the reading on the ammeter which measures the current in the circuit.
 7. Open the switch.
 8. Set the variable resistance to its lowest resistance.
 9. Close the switch. What do you observe?
-

QUESTIONS:

1. Draw a circuit diagram for your circuit.



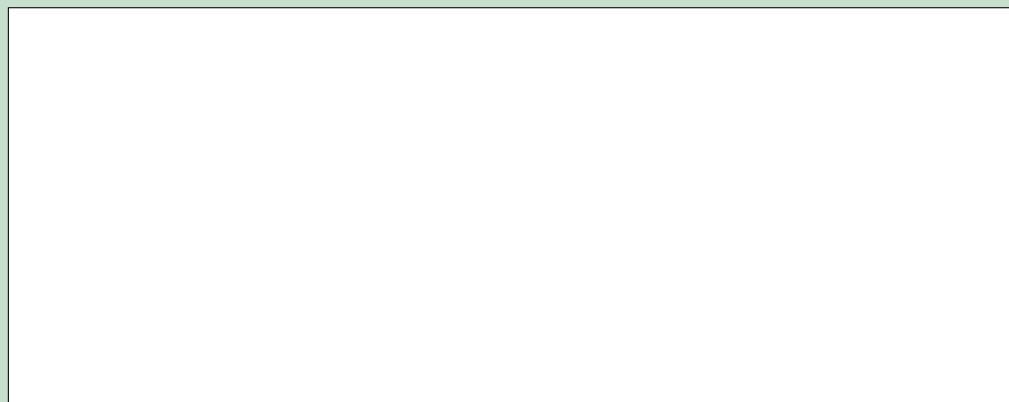
This is the symbol for an ammeter.



TAKE NOTE

An ammeter is used to measure the electric current in a circuit.





2. Why is the light bulb included in the circuit?

3. When you decreased the resistance, what happened to the current? In other words, what happened to the reading on the ammeter?

4. What do you think happens to the electric current when the steel wool has burnt? Explain your answer.



In this activity, we just demonstrated how a **fuse** works. The steel wool acted as a fuse. When the current was too high, the steel wool melted and prevented any further current in the circuit.

What are fuses?

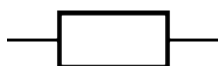
The heating effect of an electric current can be dangerous. If a circuit overheats it could cause a fire. To avoid overheating, circuits often contain a fuse. Fuses contain a low resistance wire made of a metal with a low melting point. Therefore, the piece of wire melt if it gets too hot, just like the steel wool in our activity.



An example of a fuse. Can you see the low melting point wire inside?

Different circuits need different strength currents and so we need different types of fuses. Some fuses can only handle a little bit of heat, some can handle a lot. We choose the fuse that suits the safety needs of our circuit. If the circuit overheats, the fuse will melt and break the circuit to reduce the danger of fire as well as protect electronic equipment.

How did you draw the fuse that we made using steel wool in the last activity? The conventional symbol for drawing a fuse in a circuit diagram is shown here:



A fuse.

What is a short circuit?

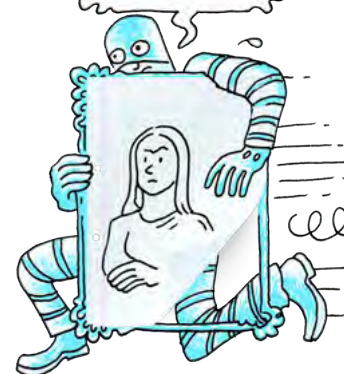
Have you ever heard that something broke because it **short circuited**? A short circuit happens when another, easier path is accidentally made in an electric circuit. What do we mean by *easier*?

We mean that the path offers very little resistance to the electric current. As there is so little resistance the current flows along the short circuit and doesn't pass through the main circuit. Short circuits can be dangerous and cause a lot of damage to appliances.

Have you ever had a piece of toast get stuck in a toaster? It's a real nuisance. Lots of people are tempted to use their butter knife to unhook the bread. Don't be tempted. Your knife is a conductor and can act as a short circuit. All the electric current will flow through your knife and, because you are touching it, through you. What would be the safe way to unhook your toast?

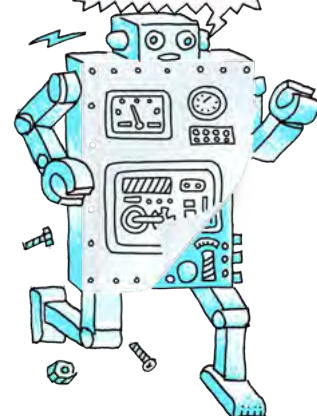
TAKE NOTE

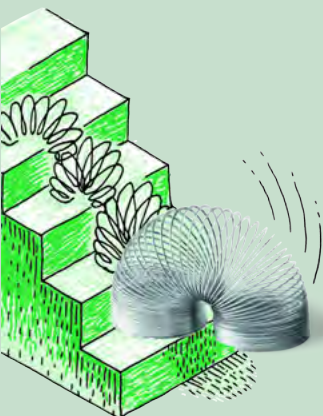
It is important to never remove a fuse from a circuit without first switching off the current. You could get a nasty shock if you do.



TAKE NOTE

There are different types of fuses. The ones we have investigated so far require you to replace the fuse if the wire melts. However, some fuses work differently to break the circuit and can just be reset once the problem in the circuit is fixed.





ACTIVITY: How are fuses used in everyday circuits?

INSTRUCTIONS:

1. Find out about common household appliances which use fuses. Choose one of these appliances on which to focus your research.
2. Write a short paragraph describing the appliance and explaining why a fuse is necessary for that appliance.



VISIT

Some fun tricks with magnets. (video)

bit.ly/1c0IQsA



Most modern homes have **circuit breakers** instead of fuses. A circuit breaker is similar to a fuse in that it is designed to protect an electric circuit from damage, due to overload or a short circuit, by stopping the current flow. However, unlike a fuse which melts and must then be replaced, a circuit breaker can be reset to start operating again. This can be done manually or take place automatically.

Magnetic effect

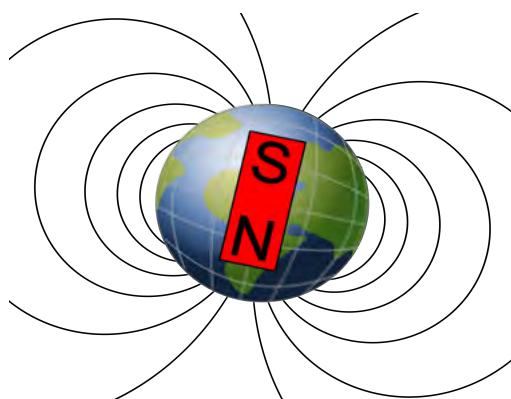
Before we look at how a current produces a magnetic field, let us first learn more about magnets. A magnet is a piece of material which produces a magnetic field. A magnet has a north pole and a south pole. Opposite poles will attract each other and the same poles will repel each other. A magnet has a magnetic field around it.



A bar magnet.

Did you know that the Earth is like a bar magnet with a North and a South Pole?

The Earth has a magnetic field. This is why we can use compasses to tell direction. A plotting compass has a needle with a small magnet. The needle points to magnetic north because the small magnet is attracted to the opposite magnetic pole and can be used to determine direction.



Earth has a magnetic field, as though there is a big bar magnet running through the core, with its South Pole under Earth's magnetic North pole.



A compass with the needle pointing North.



ACTIVITY: Playing with plotting compasses and magnets

MATERIALS:

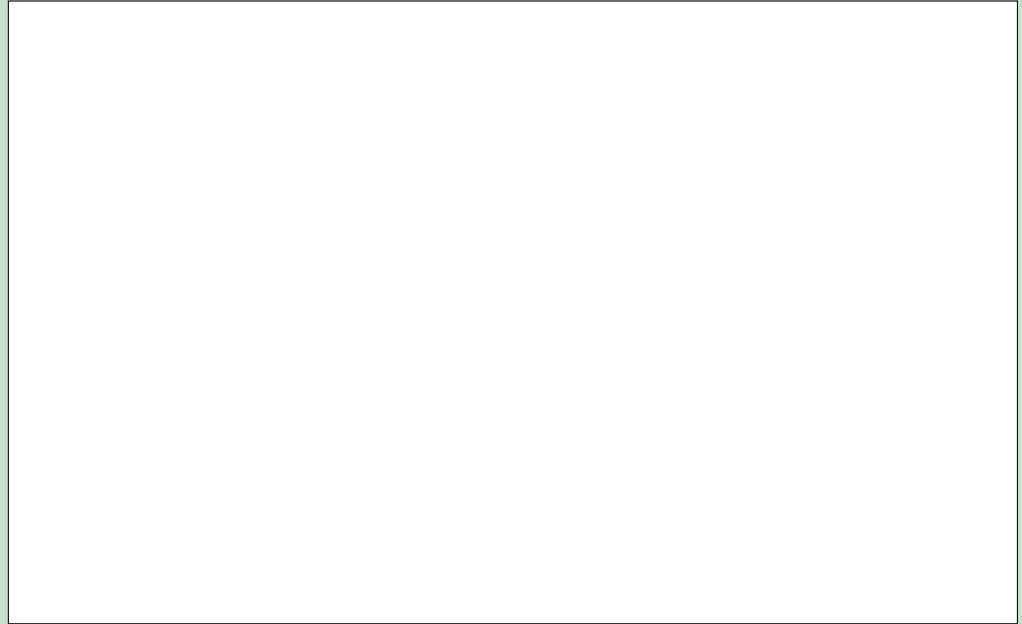
- plotting compasses
- bar magnets
- piece of white paper
- iron filings

INSTRUCTIONS:

1. Hold the plotting compass in your hand. The north end of the needle should point to magnetic north.
2. Put the bar magnet flat on the desk. Make sure you know which end is north and which is south. If you are not sure, ask your teacher.
3. Put plotting compasses in a circle around the bar magnet.



Draw what you see.



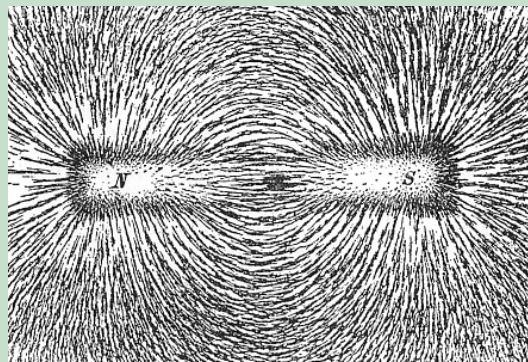
VISIT

Explore the interactions between a compass and bar magnet with this simulation.

bit.ly/19etlNQ



4. Next, place a white sheet of paper over the bar magnet and sprinkle iron filings over the sheet of paper over the magnet. Observe what happens to the iron filings. Did you see something similar to what is shown in the photograph below? Describe what you see.



Iron filings on a piece of paper over a bar magnet.

So now we know that there is a magnetic field around a magnet and that plotting compasses and iron filings can be used to visualise that field. Is there anything else that has a magnetic field around it?

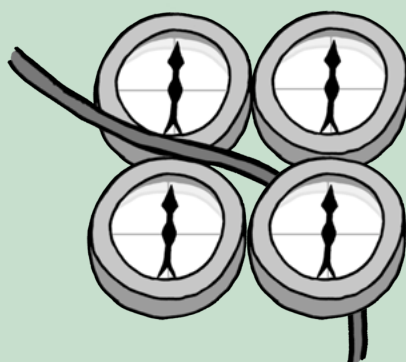
ACTIVITY: Magnetic field around a conductor

MATERIALS:

- plotting compasses
- three 1,5 V cells
- insulated copper conducting wires
- switch

INSTRUCTIONS:

1. Construct a circuit which contains the batteries, copper wires and the switch.
2. Put the plotting compasses on either side of the conducting wire as shown in the diagram, as well as below and above the conducting wire.



Plotting compasses placed around a conducting wire.

3. Keep the switch open. What do you notice about the needles of the plotting compasses?

4. Close the switch and observe what happens to the needles.
5. Draw a picture of the wire and plotting compasses in the space below:
6. What does the pattern of the compasses tell us?



VISIT

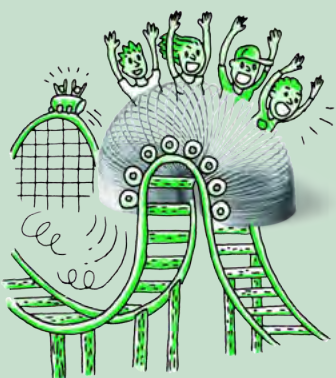
Discover how the Earth is a magnet that protects us from damaging radiation from the sun!

bit.ly/GCctjK



We saw from our first activity that plotting compasses react to magnetic fields. The plotting compasses changed direction when the current was switched on. This means there is a magnetic field around the wire. Was it there when the current was switched off? No, it was not. That means that the presence of the electric current in the wire must have produced a magnetic field.

The magnetic effect of an electric current has many useful applications.



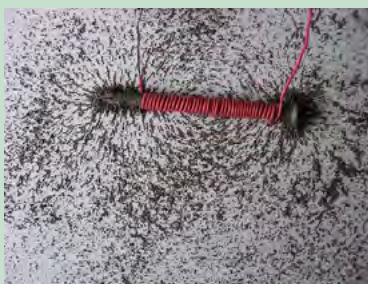
ACTIVITY: Making an electromagnet

MATERIALS:

- one iron nail (approximately 15 cm long)
- 3 metres of 22 gauge insulated copper wire
- two D cell batteries
- paper clips
- iron filings

INSTRUCTIONS:

1. Wrap the insulated copper wire tightly around the nail. Make sure that you wrap the wire in the same direction.
2. Strip some of the insulation off each end of the insulated copper wire.
3. Attach the ends of the insulated copper wire to the terminals of the battery.
4. Hold the wrapped nail above the paper clips.
5. Disconnect the wire from the battery.
6. Hold the wrapped nail above the paper clips.
7. If you have iron filings, place some on a piece of paper around the electromagnet you have made and observe the magnetic field.



The magnetic field around an electromagnet.

QUESTIONS:

1. What happened when you held the nail over the paper clips?

2. Why were the paper clips attracted to the nail?

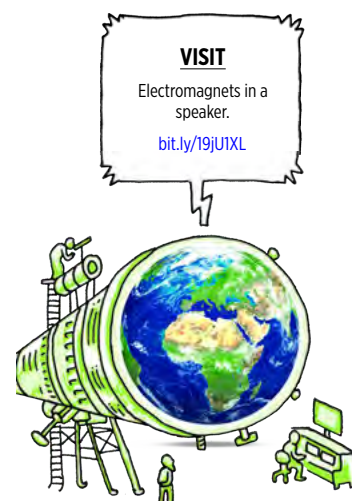
3. Did the disconnected nail attract the paper clips? Why?



Electromagnets can be used in all sorts of practical applications, including speaker and electric bells, as you can see in the photo.



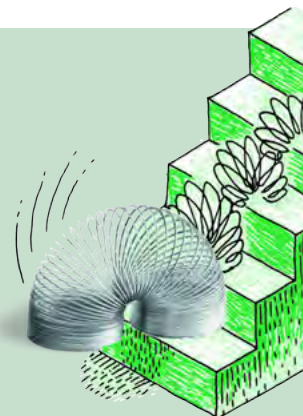
An electromagnet in a bell.



ACTIVITY: Research the use of electromagnets

INSTRUCTIONS:

1. Work in groups of 2 or 3.
2. Research one of the following applications of the magnetic effect of an electric current to explain how the device works:
 - a) speakers
 - b) electric bells
 - c) telephones
 - d) magnetic trains
 - e) industrial lifters and separators
3. Write a short paragraph showing what you've learnt. Remember to note down from where you got your information.
4. Share your paragraph with the rest of the class.



Chemical effect

The last effect of an electric current that we are going to look at is how an electric current can cause a chemical reaction in a solution.





ACTIVITY: Electrolysis

You might already have done this activity in Matter and Materials when we investigated the decomposition of copper chloride. We are going to perform it again, this time focussing on the effects of an electric current.

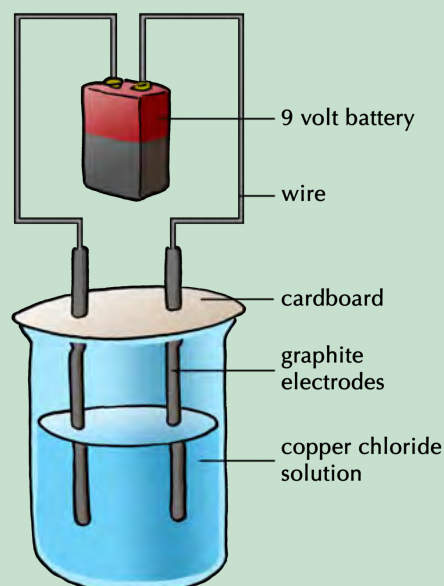
MATERIALS

- 250 ml beaker
- 2 carbon electrodes
- sandpaper
- 3 copper conducting wires (with crocodile clips)
- copper chloride solution
- torch bulb
- power pack

INSTRUCTIONS

1. Sand down the electrodes with the sandpaper to make sure they are clean.
2. Connect the conducting wire from one electrode to the torch bulb and another wire from the torch bulb to the negative terminal of the power source.
3. Connect the crocodile clip from the second electrode to the positive terminal of the power source.
4. Pour 100 ml copper chloride solution into the beaker.
5. Put the electrodes into the beaker. Make sure that they do not touch each other.
6. Look at the electrodes. What do you observe?

-
7. Turn on the power source. Leave it on for a few minutes.



The setup might look something like this, which you have seen before. You might also have a light bulb connected in the circuit.

QUESTIONS

1. When you switch on the power source, does the torch bulb glow?

2. What do you observe happening at the two different electrodes?

3. Can you smell anything? What do you think this is?

4. What is happening to the copper chloride solution when the electric current is passed through it?

5. If you switch off the power source, what happens?

6. What is causing the separation of the copper chloride?

7. Why is it important that you do not let the carbon electrodes touch each other while the current is flowing?

VISIT

Learn more about silver refining through electrolysis. bit.ly/1fZQ55W and the process of electroplating (video) bit.ly/GzH851



NEW WORDS

- electrolysis
- electrodes
- electroplating

The separation of the copper chloride means that an electric current can cause chemical reactions to occur. There are many ways in which we can harness this chemical effect for practical uses.

Electrolysis is the breaking down of a substance into its component elements by passing an electric current through a liquid or solution. We can also use electrolysis to purify substances.

Impure copper can be purified using electrolysis. Instead of using carbon electrodes in a copper sulphate solution we can use copper electrodes. If one of the copper electrodes is pure copper and the other is impure copper, then the impure electrode will break down and deposit pure copper on to the already pure copper electrode.



One of the most important uses of electrolysis is **electroplating**.

Electrolysis is used to electroplate metals. In the last activity, one of the carbon electrodes was coated with an even layer of pure copper. We say that the carbon electrode was electroplated with copper.

Why do we electroplate? An example is in the making of jewellery where an inexpensive metal is made into a ring, for example, and then coated with gold by electroplating. This makes it less expensive than if it were made from pure gold. Iron rusts easily and so it is useful to coat it with a layer of a zinc to protect it from corrosion. Many car parts, bathroom taps and wheel rims are electroplated with chromium.



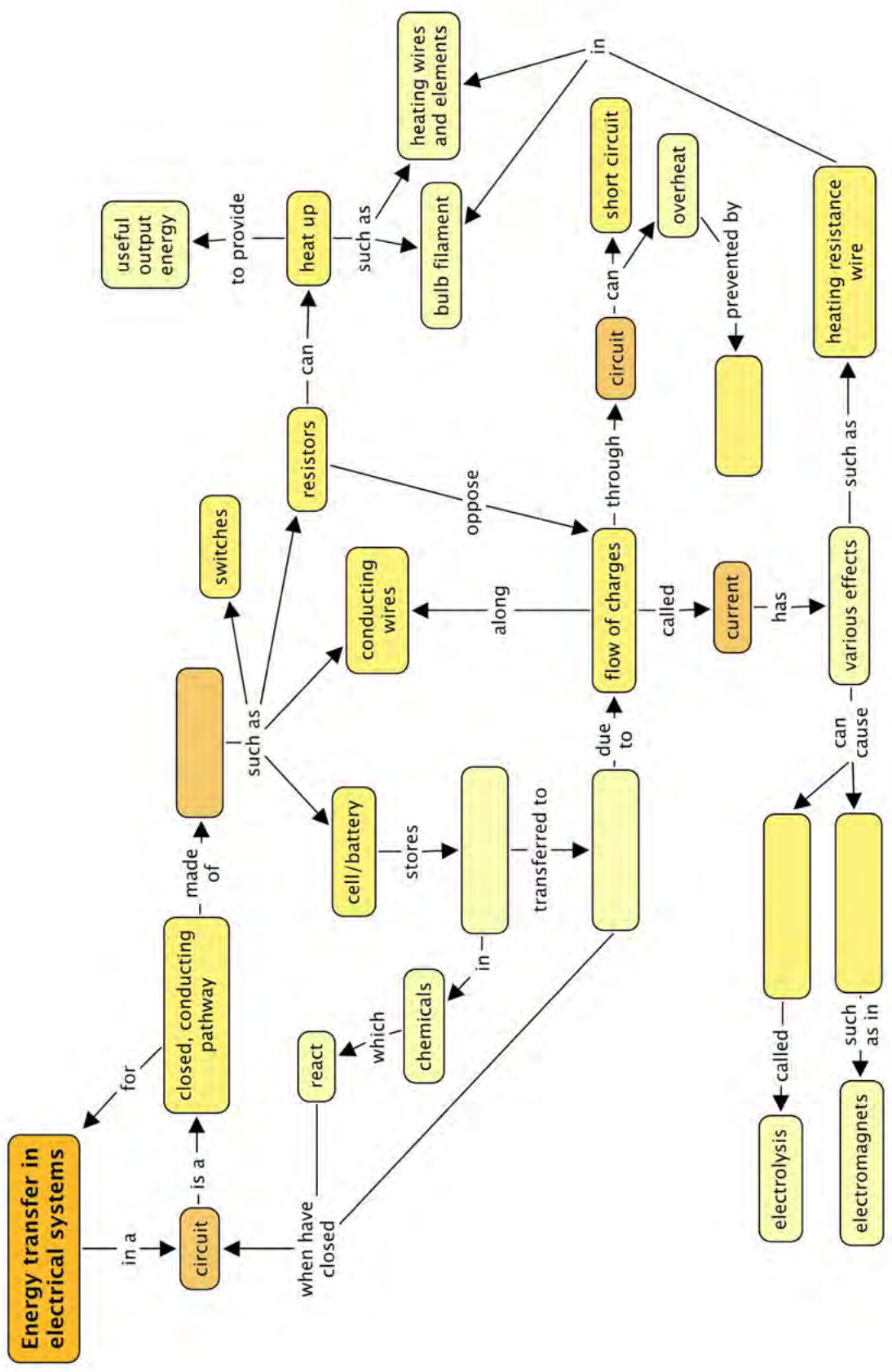
SUMMARY:

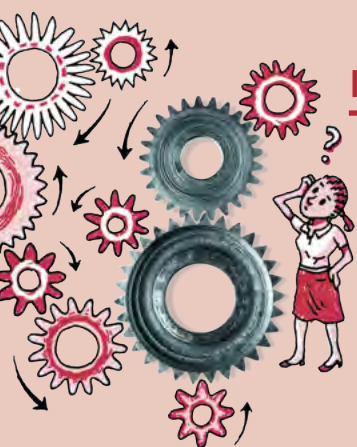
Key Concepts

- A circuit is a system for transferring electrical energy.
- For a circuit to function there must be a complete, unbroken pathway for the electrons to follow, a source of energy (cell or cells) and a load (lightbulb or any other resistor).
- We use symbols to represent components of an electric circuit so that everyone can interpret the diagrams.
- A resistor is a component in a circuit which resists the movement of electrons through the circuit.
- An electric current can heat a resistance wire. This heating effect is used in many everyday appliances, such as kettles and irons.
- An electric current causes a magnetic field. This magnetic effect is used in electromagnets.
- An electric current can cause a chemical reaction in solutions. This is called electrolysis, and is used to electroplate objects.

Concept Map

Complete the concept map to summarise what you have learned about electric circuits and the effects of an electric current in this chapter.





REVISION:

1. Write your own definition for an electric circuit. [2 marks]

2. What type of energy does a battery have? [1 mark]

3. When a battery is connected to a circuit, it causes an electric current in the circuit. Explain what an electric current is and why it is possible in metals. Use the word 'delocalised' in your explanation. [3 marks]

4. List 3 materials which conduct electricity. [3 marks]

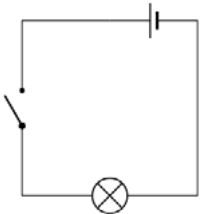
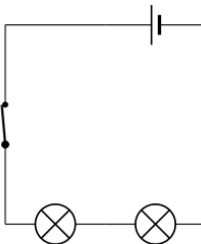
5. List 3 materials that do not conduct electricity. [3 marks]

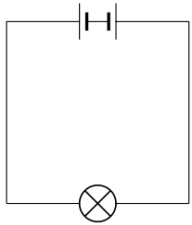
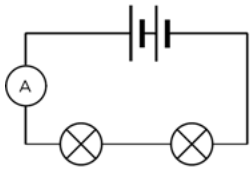
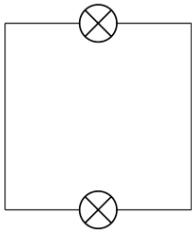
6. You have a battery, insulated copper conducting wires and a light bulb. Draw a setup which would allow you to test whether the materials you listed in questions 1 and 2 are conductors or not. [4 marks]

7. Draw the symbols for the following components. [6 marks]

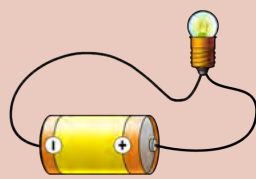
A cell	A light bulb
A conducting wire	An open switch
A resistor	A variable resistor

8. Look at the circuits below. If the bulb(s) will glow, place a tick next to the picture and explain why it will glow. If the bulb(s) will not glow, place a cross next to the picture and explain why it will not glow. [10 marks]

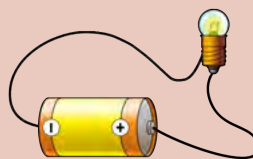
Circuit	Glow/Not Glow	Explanation
		
		

Circuit	Glow/Not Glow	Explanation
		
		
		

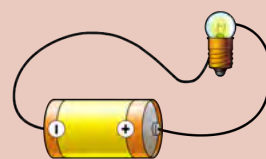
9. Which of the following setups shows the correct way to connect a light bulb to a battery? Explain your answer. [2 marks]



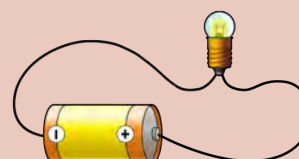
A



B

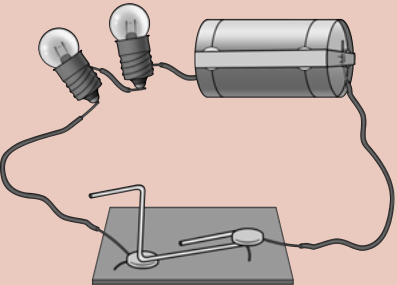


C



D

10. Draw a circuit diagram to illustrate the following circuit: (3 marks)

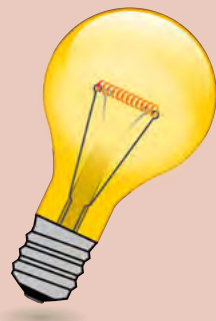
Image	Circuit diagram
	

11. An electrician wants to replace a faulty fuse with a normal piece of conducting wire. Should you let him? Why or why not? [3 marks]

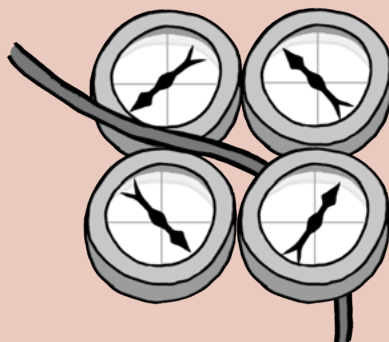
12. A child, while inserting an electric plug into the socket, did not see that there was a thin piece of aluminium foil stuck between the pins of the plug. When he turned the switch on, he noticed a spark at the plug, and at the same time, the lights went out. What could have happened to cause the spark and to make the lights go out? [4 marks]

13. What is the benefit of using a circuit breaker rather than a fuse? [2 marks]

14. Look at the following photo of a light bulb. Label the filament and explain why it glows. [4 marks]



-
-
15. You place some plotting compasses around an electric wire and observe the following.



- a) Is there are current in the conducting wire? [1 mark]

- b) Explain your answer. [2 marks]

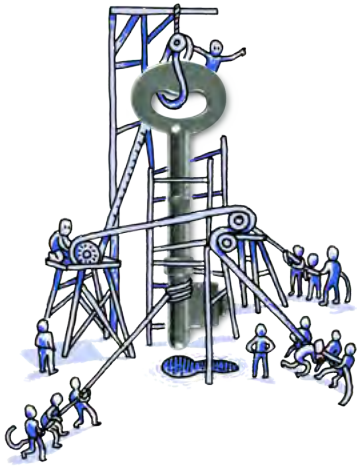
16. Give two advantages of electroplating iron metal. [2 marks]

Total [55 marks]



Curious? Discover the possibilities with a magnifying glass.





KEY QUESTIONS:

- Are there different types of electric circuits?
- If all the light bulbs in a house are part of the same circuit, how can you switch one light off without the rest also turning off?
- What is a series circuit?
- What is a parallel circuit?
- What happens when you connect more components in series or in parallel?

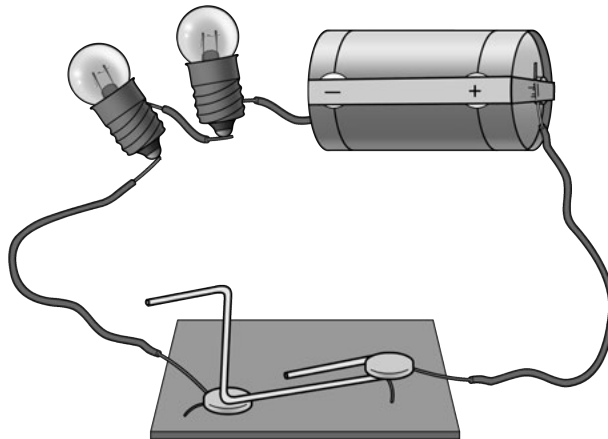
In the last chapter, and in Gr 6 and 7, we have been looking at electric circuits. These have mostly been series circuits. What does this mean? And how else can a circuit be arranged?

3.1 Series circuits

NEW WORDS

- series
- ammeter
- ampere
- resistance

A **series circuit** is one in which there is only one pathway for the electric current to follow. The components are arranged one after another in a single pathway. When we connect the components we say that they are **connected in series**. We have already seen examples of series circuits in the last chapter.



A series circuit with one pathway for the current, from the negative to the positive terminal of the battery.

Ammeter

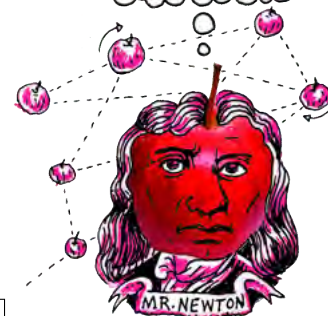
An ammeter is a measuring device used to measure the electric current in the circuit. It is connected into the circuit in series. The current is measured in **amperes (A)**.



An ammeter.

DID YOU KNOW?

The ampere is named after André-Marie Ampère (1775-1836), a French mathematician and physicist. He is considered the father of electrodynamics, which is the study of the effect of electromagnetic forces between electric charges and currents.



TAKE NOTE

The ampere is often shortened to 'amp'.



What is the symbol for an ammeter? Draw it here.

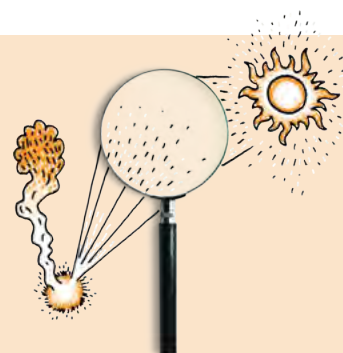
Do you think that an ammeter would have a high resistance or a low resistance to the current? Explain your choice.

A series circuit only provides one pathway for the electrons to follow. Let's investigate what happens when we increase the resistance in a series circuit.

INVESTIGATION: What happens when we add more resistors in series?

AIM: To investigate the effect of adding resistors to a series circuit.

HYPOTHESIS: Write a hypothesis for this investigation.

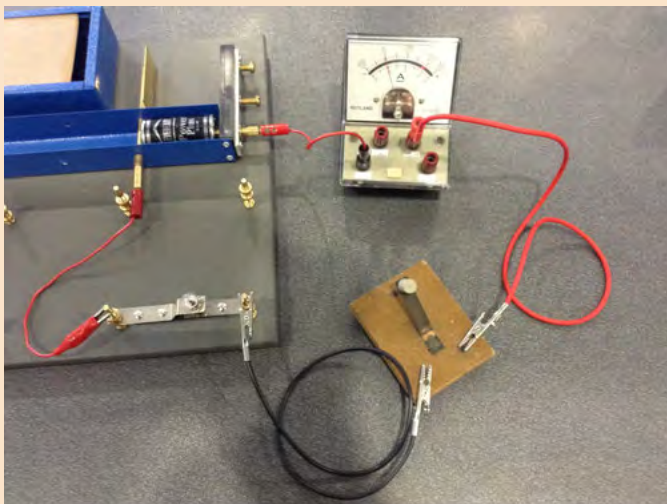


MATERIALS AND APPARATUS:

- 1,5 V cells
- 3 torch bulbs
- insulated copper conducting wires
- switch
- ammeter

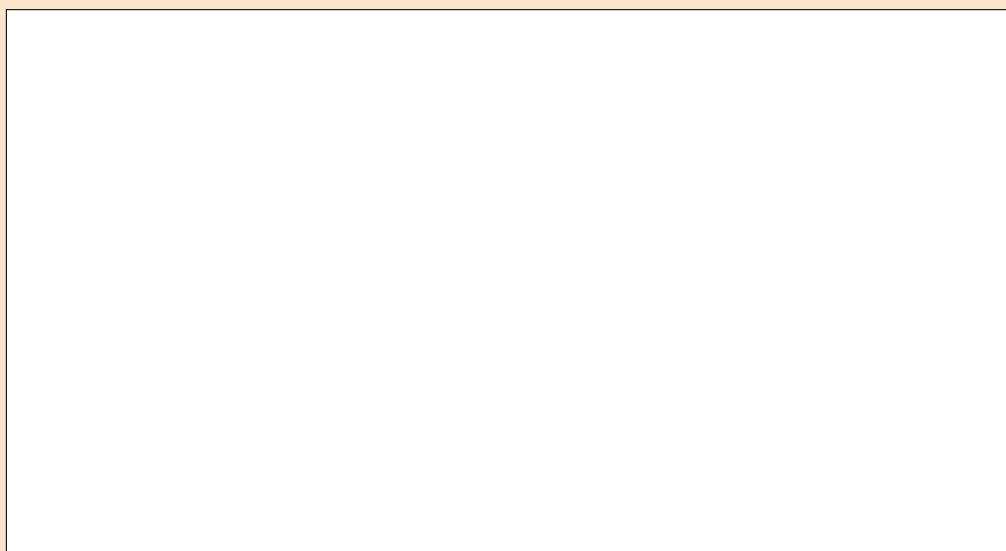
METHOD:

1. Construct the circuit with the cell, the ammeter, 1 bulb and the switch in series.



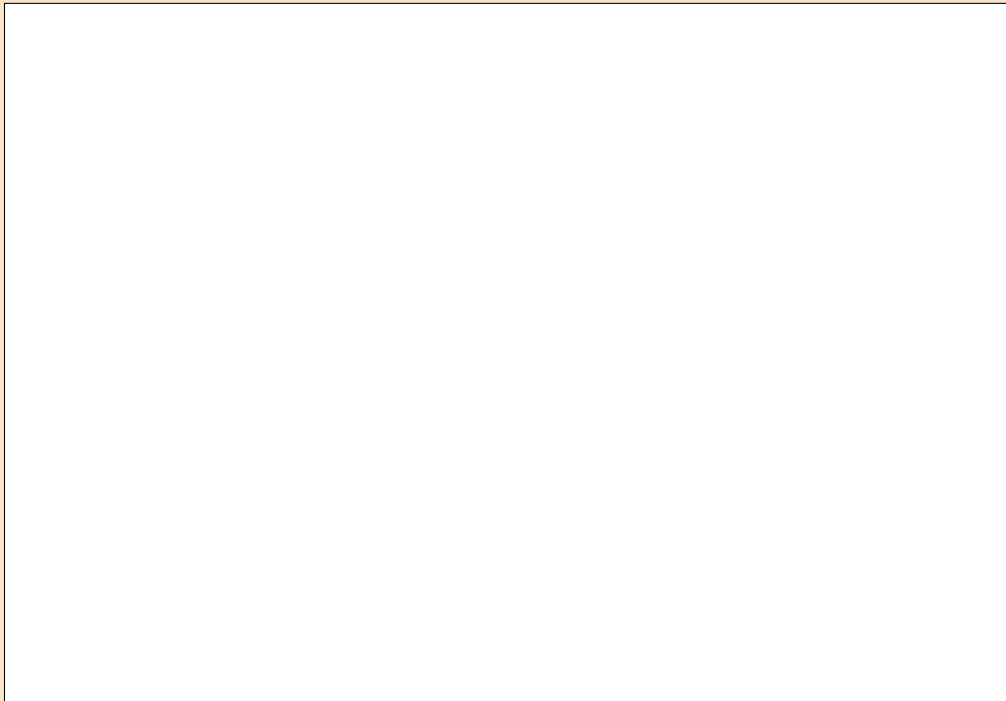
A photo showing the setup.

2. Close the switch, or the circuit if you are not using a switch.
3. Note how brightly the bulb is shining and write down the ammeter reading. Draw a circuit diagram.

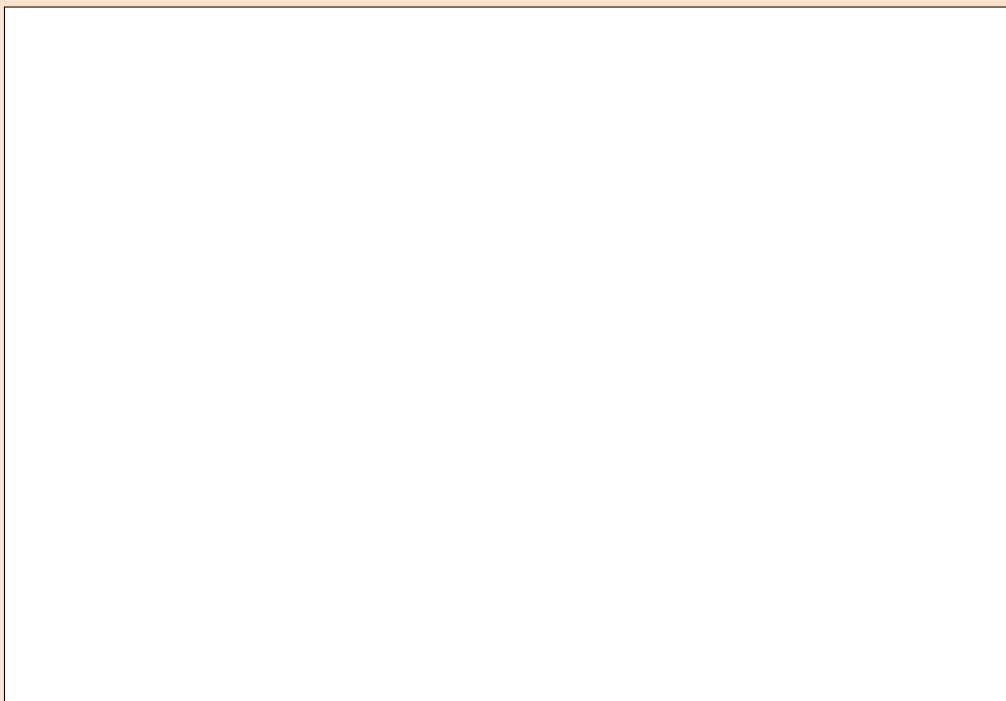


4. Open the switch.
5. Add another light bulb into the circuit.
6. Close the switch.

7. Note how brightly the bulbs are shining and write down the ammeter reading. Draw a circuit diagram.



8. Open the switch.
9. Add the third light bulb into the circuit.
10. Close the switch.
11. Note how brightly the bulbs are shining and write down the ammeter reading. Draw a circuit diagram for the last circuit you built.

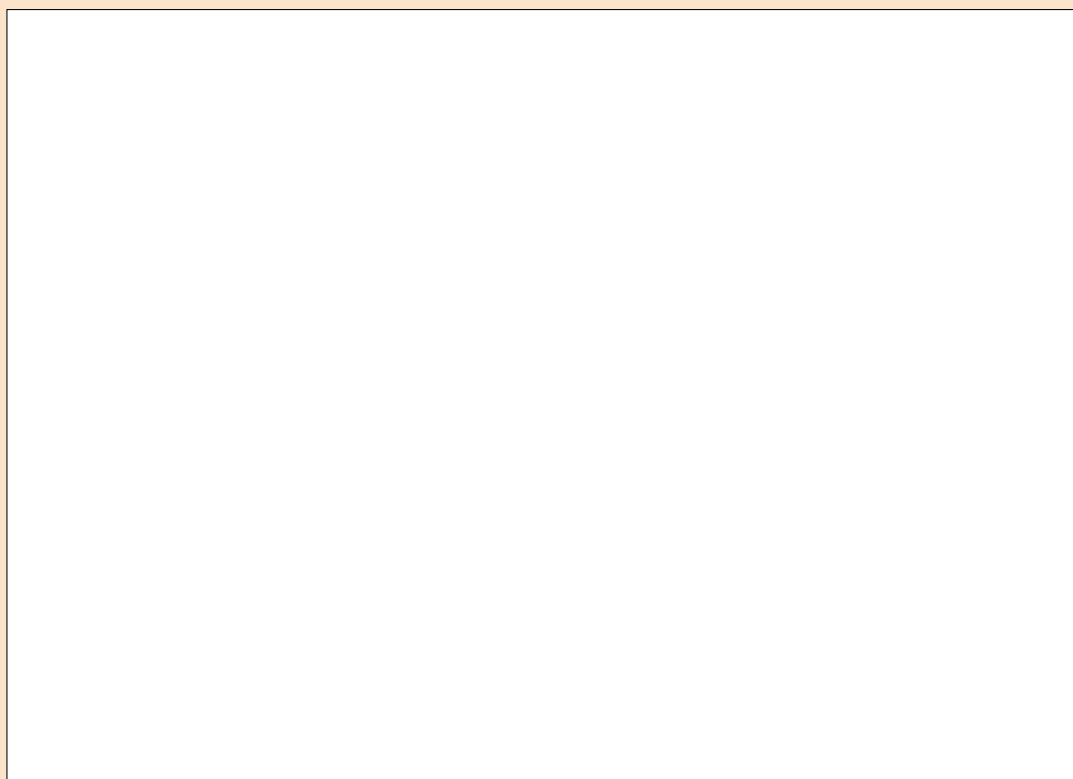


RESULTS:

Complete the table:

Number of bulbs in series	Brightness of bulbs	Reading on ammeter (A)
1		
2		
3		

Draw a graph to show the relationship between the number of bulbs and the current.



ANALYSIS:

1. What happened to the brightness of the bulbs as the number of bulbs increased?

2. When you had two bulbs, did they glow with the same brightness, or was one brighter than the other?

3. When you had three bulbs, did they glow the same as each other or was one brighter than the others?

4. What do your answers to the previous questions tell you about the current in the series circuit?

5. What happened to the reading on the ammeter as you added more bulbs in series?

CONCLUSION:

1. Based on your answers, what happened to the current when more bulbs were added in series?

2. Is your hypothesis accepted or rejected?



As more resistors are added in series, the total resistance of the circuit increases. As the total resistance increases, the current strength decreases. What would happen if we increased the number of cells connected in series? Would the current become larger or smaller? Let's investigate.

INVESTIGATION: How does adding more cells in series affect the current?

AIM: To investigate the effect of increasing the number of cells connected in series on the electric current strength.

HYPOTHESIS: Write a hypothesis for this investigation. Remember to mention how the increase in the number of cells will affect the current strength.

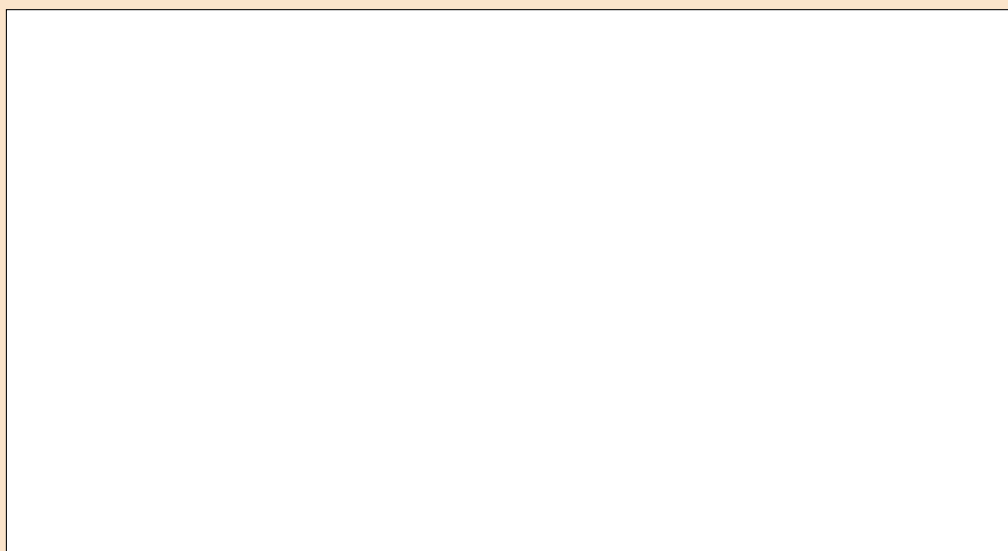


MATERIALS AND APPARATUS

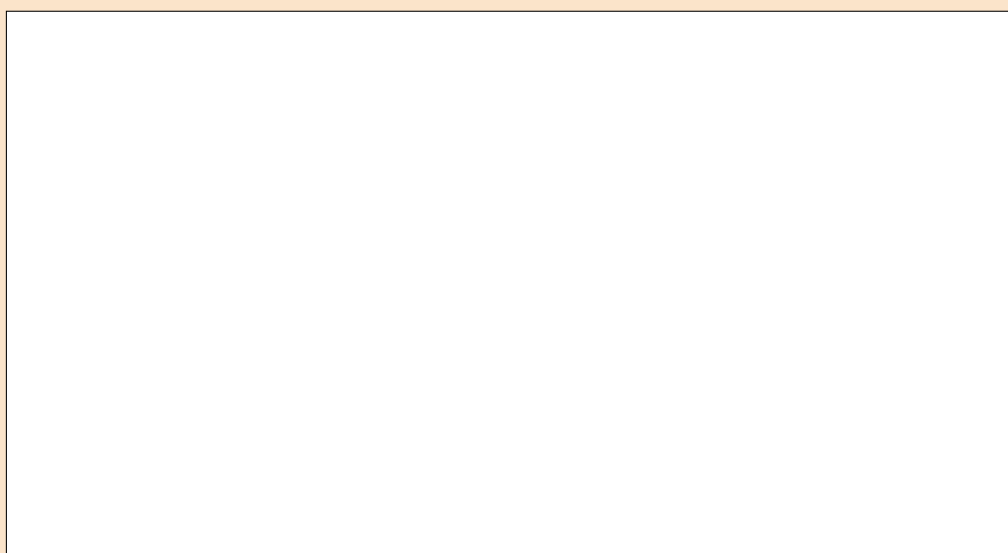
- three 1,5 V cells
- insulated copper conducting wires
- ammeter
- 2 torch light bulbs (or 1 torch light bulb and one resistor)

METHOD:

1. Construct a circuit with 1 cell, the ammeter and the two torch light bulbs.
2. Observe the brightness of the bulbs and record the ammeter reading in the table of results. Draw a circuit diagram.

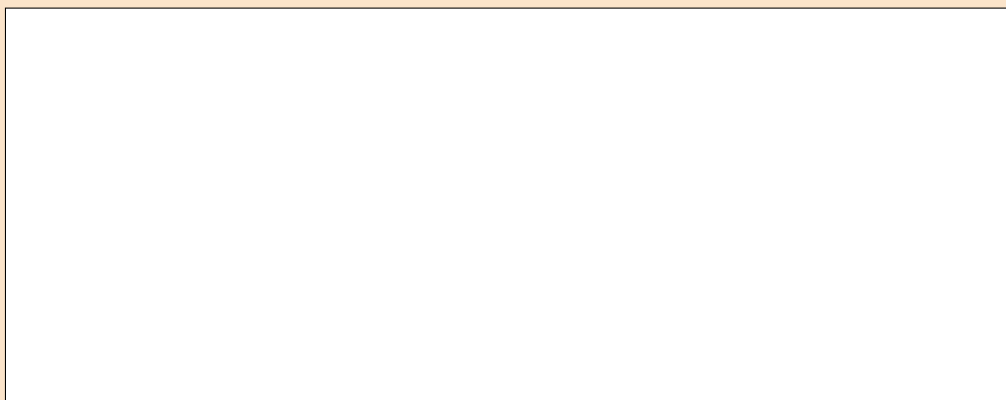


3. Add a second cell in series and observe the brightness of the bulbs. Draw a circuit diagram of your circuit.
4. Record the ammeter reading in the table of results. Draw a circuit diagram.



5. Add a third cell in series and observe the brightness of the bulbs. Draw a circuit diagram of your circuit.

6. Record the ammeter reading in the table of results. Draw a circuit diagram.



RESULTS:

Complete the table:

Number of cells in series	Brightness of bulbs	Reading on a mmeter (A)
1		
2		
3		

CONCLUSION:

1. What can you conclude from the shape of the graph?

2. Is your hypothesis true or false?

We have seen that increasing the number of cells in series increases the current, but increasing the number of resistors decreases the current.

We will now investigate the current strength at different points in a series circuit.





INVESTIGATION: Testing the current strength

INVESTIGATIVE QUESTION: Is the current strength the same at all points in a series circuit?

HYPOTHESIS: Write a hypothesis for this investigation. What do you think will happen in this investigation?

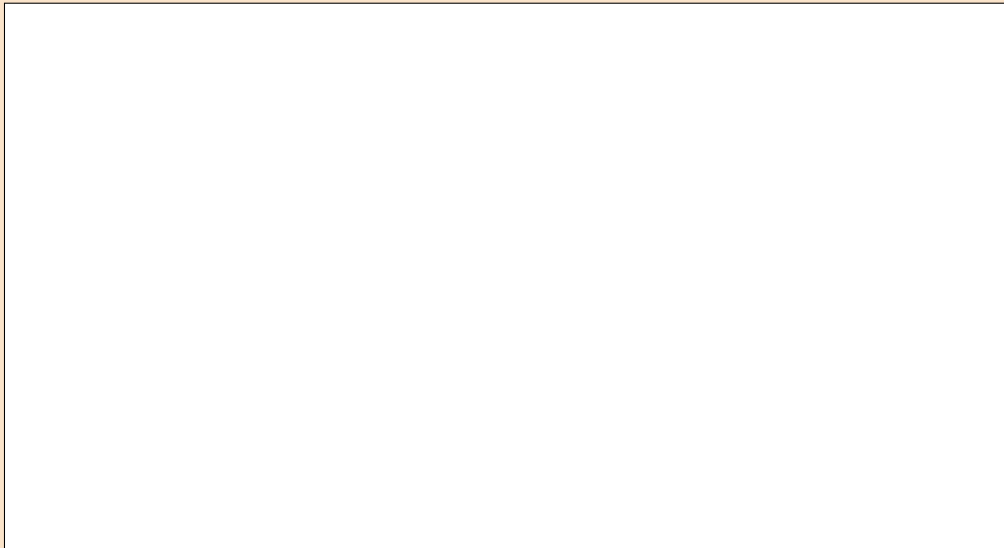
MATERIALS AND APPARATUS:

- insulated copper connecting wires.
- two 1,5V cells
- two torch light bulbs
- ammeter

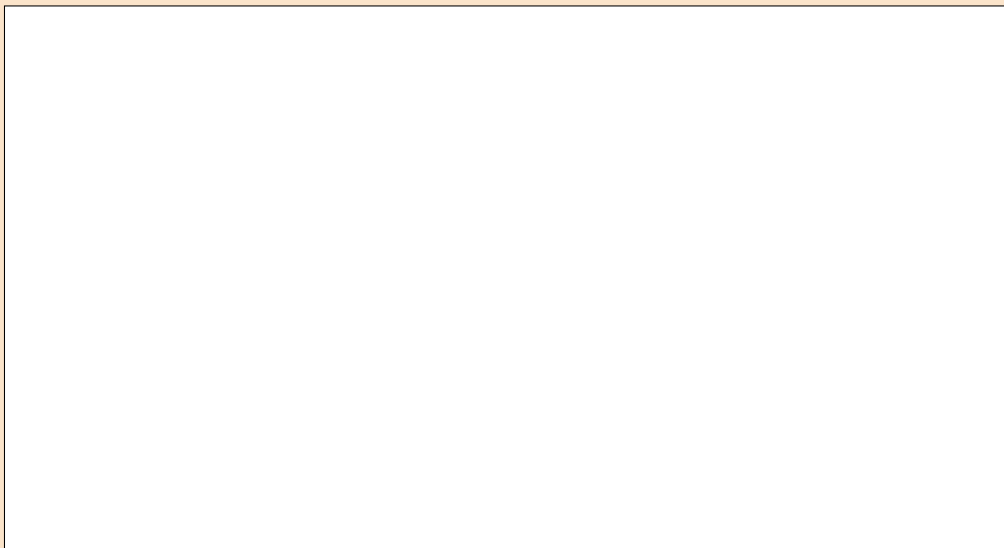
METHOD:

1. Set up a series circuit with two cells and two torch light bulbs in series with each other.
2. Insert an ammeter in series between the positive terminal of the batteries and the first torch bulb.
3. Measure the current strength using the ammeter. Draw a circuit diagram of this set up.

4. Remove the ammeter and close the circuit again.
5. Insert the ammeter in series between the two torch bulbs.
6. Measure the current strength using the ammeter. Draw a circuit diagram of this set up.



7. Remove the ammeter and close the circuit again.
8. Insert the ammeter in series between the last torch bulb and the negative terminal of the batteries.
9. Measure the current strength using the ammeter. Draw a circuit diagram of this set up.



RESULTS:

Complete the following table:

Position of ammeter in circuit	Ammeter reading (A)
Between positive terminal of cell and first bulb	
Between two bulbs	
Between negative terminal of cell and last bulb	

CONCLUSIONS:

1. Write a conclusion based on your results.

2. Is your hypothesis true or false?



VISIT

Animation showing the movement of electrons.

bit.ly/19Ww8pW



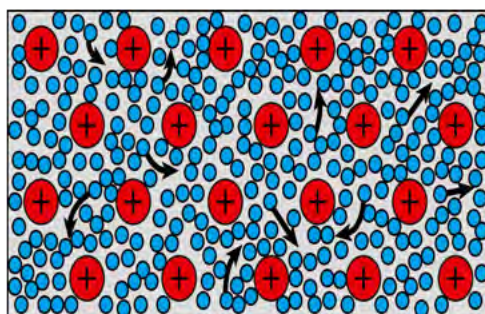
In a series circuit, there is only one pathway for the electrons to move through. The current strength is the same everywhere in that pathway.

What have we learned about series circuits?

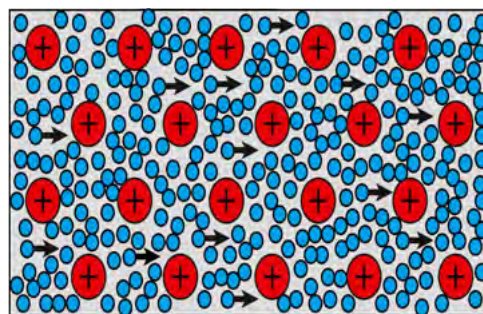
- There is only **one** pathway for the electrons to follow.
- The current flows at the **same strength** everywhere in a series circuit, because there is only one pathway. We say that the current is the same at all points in the circuit.
- If you add more resistors in series, the current in the whole circuit **decreases**.

Why does the current stay the same at all points? Let's think about how electric current moves through a circuit. Do you remember that we spoke about the delocalised electrons in metals in the last chapter?

The electrons in a conductor normally drift in various different directions within a metal, as shown in the diagram.



Delocalised electrons move freely in a conducting wire.



When the wire is connected in a closed circuit, the electrons move towards the positive terminal of the battery.

VISIT

Flip the switch and watch the electrons with this simulation.

bit.ly/15NlqBd



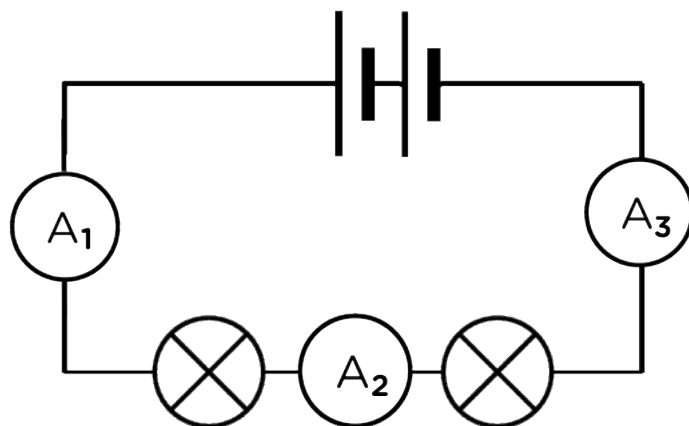
When we build a closed circuit with a cell as an energy source, the electrons will all begin to move towards the positive side of the cell. The rate at which the electrons move, is determined by the resistance of the conductor.

There are electrons everywhere in the conducting wires and electrical components. When the circuit is closed, all the electrons start moving in the same general direction **at the same time**. This is why a light bulb turns on immediately when you close the switch.

In a series circuit, all the electrons travel through every component and wire as they travel through the circuit. All the electrons experience the same resistance

and so they all move at the same rate.

This means that in the diagram below, the readings on all three ammeters will be the same, so: $A_1 = A_2 = A_3$



NEW WORDS

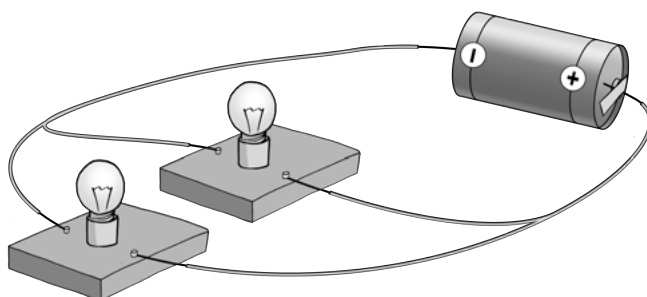
- parallel circuit



3.2 Parallel circuits

Parallel circuits offer more than one pathway for the electrons to follow. When constructing a parallel circuit, we say that components are connected **in parallel**.

Look at the diagram which shows how two light bulbs are connected in parallel.



There are two paths for the current in this parallel circuit, one path through each of the bulbs.

VISIT

Watch a video that explains the difference between series and parallel circuits

bit.ly/1f5hZ0W



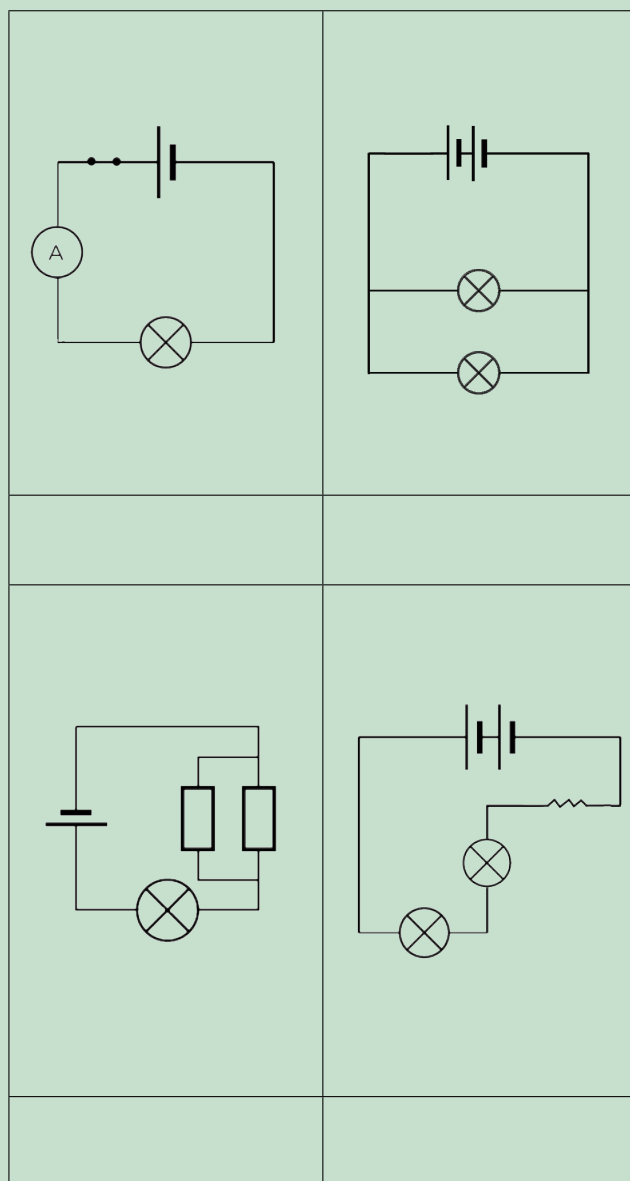
How can you tell whether or not a circuit is connected in series or in parallel? Let's look at some circuit diagrams to tell the difference.

ACTIVITY: Series or parallel?

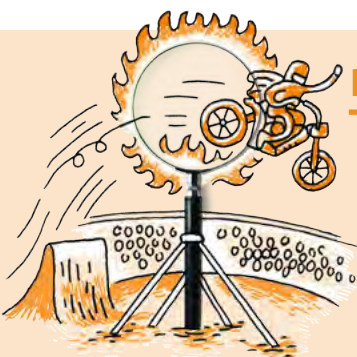
INSTRUCTIONS:

Look at the following circuits and write down which are in series and which are in parallel. The series circuits will only offer one pathway, but the parallel circuits will have more than one pathway for the electrons to follow.





Let's investigate how parallel circuits work.



INVESTIGATION:

How does adding resistors in parallel affect the current strength?

AIM: To investigate the effect of adding resistors in parallel on the current strength.

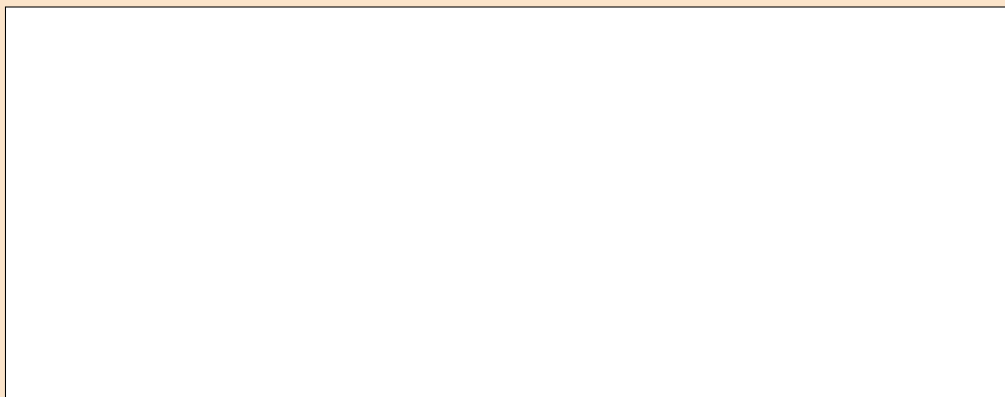
HYPOTHESIS: Write a hypothesis for this investigation.

MATERIALS AND APPARATUS:

- 1,5 V cell
- three identical torch bulbs
- insulated copper conducting wires
- switch
- ammeter

METHOD:

1. Construct the circuit with the cell, ammeter, one bulb and the switch in series.
2. Close the switch.
3. Note how brightly the bulb is shining and record the ammeter reading.
Draw a diagram of your circuit.



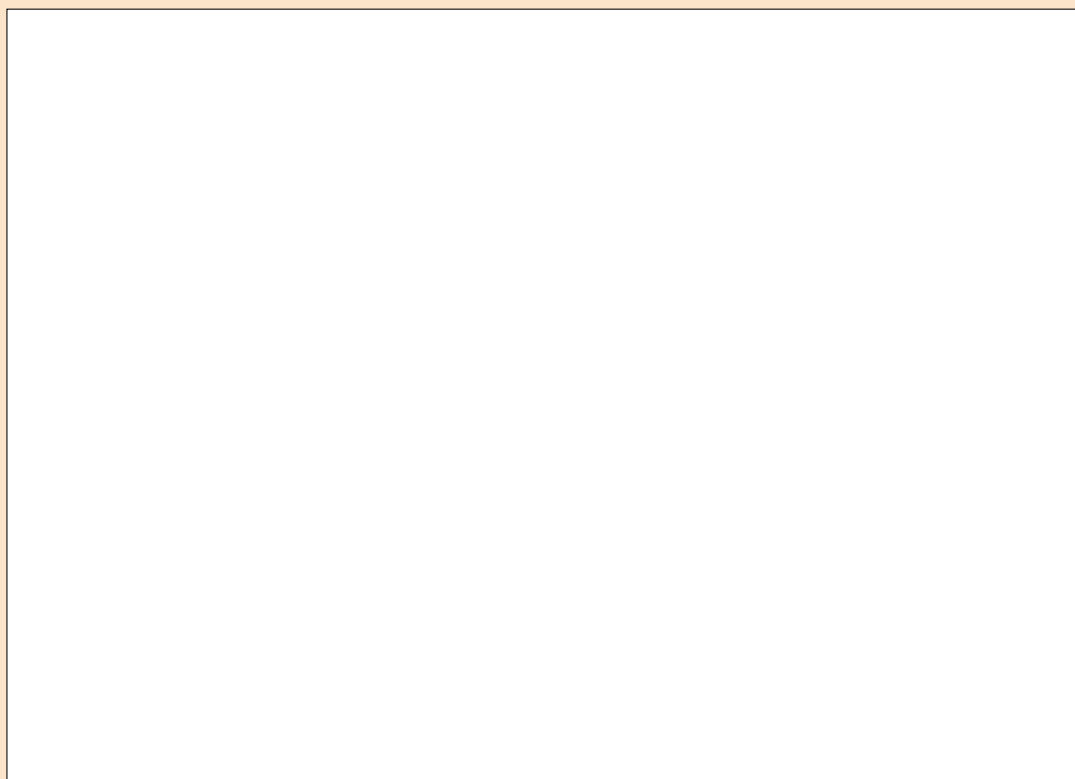
4. Open the switch.
5. Add another light bulb, in parallel to the first, into the circuit.
6. Close the switch.
7. Note how brightly the bulbs are shining and record the ammeter reading.
8. Open the switch.
9. Add the third light bulb, in parallel to the first two, into the circuit.
10. Close the switch.
11. Note how brightly the bulbs are shining and record the ammeter reading.

RESULTS:

Complete the table:

Number of bulbs in parallel	Brightness of bulbs	Reading on ammeter (A)
1		
2		
3		

Draw a graph to show the relationship between the number of bulbs and the current.



ANALYSIS:

1. What happened to the brightness of the bulbs as the number of bulbs increased?

2. When you had two bulbs, did they glow with the same brightness or was one brighter than the other?

3. When you had three bulbs, did they glow the same brightness or was one brighter than the others?

4. What do your answers to the previous questions tell you about the current in the parallel branches of the circuit?

5. What happened to the reading on the ammeter as you added more bulbs in parallel?

CONCLUSION:

1. Based on your answers, what happened to the current when more bulbs were added in parallel?

2. Is your hypothesis true or false?



As more resistors are added in parallel, the total current strength increases. The overall resistance of the circuit must therefore have decreased. The current in each light bulb was the same because all the bulbs glowed with the same brightness. This tells us that the current of electrons must have split up and moved through each of the branches.

We can also connect cells in parallel. What would happen if we increased the number of cells connected in parallel? Would the current get stronger or weaker?

INVESTIGATION: What happens to the current strength when cells are connected in parallel?

AIM: To investigate how increasing the number of cells connected in parallel affects the current strength in a circuit.

HYPOTHESIS: Write a hypothesis for this investigation.

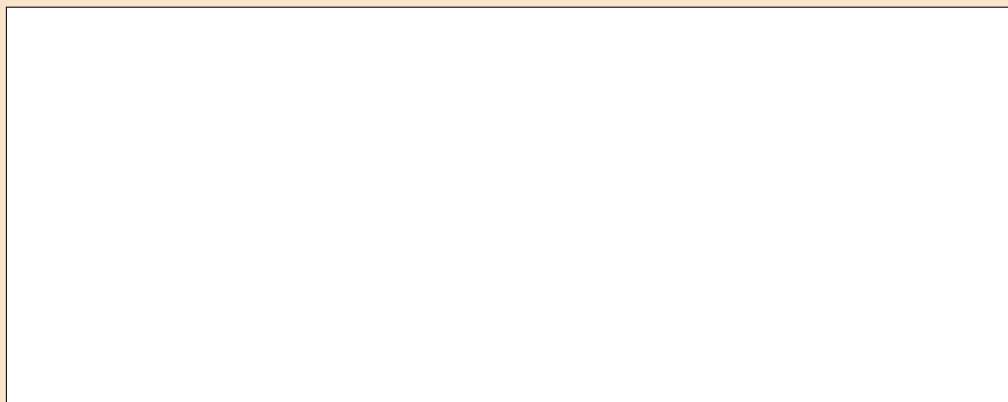


MATERIALS AND APPARATUS

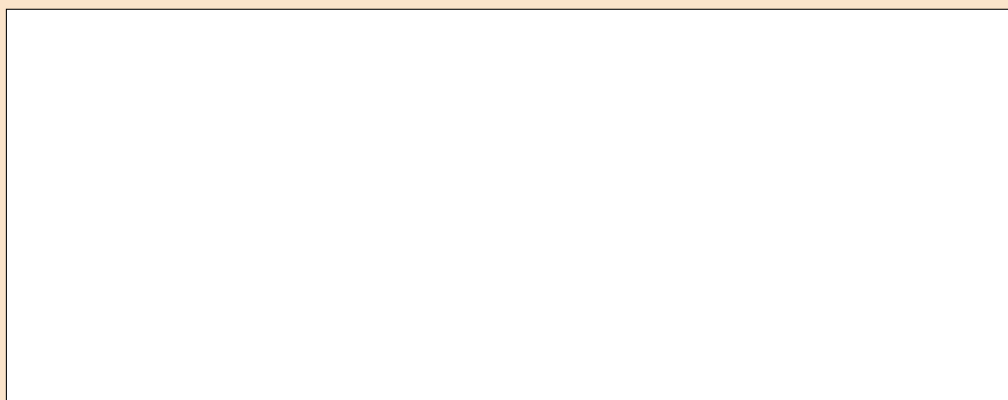
- three 1,5V cells
- one torch light bulb
- insulated copper conducting wires
- ammeter

METHOD:

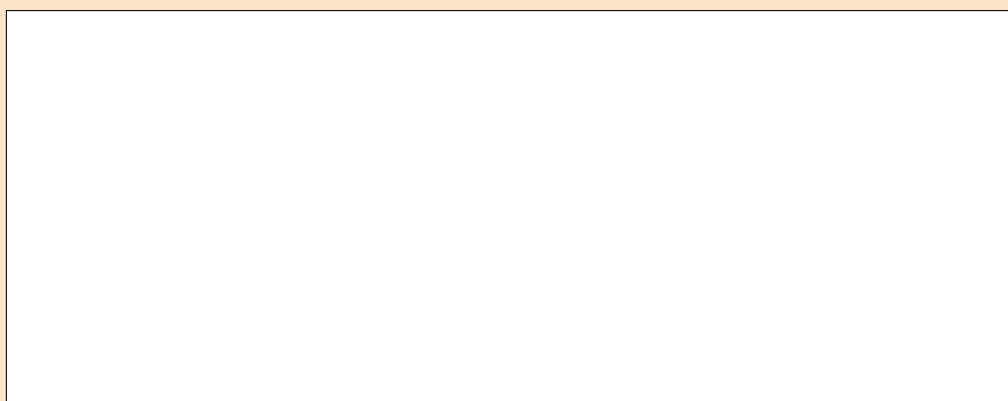
1. Set up a circuit which has one cell, the ammeter and the torch light bulb in series with each other. Draw a circuit diagram of your circuit.



2. Observe the brightness of the bulb and record the ammeter reading.
3. Connect another cell in parallel with the first cell. To connect the second cell in parallel, connect a wire from the positive terminal of the first cell to the positive terminal of the second cell. Connect another wire between the negative terminal of the first battery and the negative terminal of the second battery. Draw a circuit diagram of your circuit.



4. Observe the brightness of the bulb and record the ammeter reading.
5. Connect a third cell in parallel to the other two cells. Draw a circuit diagram of your circuit.



6. Observe the brightness of the bulb and record the ammeter reading.

RESULTS:

Complete the table:

Number of cells in parallel	Brightness of bulb	Reading on ammeter (A)
1		
2		
3		

CONCLUSION:

1. What did you notice about the brightness of the bulbs?

2. What did you notice about the ammeter readings?

3. What conclusion can you draw from your results?



Adding cells in parallel has no overall effect on the current strength. The current strength stays the same if you add cells in parallel.

We saw that the current strength increased when bulbs were connected in parallel. However, we were only testing the current strength at one point in the parallel circuit. How does the current compare in the different pathways of the circuit? Let's do an investigation to find out.

INVESTIGATION: Testing the current strength

INVESTIGATIVE QUESTION: Is the current strength equal at all points in a parallel circuit?

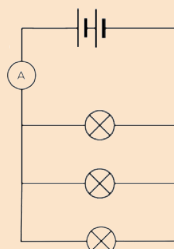
MATERIALS AND APPARATUS:

- insulated copper connecting wires.
- two 1,5V cells
- three identical torch light bulbs
- ammeter

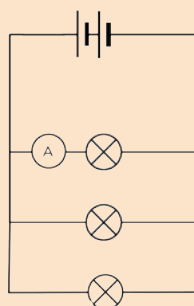


METHOD:

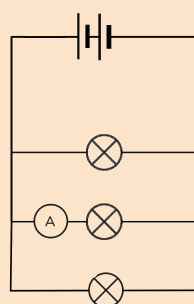
1. Set up a parallel circuit with two cells in series with each other and three torch light bulbs in parallel with each other.
2. Insert an ammeter in series between the cells and the first pathway, as shown in the diagram.



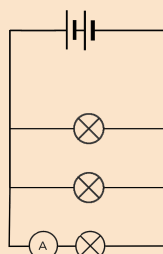
3. Measure the current strength using the ammeter.
4. Remove the ammeter and close the circuit again.
5. Insert the ammeter in series in the first pathway.



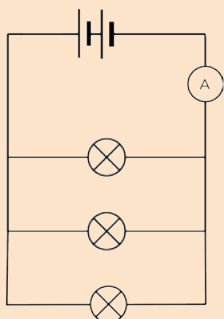
6. Measure the current strength using the ammeter.
7. Remove the ammeter and close the circuit again.
8. Insert the ammeter in series in the second pathway.



9. Measure the current strength using the ammeter.
10. Remove the ammeter and close the circuit again.
11. Insert the ammeter, in series, in the third pathway.



12. Measure the current strength using the ammeter.
13. Remove the ammeter and close the circuit again.
14. Insert the ammeter in series between the first pathway and the cells on the opposite side to the first reading.



15. Measure the current strength using the ammeter.

RESULTS:

Position of ammeter in circuit	Ammeter reading (A)
between the cell and first pathway	
in the first pathway	
in the second pathway	
in the third pathway	
between the cell and the first pathway	

CONCLUSION:

1. Write a conclusion based on your results.

2. Is your hypothesis true or false?



What have we learned about parallel circuits?

- There is **more than one** pathway for the current to follow.
- The current divides between the different branches so that each branch gets some of the current. As the torch bulbs in each branch in our example were identical, the current divided equally between them.
- If you add more resistors in parallel, the total current supplied by the cell in the circuit **increases**.

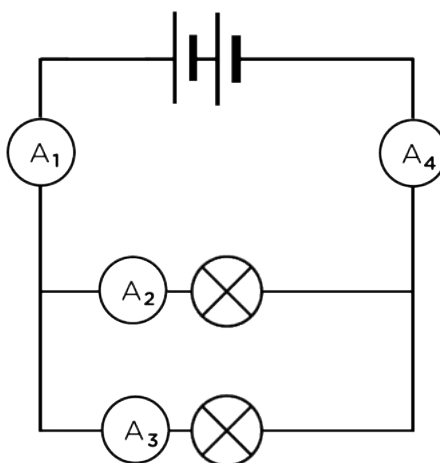
Why does the current divide when offered an alternative pathway?

Imagine that you are sitting in a school hall during assembly. You are bored and waiting for it to end so that you can go out to break to chat to your friends. There is only one exit from the hall. When you are dismissed, everyone has to exit through the same door. It takes a while because only some learners can leave at a time.

Now imagine that there is a second door that is the same as the first door. Now you and your friends have a choice of which door to go through. The speed at which the learners exit the hall will increase and some of you will exit through the first door while others will exit through the second door. No one can go through both doors at the same time.

This is similar to the way current behaves when in a parallel circuit. As the electrons approach the branch in the circuit, some electrons will take the first path and others will take the other path. The current is divided between the two pathways.

In the following circuit $A_1 = A_4$ and $A_1 = A_2 + A_3$ and $A_4 = A_2 + A_3$



We have looked at how resistors and cells behave in series and parallel circuits. Let's look at how different metals conduct electricity. All conductors have some resistance in a circuit. Are some metals better conductors of electricity than others?

Let's have a look at which metals offer more resistance than others to the flow of charge (current) through an electric circuit.

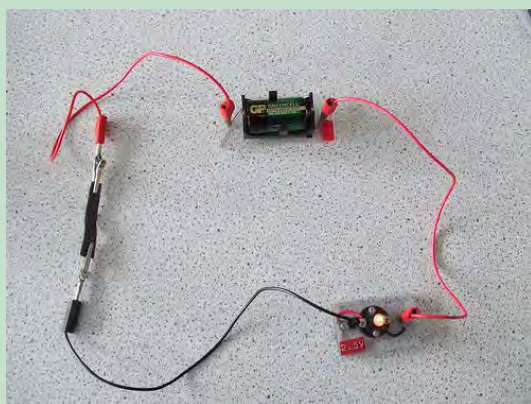
ACTIVITY: Which metals offer the most resistance?

MATERIALS:

- a cell
- torch light bulb
- insulated copper wires
- lengths of copper, aluminium, zinc and nichrome wire
- crocodile clips (if available)

INSTRUCTIONS

1. Build a circuit with the cell and the torch light bulb and leave a gap for the metal to be tested. You can use crocodile clips at the end of each piece of metal for easy insertion.
2. Insert each metal into the circuit (one at a time).

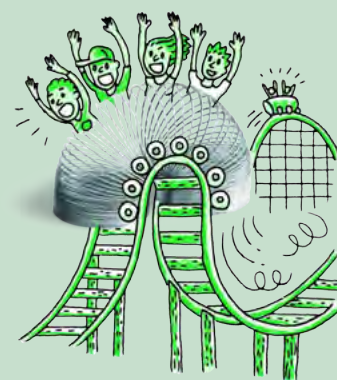
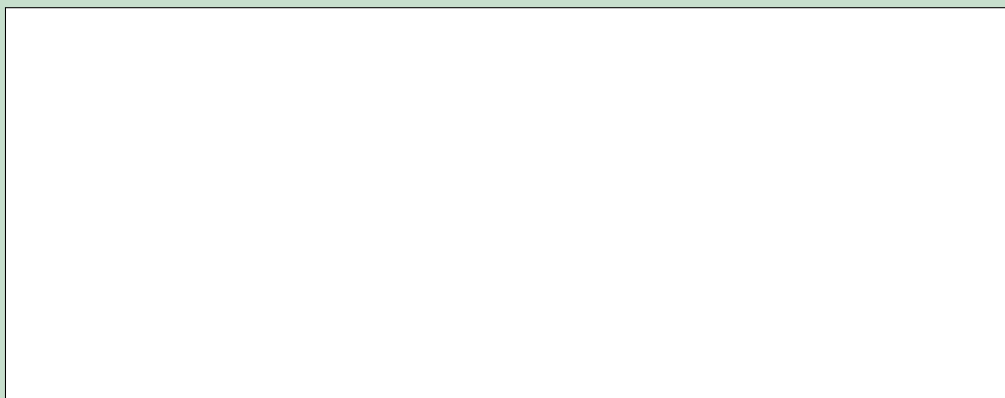


An example circuit with a cell, a light bulb and the piece of metal being tested.

Observe the brightness of the bulb.

QUESTIONS:

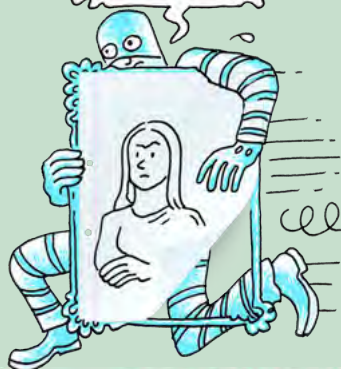
1. Draw a circuit diagram of your apparatus.



TAKE NOTE

In Gr. 9 we will look at the other factors that influence resistance. If you want to see the content in other grades, remember that you can visit

<http://www.curious.org.za>



2. Why can we use the brightness of the bulb to qualitatively measure resistance?

3. List the metals in order of increasing resistance.

4. Why do you think copper is used for connecting wires in electrical circuits?

There are several factors which influence the amount of resistance a material offers to an electric current. We have seen that the type of material is one of those factors.

3.3 Other output devices

VISIT

Watch this video about the history of the LED

bit.ly/1bC5qKc



Light bulbs are not the only devices used in electrical circuits. Devices that use electrical energy to function, including light bulbs, are called output devices. Let's look at some other common examples of output devices.

LEDs (Light-Emitting Diodes)

LEDs are widely used electronic devices. They are small lights but they do not have a filament like an incandescent bulb has. They therefore cannot burn out, as there is no filament to wear out, and they do not get as hot. LEDs are used in electronic timepieces, high definition televisions and many other applications. Larger LEDs are also replacing traditional light bulbs in many homes because they do not use as much electricity. They last longer than incandescent bulbs and are more efficient.



Different LED bulbs.



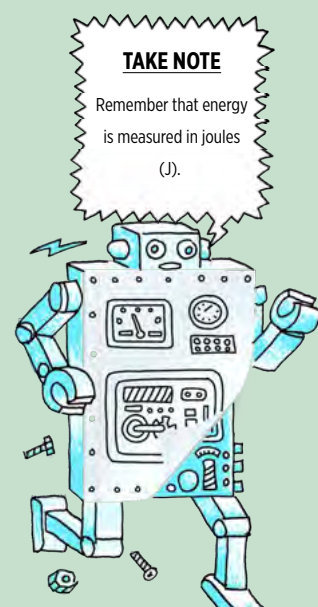
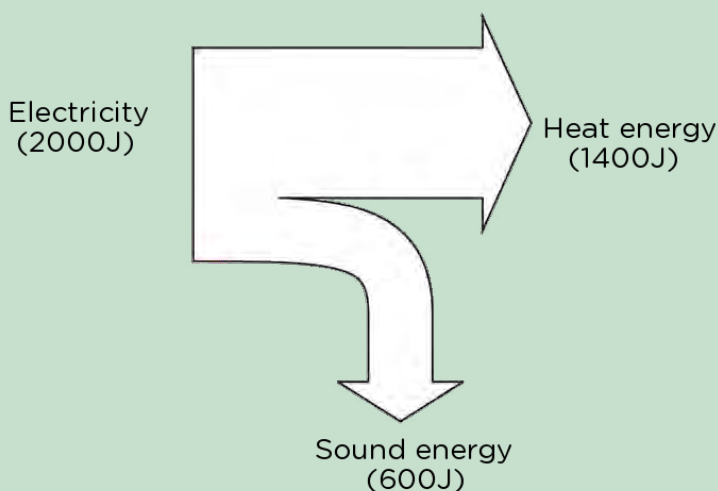
In the last chapter, we looked at the energy transfers in an electrical system. We will now represent energy transfer within electrical systems in a different way. We will apply this new representation to the difference between energy outputs in an LED and an incandescent light bulb.

ACTIVITY: Sankey diagrams

You might have drawn Sankey diagrams in Grade 7. If not, here is some quick revision.

In an energy system, input energy is transferred to useful output energy and wasted output energy. A Sankey diagram is a visual and proportional representation of the energy transfers that happen in a system.

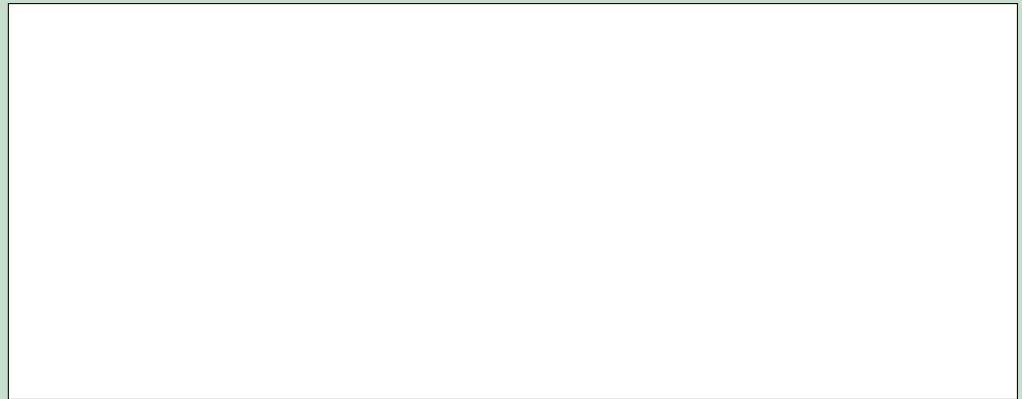
For example, a kettle uses about 2000 J of input energy, but only about 1400 J is used to heat the water. The remaining 600 J is wasted as sound. Here is the Sankey diagram to represent the energy transfer.



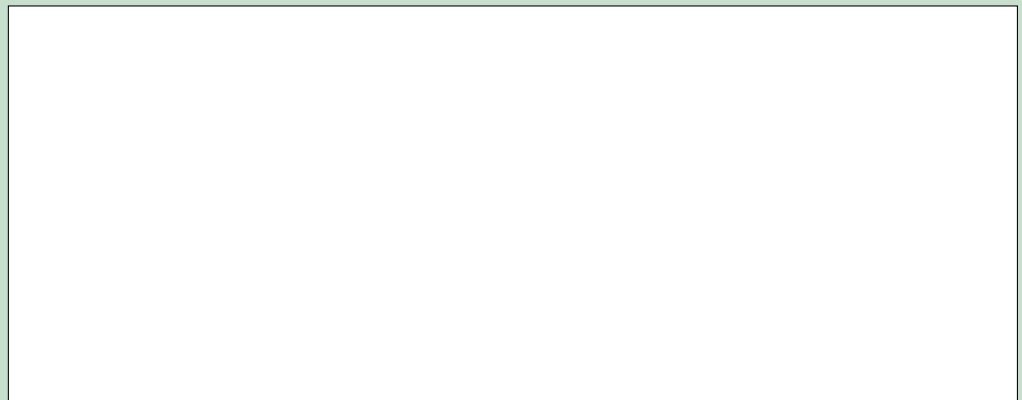
QUESTIONS:

We will now compare an LED with an incandescent light bulb.

1. Draw a Sankey diagram for an LED if the input energy is 100 J, 75 J of energy is used to produce light and the rest is lost as heat.



2. Draw a Sankey diagram for a filament light bulb if the input energy is 100 J, the wasted heat energy is 80 J and the rest produces light.



3. Which bulb do you think is more efficient? Explain your answer.

Can you think of any other output devices? Make a list of as many as you can.

VISIT

An electricity timeline
animation.

bit.ly/1fkZb8E



ACTIVITY: History of electricity production

INSTRUCTIONS:

1. Work in groups of three or four.
2. Research the history of electricity production: How was electricity discovered and how did electricity become widely used?
3. Create a basic timeline for the discovery of electricity and it's production.



ACTIVITY: Careers

INSTRUCTIONS:

1. Choose a career related to electricity production.
2. Write a short paragraph describing the career. Include information on how one can study or prepare for your chosen career.

[illegible]



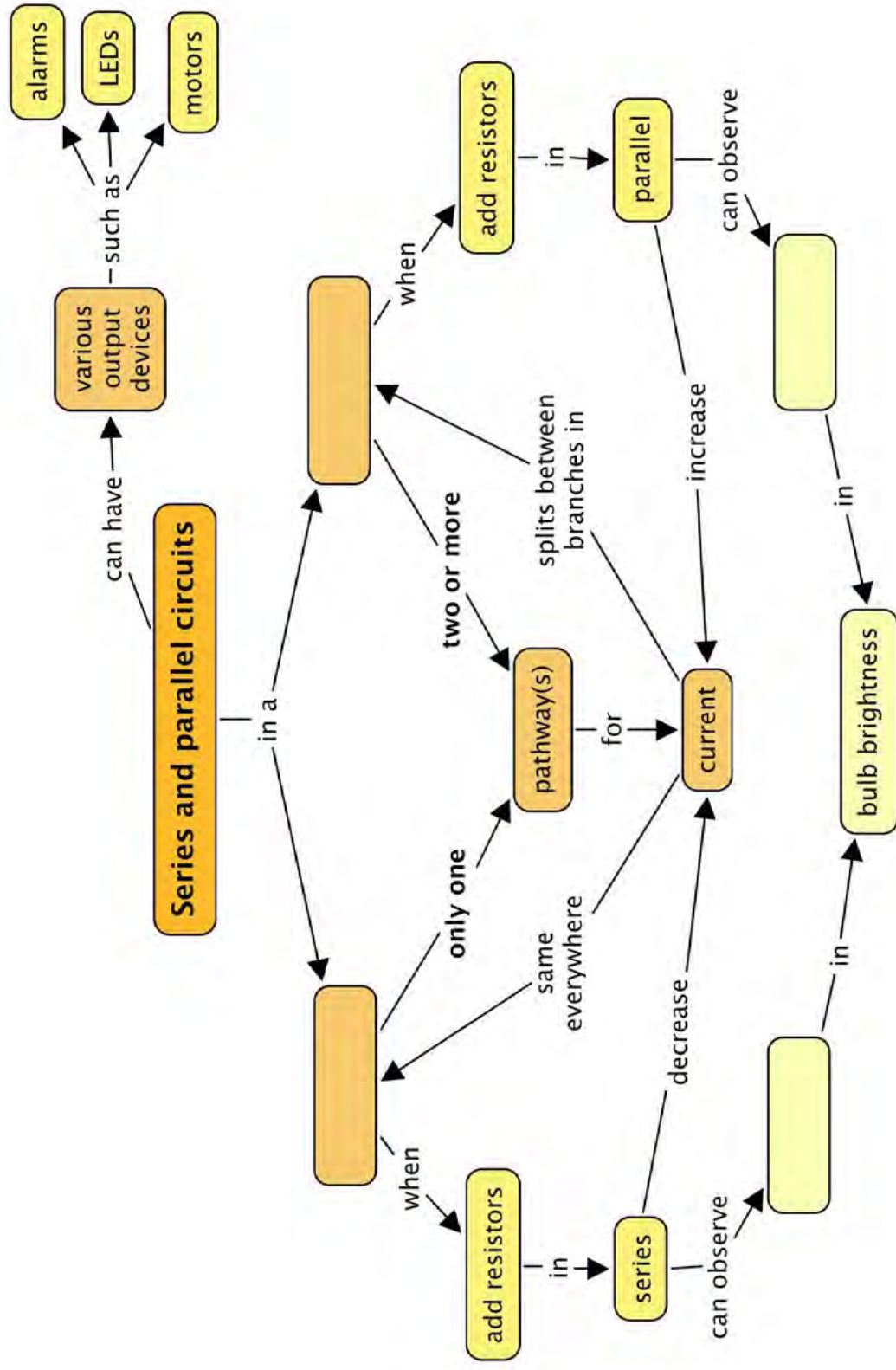
SUMMARY:

Key Concepts

- A series circuit has only one pathway for the electrons to travel through.
- A parallel circuit has more than one pathway for the electrons to travel through.
- In a series circuit, the current is the same at all points in the circuit.
- In a series circuit, the resistance increases as more resistors are added in series.
- In a parallel circuit, the current splits between the available paths.
- In a parallel circuit, the resistance decreases as more resistors are added in parallel.

Concept Map

Complete the concept map on the following page to summarise what you have learned about series and parallel circuits.

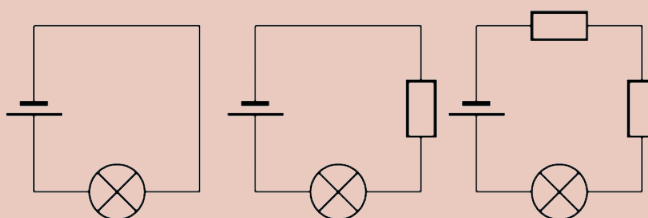




REVISION:

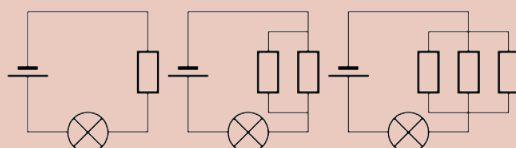
1. Look at the following circuit diagrams and decide whether they are series circuits or parallel circuits. Write the correct answer in the space below each diagram. [6 marks]

2. Look at the three circuit diagrams. Rank the circuits from brightest bulb to dimmest bulbs. [3 marks]



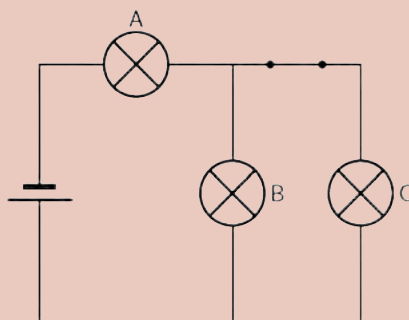
3. Explain your choices in the previous question. [5 marks]

4. Look at the three circuit diagrams. Rank the circuits from brightest bulb(s) to dimmest bulb(s). [3 marks]



5. Explain your choices in the previous question. [5 marks]

6. Look at the circuit diagram below. Each light bulb is identical.

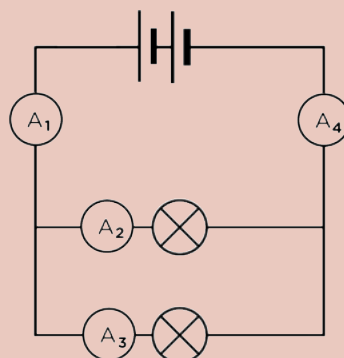


a) Is this a series or parallel circuit? Explain your answer. [2 mark]

b) How do the brightness of bulbs A, B and C compare? (which is the brightest?) [3 marks]

- c) What would happen to the brightness of the bulbs if the switch was opened? Explain your answer. [5 marks]

7. Study the following diagram.



- a) What is the relationship between the ammeter readings on A1 and A4? In other words, how do the current strengths compare at these points in the circuit? Explain your answer. [3 marks]

- b) What is the relationship between the ammeter readings on A1, A2 and A3? In other words, how do the current strengths compare at these points in the circuit? Explain your answer. [3 marks]

Total [38 marks]



Draw and discover the possibilities of what a slinky can be.





KEY QUESTIONS:

- Where does light come from?
- How does light travel?
- How do we see?
- Why do leaves look green?
- How do mirrors work?
- Why do my legs look crooked underwater?

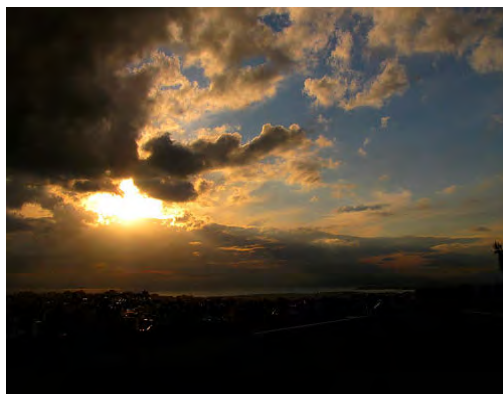
In this chapter we will learn about **visible light**. We call it visible light because we can see it with our own eyes. There are different forms of light which we cannot see with our naked eyes. Ultraviolet light is an example of a form of light which we cannot see with just our eyes. We will focus our attention on the visible light spectrum and investigate how we are able to see different colours and how light behaves.

NEW WORDS

- luminous
- radiation
- rectilinear
- propagation

4.1 Radiation of light

Where does light come from? Natural light comes from luminous objects such as the Sun and light bulbs. We say that these objects emit light.



The Sun is our main source of light on Earth.



A light bulb is a luminous object as it emits light.

VISIT

The speed of light (video)
bit.ly/GAMgFW

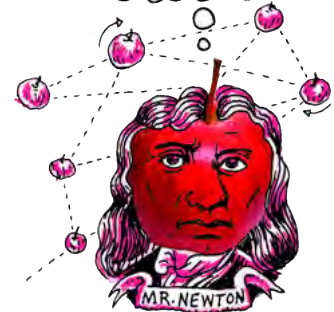




This image from NASA shows the Earth's lights at night. You can see how much we rely on light nowadays.

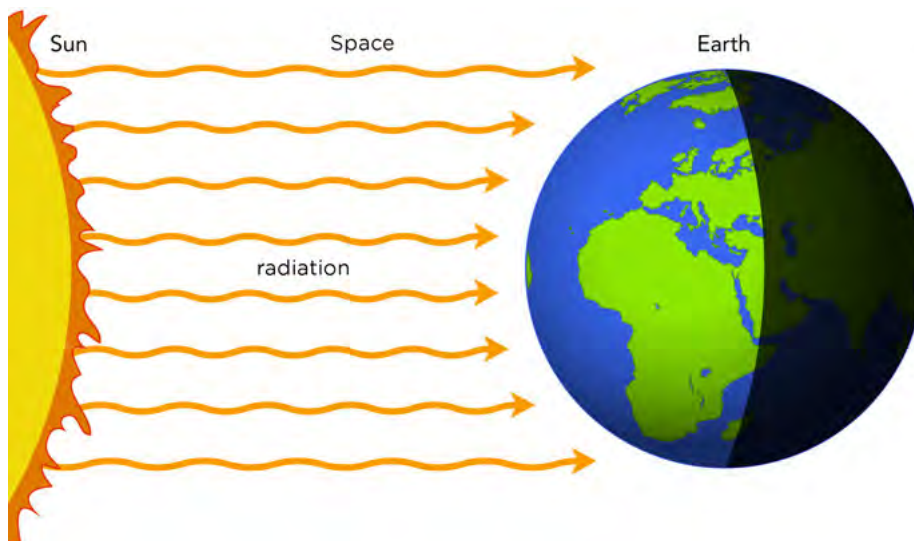
DID YOU KNOW?

If you could travel at the speed of light you could travel around the equator 7,5 times in 1 second!



TAKE NOTE

The Moon is NOT a luminous object as it does not emit its own light. It reflects the light from the Sun.



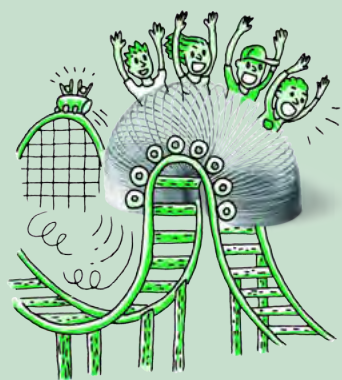
Light and heat are transferred to Earth through space from the Sun by radiation.

DID YOU KNOW?

It takes light 8 minutes to travel from the Sun to the Earth.



Let's look at how light travels. We will make a simple camera to investigate how light travels.



ACTIVITY: Make a pinhole camera

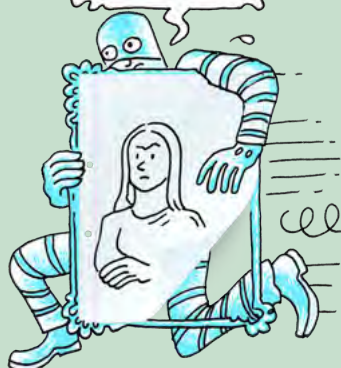
MATERIALS:

- Pringles chip can
- craft knife
- aluminium foil
- tape
- ruler
- drawing pin

INSTRUCTIONS:

TAKE NOTE

The Sun emits radiation in **all directions**, but in the diagram here, only the radiation which reaches Earth has been shown.



1. Measure 5 cm from the bottom of the can (opposite end to the plastic lid) and make a mark all around the can.



2. Cut through the can along the line so that you have cut the can into 2 pieces.



3. If you have a clear lid, put a piece of wax paper on top of the lid before sticking everything together.

4. Place the lid between the 2 pieces and stick it all together using tape.



5. Wrap the aluminium foil around the can to prevent any light from coming in from the sides.



6. Use a drawing pin to make a hole in the centre of the metal base of the can.
7. Go outside with your pinhole camera.
8. Point the metal end with the hole at an object which is in bright sunlight.
9. Cup your hands around the other end and look through the open end.

QUESTIONS:

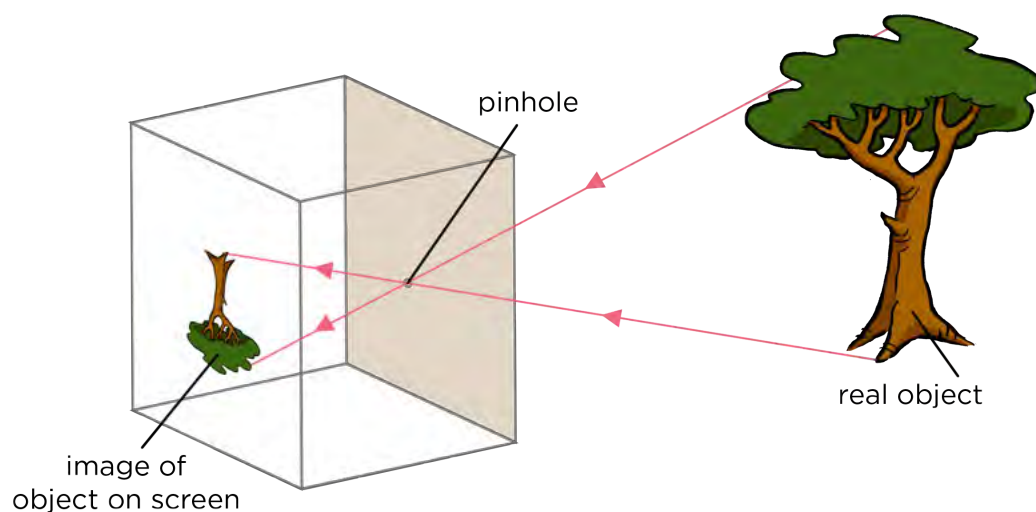
1. What did you see when you looked through the open end of the tube?

2. What happens when you move closer or further away from an object?

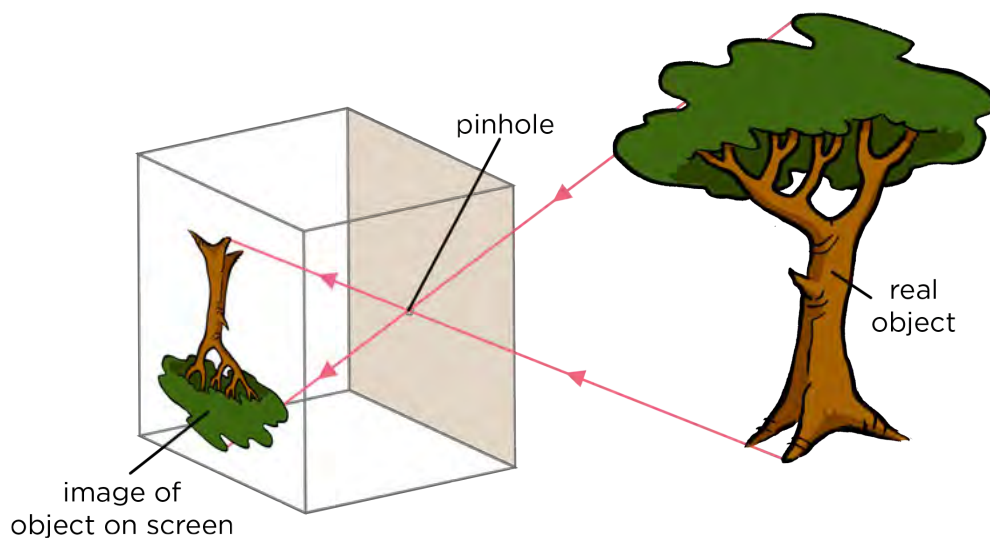


Did you see an upside down image? Why is it upside down?

We see objects because light reflects off them and enters our eyes. If the image is upside down it means that the light from the bottom of the object has arrived at the top of the screen and the light from the top of the object has reached the bottom of the screen, as shown in the following diagram.



When you moved closer to the object, the image appeared bigger, as shown in the following diagram.



VISIT

Can you use what you have learnt to understand how this shadow illusion works?

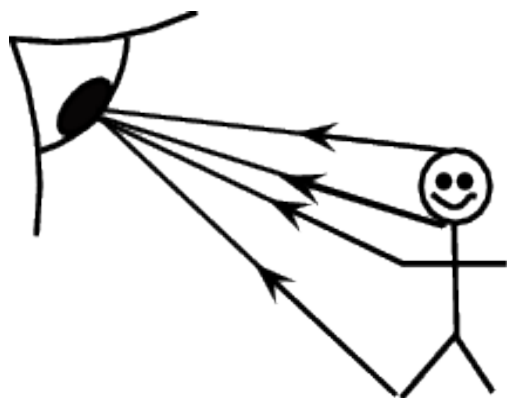
bit.ly/156mxly



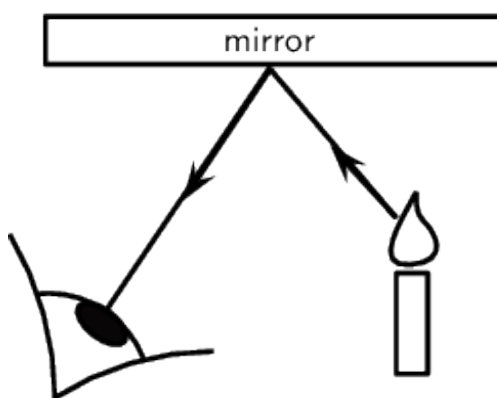
What does this mean? It means that light must be travelling in straight lines. This is called the **rectilinear propagation** of light.

Ray diagrams

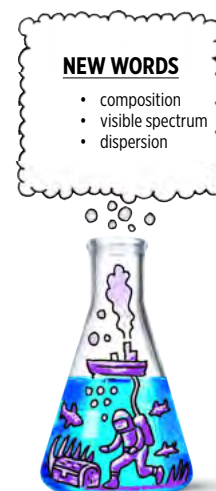
A ray diagram is a drawing that shows the path of light. Light rays are drawn using straight lines and arrowheads, because light travels in straight lines. The figure below shows some examples of ray diagrams.



A ray diagram showing how you see another person.



A ray diagram showing how you see a reflection in a mirror.



4.2 Spectrum of visible light

The visible light spectrum is the light that we are able to see with our naked eyes. Have you ever wondered why everything is colourful and not just black and white? Have you ever seen a rainbow and wondered where the colours have come from? The colours that we see everyday are part of the visible light spectrum. Let's investigate the visible light spectrum.

ACTIVITY: Splitting white light

MATERIALS:

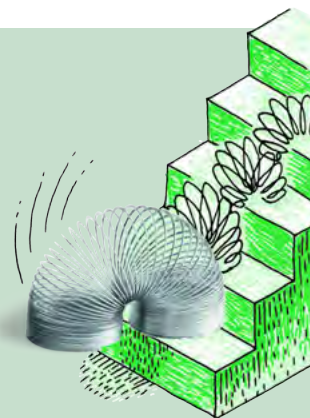
- triangular perspex prism
- ray box and power source

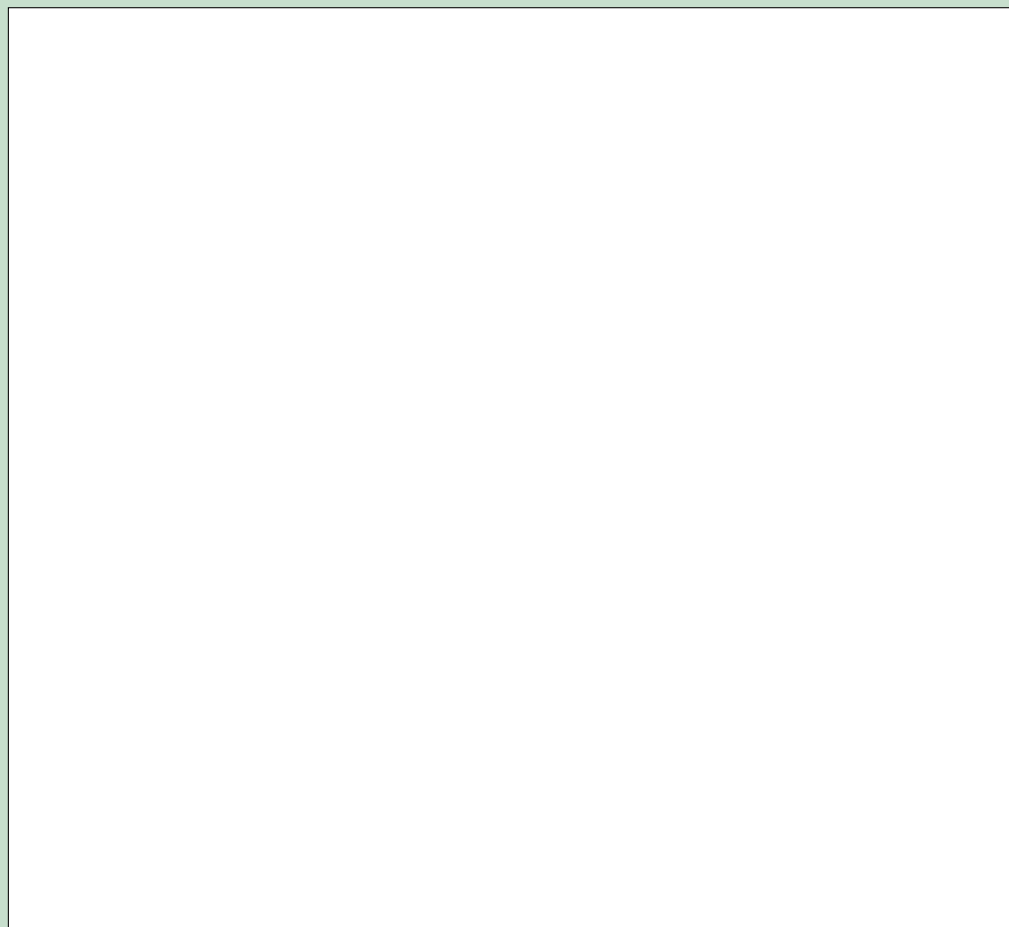
INSTRUCTIONS:

1. Connect the ray box to the power source. If you do not have a ray box, your teacher will show you how to use a piece of cardboard with a slit cut into it.
2. Place the triangular prism on a white background.
3. Shine a beam of white light through the side of the prism.

QUESTIONS:

1. Draw a picture showing what you observe.





2. Write a description of what you observed.

3. Write down the order in which the colours appear.

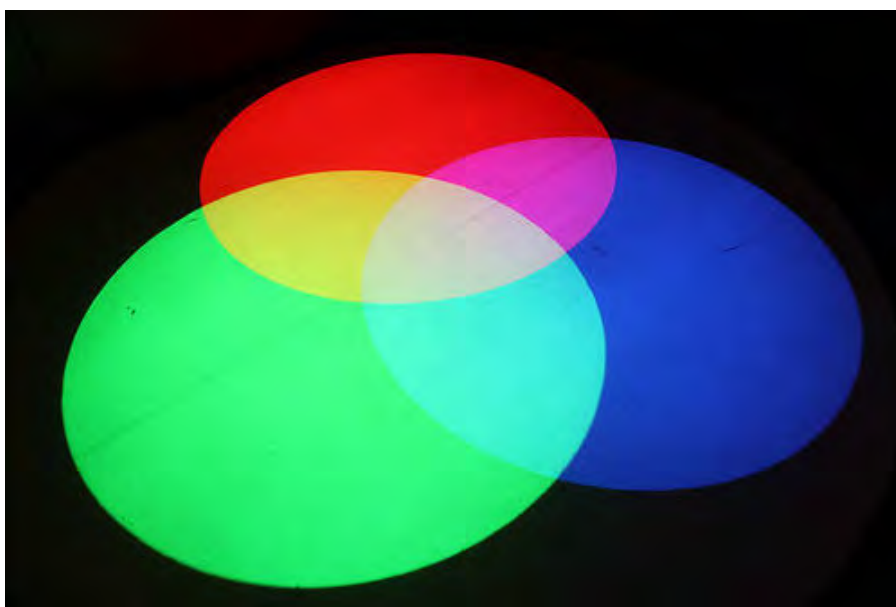
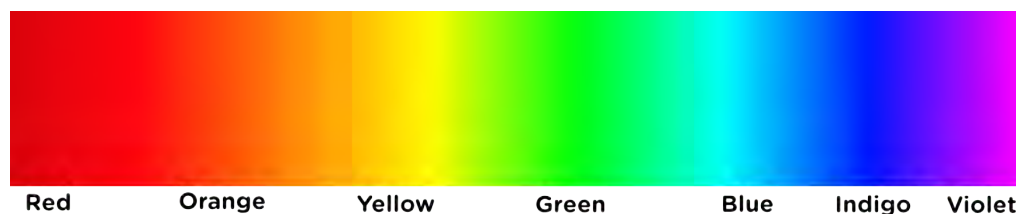
4. If you repeat the experiment, does the order of the colours change?

5. What do the different colours we see tell us about the composition of white light?



So, what have we learned so far? Light radiates from luminous objects and always travels in straight lines. The white light that we see is made up of the 7 different colours of the spectrum. When the 7 colours are travelling together we see them as white light.

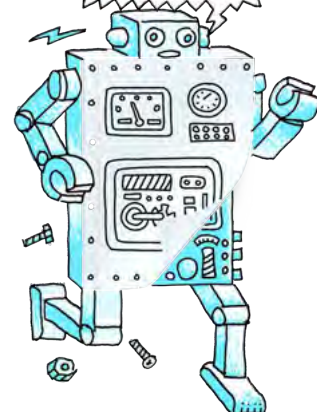
The 7 colours of the visible spectrum are **R**ed, **O**range, **Y**ellow, **G**reen, **B**lue, **I**ndigo and **V**iolet. Each colour has a different wavelength and frequency. Have a look at the following image which shows the spectrum of visible light.



The colours combine to form white light.

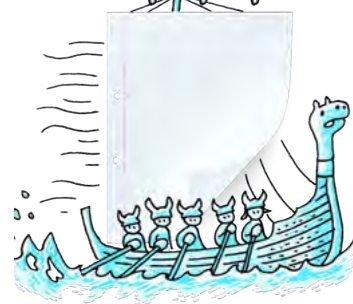
TAKE NOTE

You can use the abbreviation ROYGBIV to remember the order of the colours.



TAKE NOTE

The primary colours of light are red, green and blue.



ACTIVITY: Colour spinning wheels

MATERIALS:

- white cardboard
- coloured pens or pencils (red, orange, yellow, green, blue, indigo, violet)
- string
- scissors
- round object

INSTRUCTIONS:

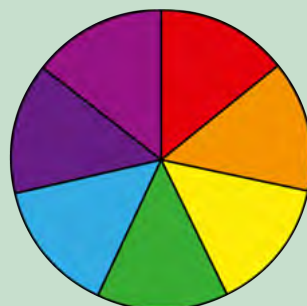
1. Draw a circle on the cardboard. You can trace around a round object such as a cup or saucer to do this. Cut out the circle.



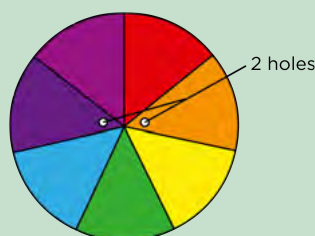
- Now divide the circle into 7 equal segments. If you do not have indigo and violet colours, but just one purple pen or crayon, then you can divide the circle into 6 equal segments rather.
- Shade in each segment a different colour, in the order red, orange, yellow, green, blue, indigo, violet (or just purple if you do not have indigo and violet).

DID YOU KNOW?

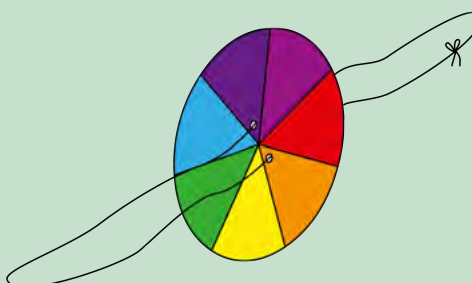
An artist might tell you that the primary colours of paint are red, yellow and blue. This is different to the primary colours of light. This is because the pigments yellow, blue and red cannot be mixed from other pigments. In printing, the primary colours are magenta, yellow and cyan.



- Next, make two holes, one on either side of the centre as shown below.



- Thread the string through the holes and tie it in a loop.



VISIT

There is no pink light.
bit.ly/1b2gFXU

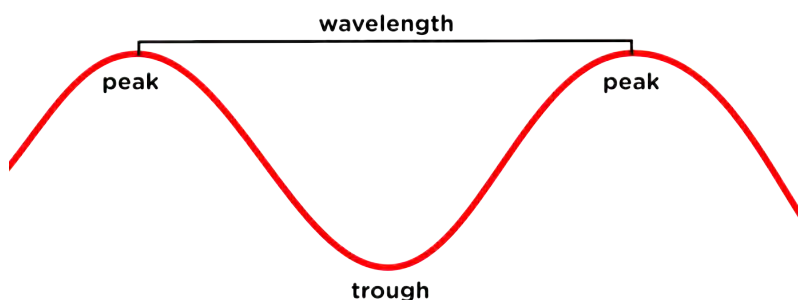


- You are now ready to spin the wheel. Holding the ends of the loop in each hand, twirl the string over, like you would a skipping rope, so that the string twists. Once the string is tightly twisted, pull your hands apart, then bring them back together. Continue bringing your hands in and out and watch the circle spin.
- What do you observe about the colour of the wheel as it spins faster?

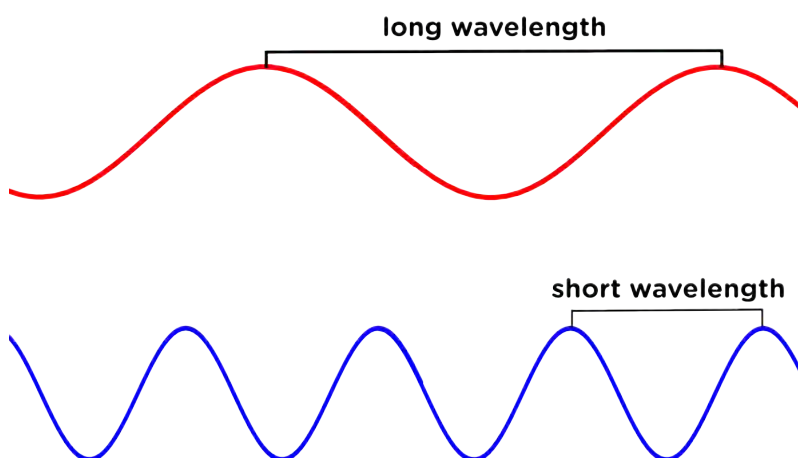


So far we have been talking about the **visible** light spectrum. As we mentioned in the beginning, this is the light that we can see. We also spoke about how light travels in **electromagnetic waves**. We can only see light with a certain range of **wavelengths**. What does this mean?

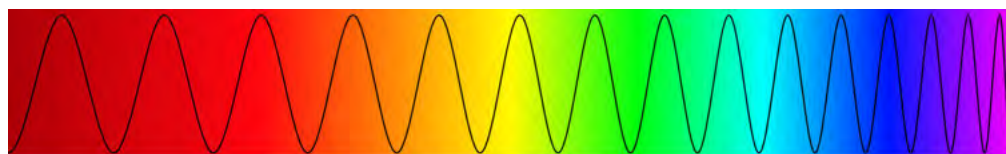
The size of a wave is measured in wavelengths. A wavelength is the distance between two corresponding points on two consecutive waves. Normally this is done by measuring from peak to peak or from trough to trough. Have a look at the following diagram which illustrates a wavelength.



The wavelengths of the different colours of visible light are different lengths, as shown in the following diagram.



We can also talk about the frequency of a wave. If a wave has a long wavelength, then it has a low frequency; if it has a short wavelength, then it has a high frequency.



Of visible light, orange and red light have the longest wavelengths (and lowest frequency) and violet, indigo and blue have the shorter wavelengths (and highest frequency).

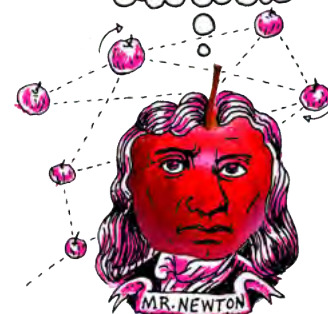
DID YOU KNOW?

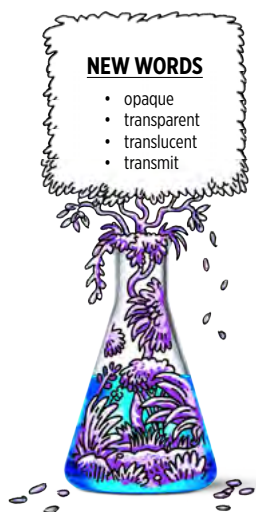
Wavelengths can be as small as one billionth of a meter, as with gamma rays. Wavelengths can even be as long as meters, for example in radio waves.



DID YOU KNOW?

In police forensics, ultraviolet light can be used along with a special powder to detect finger and shoe prints that can help solve crimes.





When it comes to visible light, we only see wavelengths of 400 to 700 billionths of a meter. This is called the visible spectrum. But, light waves are just part of the wave spectrum. There is invisible light with shorter wavelengths, such as ultraviolet light, and there are longer wavelengths, such as infrared light.

Have you ever looked through a window and wondered why it is made of glass? Let's find out how light behaves when it strikes the surface of different types of materials in the next section.

4.3 Opaque and transparent substances

Three different things happen when light hits a surface, it can be **reflected** (bounce off), **absorbed** or **transmitted** (pass through). Glass reflects some light but most of the light is transmitted straight through. That's why we can see objects on the other side of a closed window.

We say that glass is **transparent**. Let's find out more about what this means. If a substance is not transparent, it is **opaque**.



ACTIVITY: Shadow Play

MATERIALS:

- cardboard
- clear plastic
- plastic shopping bag
- scissors
- light source (ray box or light bulb)

INSTRUCTIONS:

1. Cut out three shapes from your cardboard. All of the shapes should be similar but three different sizes: small, medium and large.
2. Switch on the light source.
3. Hold your first shape a short distance in front of the light source.
4. Look at the shadow that forms. Write down what you observe.

-
5. Hold your second shape the same distance in front of the light source.
 6. Look at the shadow that forms. Write down what you observe.
-

-
7. Hold your third shape the same distance in front of the light source.
 8. Look at the shadow that forms. Write down what you observe.
-

-
9. The shadow is formed on the side furthest from the light source. It is dark

in colour and larger than the first and second shadows.

10. Use your first cardboard shape as a template and cut the shape from the clear plastic and the plastic shopping bag.
11. Hold the clear plastic shape the same distance from the light source. Write down what you observe.

12. Hold the plastic shopping bag shape the same distance from the light source. Write down what you observe.

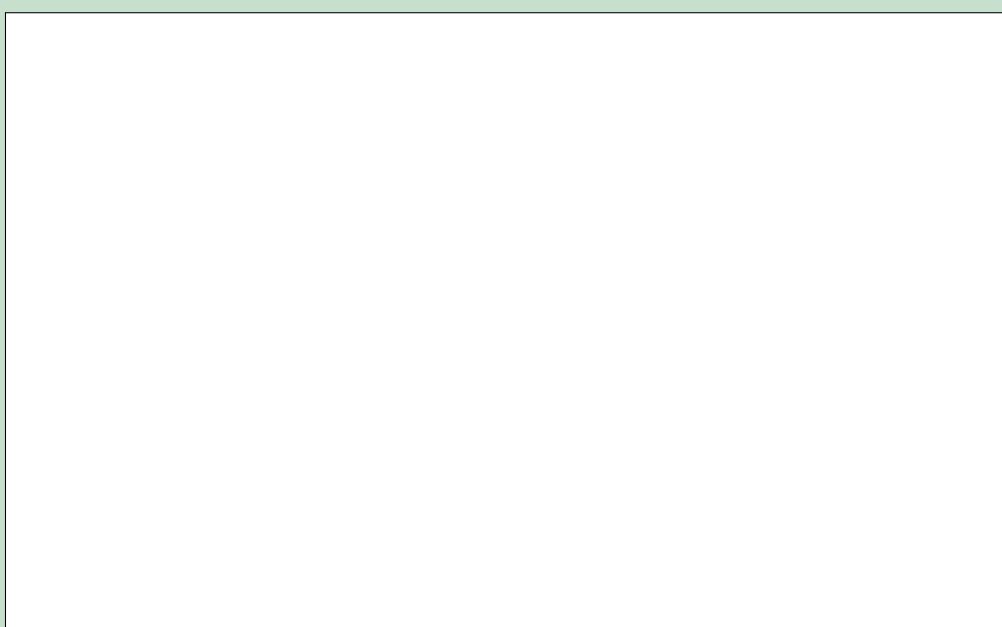
QUESTIONS

1. When you held the cardboard up to the light, did it allow light to pass through it? How do you know this?

2. Is the cardboard shape opaque or transparent?

3. What did you notice about the shadows formed by the different size cardboard shapes?

4. Draw a diagram to show how the shadow is formed behind the opaque shape. Use straight lines with arrowheads to represent the rays of light.



5. The distance between the shape and the light source was kept the same. What do you think would have happened to the shadow if the distance was increased?

6. Test your idea from question 5 by moving your cardboard shapes closer to and further away from the light source. What do you see? Were you correct in your prediction?

7. Is the clear plastic shape opaque or transparent?

8. Did the clear plastic cast a shadow?

9. Explain why the cardboard casts a shadow but the clear plastic does not.

10. Is the plastic shopping bag shape opaque or transparent?

11. Explain why the shopping bag casts a lighter shadow.



What have we learned? Shadows are formed because light travels in straight lines and cannot pass through opaque objects.

Substances which transmit most of the light and only absorb or reflect a little bit are called **transparent**. Can you list some everyday objects which are transparent?

Substances which completely reflect or absorb light without transmitting any are called **opaque**. Can you list some everyday objects which are opaque?

Some substances, such as the plastic shopping bag, allow some light to pass through, but not all of it. This substance is **translucent**, or **semi-transparent**.



Shadows can be useful. Sundials have been used since ancient times as a time-keeping device, like a watch or a clock. As the position of the Sun changes in the sky, the shadow cast by the style moves across the surface of the sundial. The surface is marked with numbers, allowing the shadow to indicate time of day.

We can use transparent objects to make filters. If we want red light we use a red glass bulb or a red plastic film placed in front of the light. Only red light is able to transmit through the red glass or plastic. The other colours are absorbed by the filter.

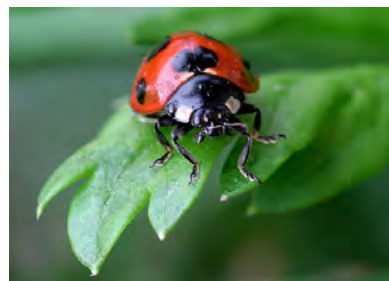


These are different colour filters for a camera. The red filter will only allow red light through and so the photograph will have a red effect applied to it. The other colours of light are absorbed by the filter.

Now that we have seen some examples of transparent and opaque substances, let's take a closer look at what it means to absorb or reflect light.

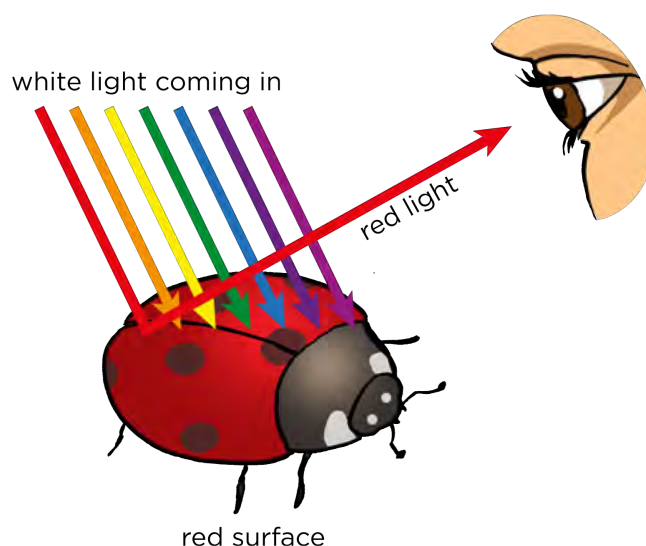
4.4 Absorption of light

Look at this picture of a ladybird. Why is it red and black? And why is the leaf so green? How do we see the different colours? It all has to do with what happens when light hits a surface. When light hits a surface, some of the light is absorbed and the rest is reflected. It is the reflected light that reaches our eyes and allows us to see the object.



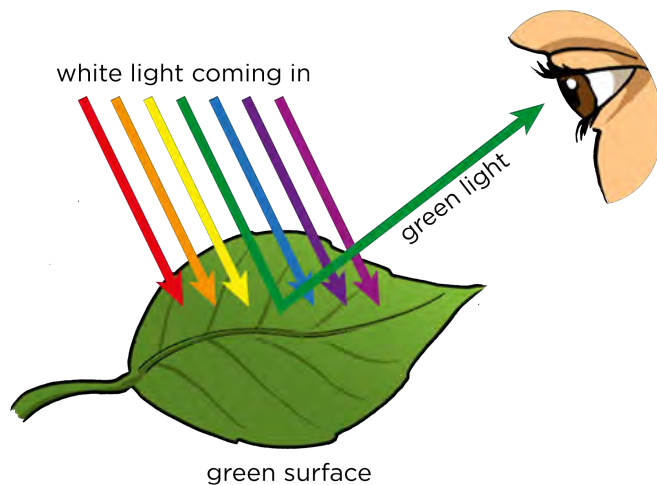
A ladybird.

Previously, we learned that white light is a mixture of different colours. When white light from the Sun hits the red shell of the ladybird all of the colours are absorbed, except red. Red light is reflected back to our eyes and so we see a red ladybird.



We see the red shell of the ladybird as red light is reflected and the other colours are absorbed.

The green leaf absorbs all the colours except green which it reflects back into our eyes.

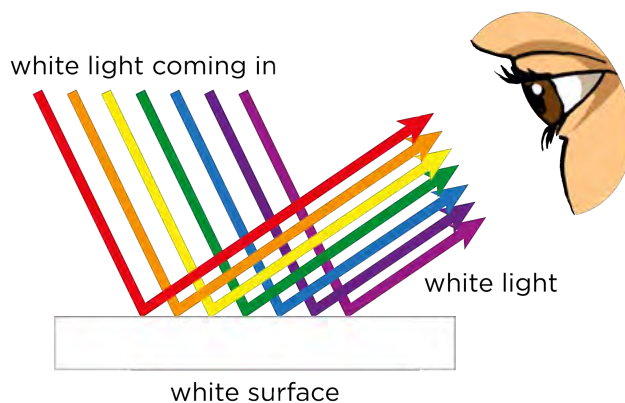


We see a green leaf as green light is reflected and the other colours are absorbed by the leaf's surface.

What about the black spots of the ladybird? Is black a colour? The black spots on the ladybird absorb **all** the colours and no light is reflected. That is why they appear black.

Do you remember learning about heat as energy transfer in Gr 7? We looked at the absorption of heat. We saw that black, matt objects absorbed all of the light energy, while white objects reflected all of it. Black, matt (not shiny) objects absorb all of the colours of light and reflect none and so appear black to our eyes.

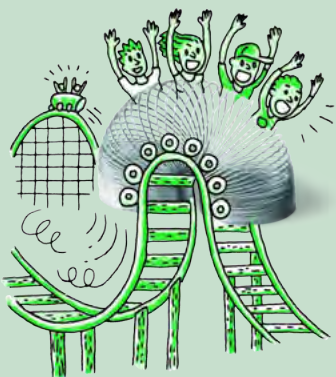
What about a white object? Why do you think white objects look white? Have a look at the following diagram for a clue.



TAKE NOTE

Although we can get black paint as a pigment, black is not a colour of light. Black is the result of the complete absorption of light.





ACTIVITY: Why do objects look red under red light?

MATERIALS:

- piece of red plastic to act as a filter
- light source (light bulb or torch)
- white object

INSTRUCTIONS:

1. Place a white object on the desk.
2. Switch on your light source and place the red plastic in front of the light.
3. Shine the light (with the red plastic in front) onto the piece of white paper.

QUESTIONS:

1. What colour was the page under normal light?

2. Why does the page appear white in normal light?

3. What did you see when the red plastic filter shone on the white page?

4. Explain why the paper changed colour.



Let's now look more at what we mean by reflection of light.

4.5 Reflection of light

When light hits a surface it is often reflected off the surface. This photograph shows how light is reflected off a still lake, creating a mirror image of the tree. The still, flat surface of the lake has acted as a mirror.



A tree reflection.

NEW WORDS

- reflect
- incident ray
- reflected ray
- normal line
- angle of incidence
- angle of reflection
- perpendicular



Have some fun with these photos of reflections in water. One photograph is the right way up and the other one is upside down! Which one is which?



Reflections on the Negro River in the Amazon.



Reflections in the Arno River in Italy.

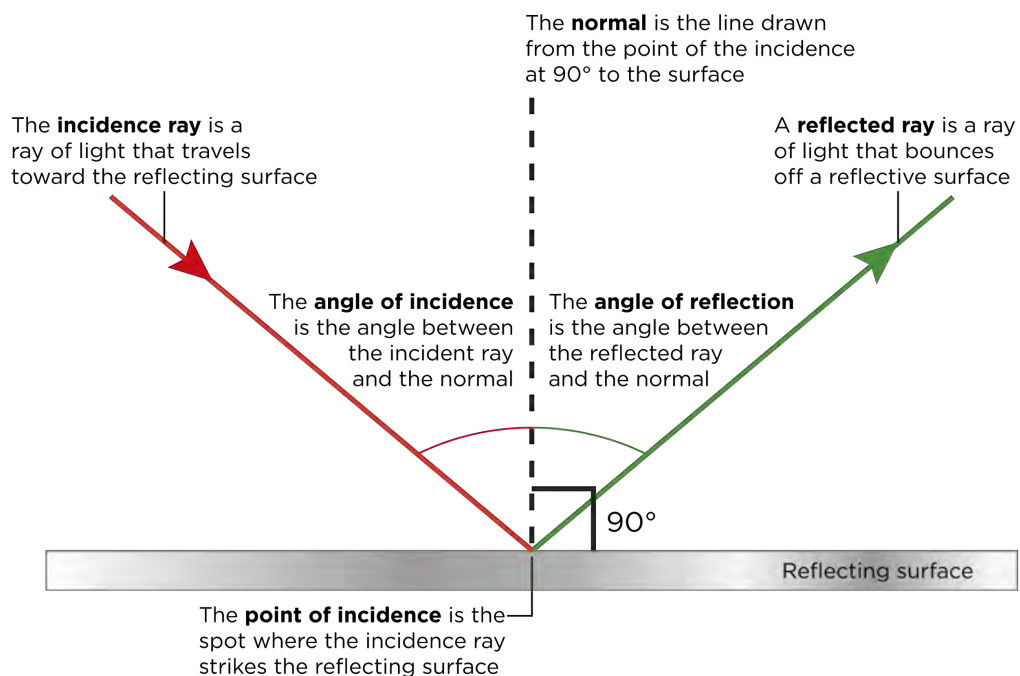
Most surfaces reflect light. When light strikes a reflective surface, it can change direction. Let's look at how this happens.

When light reflects off a surface the ray which hits the surface, it is called the **incident ray**. The ray of light which is reflected from the surface is called the **reflected ray**. When we draw diagrams of reflection we also draw in an imaginary line to help us measure different angles. This line is called the **normal**. The normal line is always drawn perpendicular to the surface.

Between the normal line and the incident and reflected rays, there are two angles. These are:

- **angle of incidence** - the angle between the incident ray and normal line
- **angle of reflection** - the angle between the reflected ray and normal line

The following diagram explains these concepts.



Let's investigate the relationship between the angle of incidence and the angle of reflection.



INVESTIGATION: Is there a relationship between the angles of incidence and reflections?

AIM: To investigate the reflection of light from a surface.

INVESTIGATIVE QUESTION:

Look at the diagram above and try to formulate an investigative question for this investigation.

HYPOTHESIS: The angle of incidence is equal to the angle of reflection

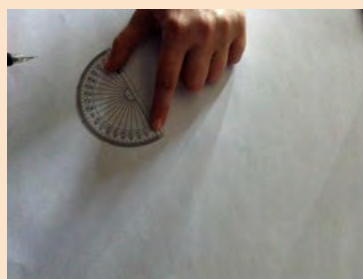
MATERIALS AND APPARATUS:

- mirror
- white paper
- pencil
- protractor
- ruler
- ray box

METHOD:

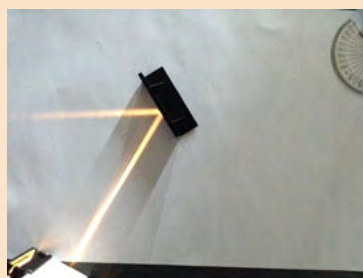
1. Put a white piece of paper on the desk.
2. Use your ruler to draw a straight line near the top of the white paper.

3. Use your protractor to make a right angle in the middle of your pencil line. This is the **normal** line.



Marking a right angle with a protractor.

4. Place your mirror upright along the first line.
5. Shine a light from the ray box along the paper so that it "hits" the mirror where your normal line and your mirror meet.



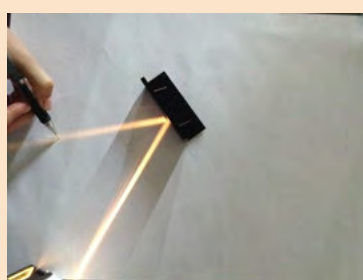
A mirror is placed on the line and a ray shone to strike the mirror at the normal line.

6. Use a pencil to mark the incident light ray.



Marking the incident light ray.

7. Use a pencil to mark the reflected light ray.

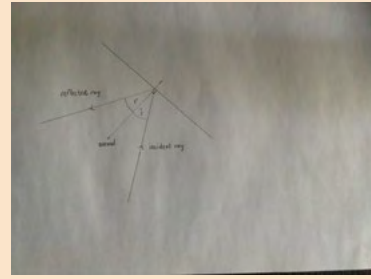


Marking the reflected ray.

8. Remove the mirror and switch off the ray box.
9. Use a ruler and pencil to draw a line from the points you have marked on each ray to the normal line.

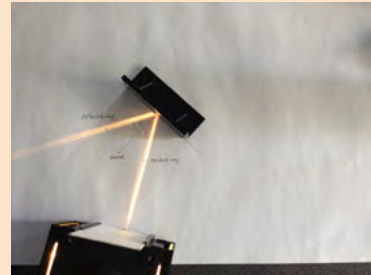


10. Mark the angle of incidence (i) and angle of reflection (r).



Your ray diagram should look similar to this.

11. Turn the ray box on again to confirm that your pencil lines follow the rays.



The ray diagram overlaps the actual rays.

TAKE NOTE

Keep one of the sheets with your drawn ray diagram for the next activity.

12. Use a protractor and measure the angle of incidence and the angle of reflection and record your results in the table.
13. Repeat this method 3 more times, each time using a different angle of incidence.



A different angle of incidence.

RESULTS:

Fill your results into the following table.

Repeat	Angle of Incidence	Angle of Reflection
1		
2		
3		
4		

ANALYSIS:

1. Has your investigation provided everything you need to answer your investigative question?

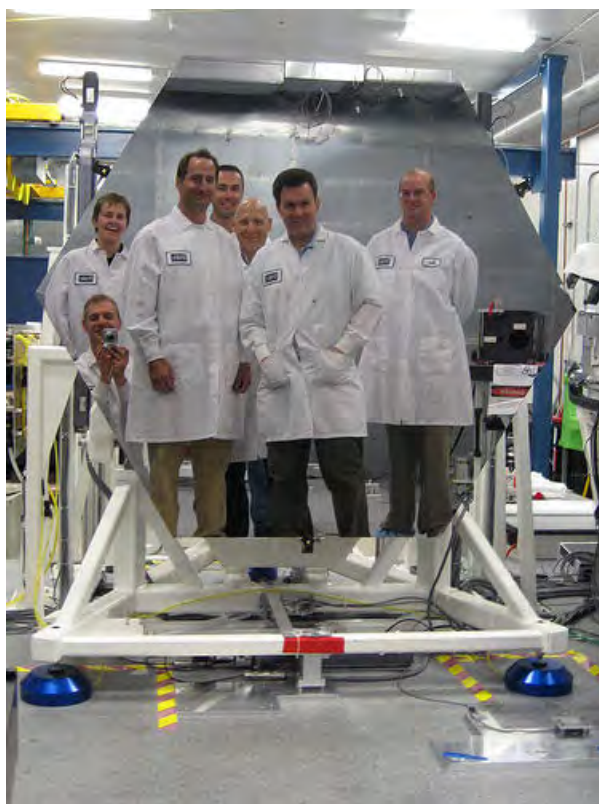
2. How could you improve this investigation to get more accurate results?

CONCLUSION:

What can you conclude based on your results?

Whenever light is reflected from a surface, the angle of incidence is equal to the angle of reflection. On a smooth surface all the light rays are reflected in the same way and so the image is clear and focused.

A mirror is an example of a smooth surface. The image you see is focused and clear. As you can see in the photograph, the scientists and engineers are clear and focused in the mirror image.

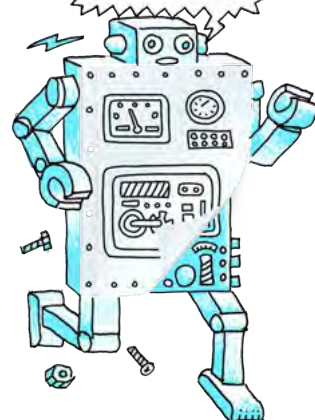


A mirror segment from one of NASA's telescopes provides a clear and focused reflection.

What happens when we do not have a smooth surface? Have a look at the photo.

TAKE NOTE

In reflection, not only is the angle of incidence equal to the angle of reflection, but the incident ray and reflection ray are also in the same **plane**.



VISIT

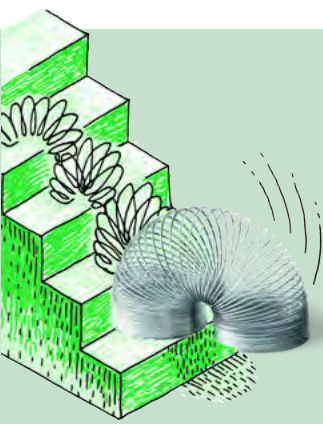
What colour is a mirror?
(video)

bit.ly/GABdNZ





Why is the reflection of the grass and reeds not clear, but rather blurred?



ACTIVITY: Light reflection off aluminium foil

MATERIALS:

- aluminium foil
- white paper
- ray box

INSTRUCTIONS:

1. If possible, use the white sheets of paper from the last investigation where you drew your ray diagrams.
2. Similar to what you did in the last investigation, set up a ray box and direct the ray along the line of incidence which you drew.
3. Crumple a piece of aluminium foil and place this in the spot instead of the mirror.
4. Observe the reflected ray.

QUESTIONS:

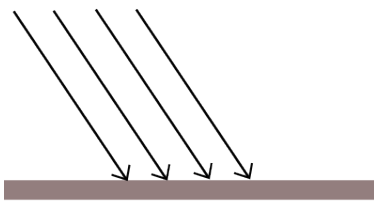
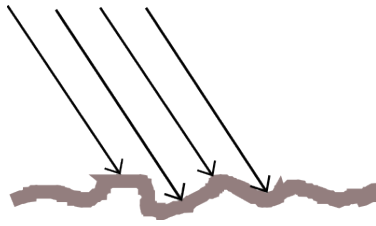
1. Describe the reflected ray off the aluminium foil and how this compares to the reflected ray off the mirror.

2. Why do you think you observed these differences?



Can you now see why reflections off rippled water are not clear, but rather blurred? This is because the light rays have not reflected parallel to each other as they do from a smooth surface, but have scattered in different directions.

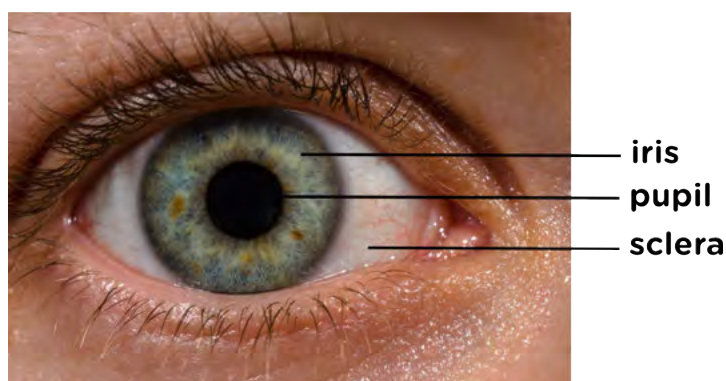
The following table shows the difference between a smooth surface and a rough surface. Straight parallel rays are approaching the surface. You need to draw in the reflected rays to show specular (clear) reflection from a smooth surface and diffuse (unclear) reflection from a rough surface.

Specular diffusion from a smooth surface	Diffuse reflection from a rough surface.
	

Visible light is the range of frequencies of light that are visible to the human eye, and is responsible for the sense of sight. Are you curious to find out how we actually see light? Let's discover more in the next section.

4.6 How do we see light?

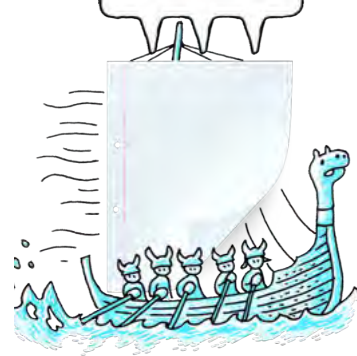
How is it that we are able to see light? Light that is absorbed by objects does not enter the eye. Only reflected light or direct light from luminous objects can enter the eye and be interpreted. Have a look at the following image which shows the outer structure of the eye.



We can see the iris, the pupil and the sclera. The sclera is the tough white, outer part of the eye, which acts as protection. The iris is the coloured part of the eye which differs from person to person. It is circular and surrounds the pupil. Light enters the eye through the pupil.

TAKE NOTE

'Diffuse' can mean unclear as well as spread out. In this example, the reflection is unclear because the rays are spread out or diffuse.



NEW WORDS

- retina
- stimulate



VISIT

2012 Nobel Prize: How do we see light?

bit.ly/1a4zs2D



TAKE NOTE

The fovea is the part of the eye located in the centre of the retina where the clearest image is formed.

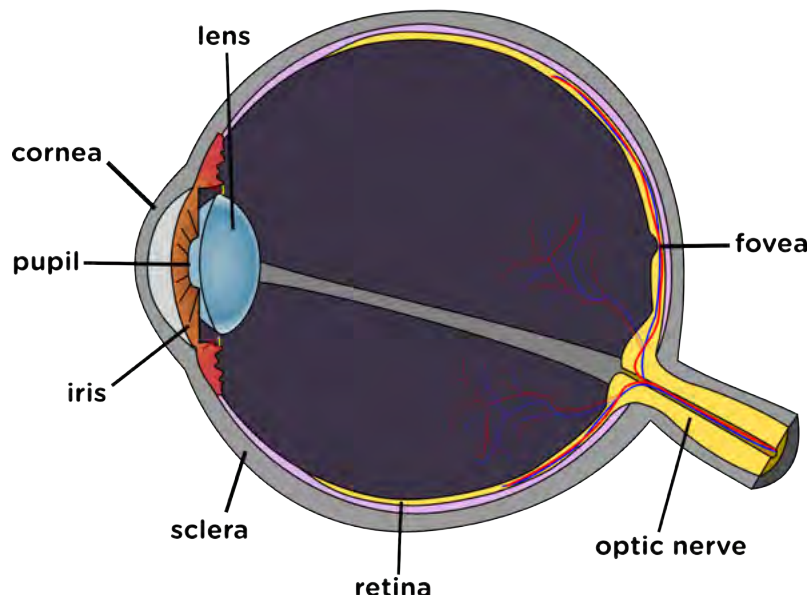


The size of your pupil changes in different light conditions. In bright light, the pupil contracts (gets smaller) to let less light through (as on the left), and in low light your pupil dilates (gets bigger) to let more light through (as on the right).

Let's take a look at the internal structure of the human eye. The following diagram shows a cross section through the eye. The eye is actually a large ball, and only a small part is visible on the outside. Covering the iris is a tough, transparent layer called the cornea. Behind the iris is the lens. Both the cornea and the lens help you to focus the light entering your eyes, as we will learn about in the next section.

TAKE NOTE

The cell is the basic structural and functional unit of all living things. We will be learning more about the cell next year in Gr 9 Life and Living.



A diagram of the eye.

VISIT

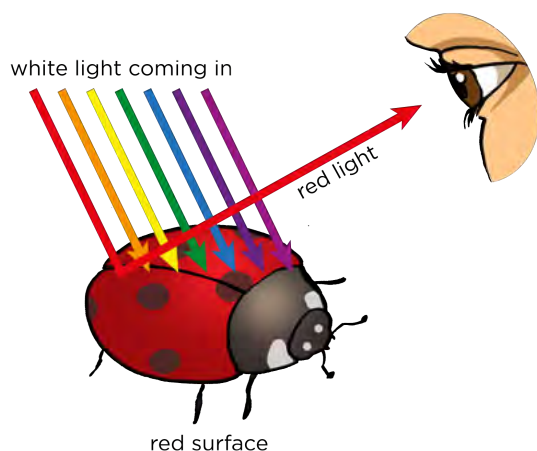
Find your blind spot with this optical illusion.

bit.ly/19jumEr



The light travels through the eye and hits the retina at the back of the eyeball. The retina is a layer of tissue lining the back of the eyeball, as indicated in the diagram, it is the yellow layer. The retina consists of cells which are sensitive to light. Light enters the eye and forms an image on the back of the eyeball. The way in which light hits the back of the eye, is similar to what happens in a pinhole camera. The receptor cells convert the light energy into electrical nerve impulses. These impulses travel out of the eye through the optic nerve and to the brain where they are interpreted as sight.

So how do we see colour? Do you remember when we spoke about why the ladybird appears red and black? Look at the following diagram again.



The white light hits the ladybird's surface. The white light has all the colours of light, but when it hits the red surface, only the red light is reflected. The other colours are absorbed by the red surface. This means that when we look at the red parts of the ladybird, we only get red light reflected into our eyes. Therefore, when this reflected light hits our retina and the electrical impulse is sent to our brains, we see the red colour.

DID YOU KNOW?

Each of your eyes has a small blind spot at the back of the retina where the optic nerve attaches. You do not normally notice the hole in your vision because your eyes work together to fill in each other's blind spot.



ACTIVITY: Seeing colours

MATERIALS:

- coloured pens or pencils

INSTRUCTIONS:

- Answer the following questions about how we see objects.
- Draw a ray diagram to accompany your written answer.
- An example has been done for you.

Look at the picture of a sunflower.



A black and yellow sunflower.

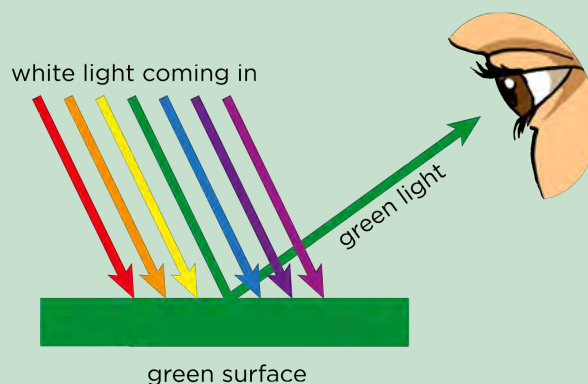


DID YOU KNOW?

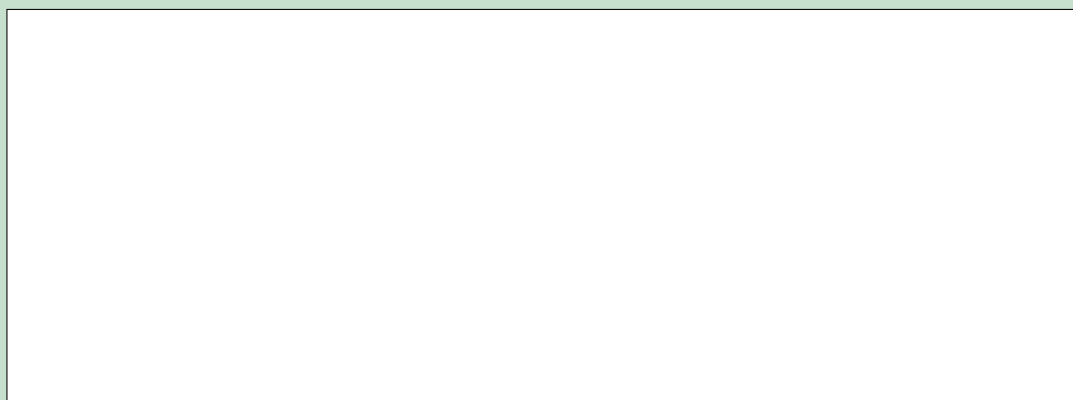
The cells in your eye come in different shapes. Rod-shaped cells allow you to see shapes, and cone-shaped cells allow you to see colour.



We can draw a ray diagram to show why we see the green leaves as green, as shown below. The green surface of the leaves absorb all the colours of white light except green light which is reflected into our eyes.



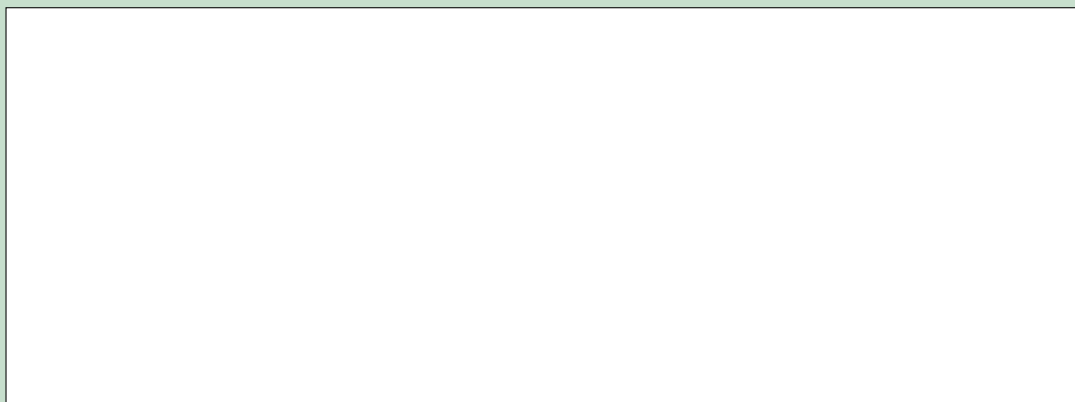
Now explain why the petals appear yellow and the centre appears black. Use the concepts of absorption and reflection in your explanation. Draw diagrams to support your answer.



Heath has bought himself a blue car. Explain why we see the car as blue by using the absorption and reflection of light. Draw a diagram to support your answer.



Heath's blue car.



VISIT

A simulation on colour vision.

bit.ly/18TbpEA



We have looked at opaque and transparent substances, absorption of light, reflection of light and how we see light. We are now going to go back to transparent substances and see how light can interact with these materials.

4.7 Refraction of light

Do you remember the last time you drank a cold drink with a straw? Did you notice that the straw did not look straight anymore once it was in the water or cool drink?



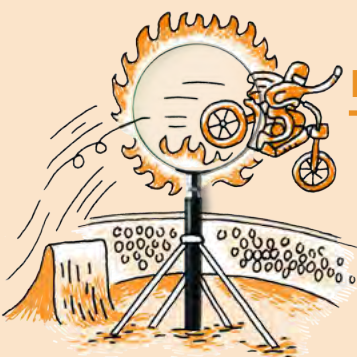
Why does the pencil in this glass of water look bent?

Let's investigate this by examining what happens to light when it passes through a glass block.

NEW WORDS

- refraction
- medium
- optical density



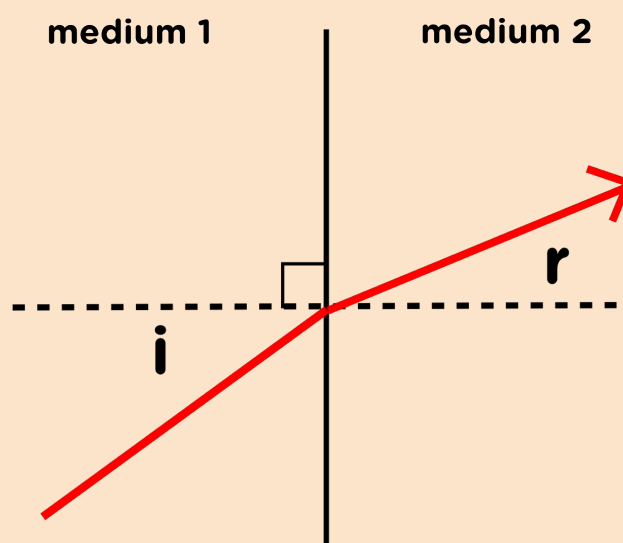


INVESTIGATION: What happens to light when it passes through a glass block

We are going to investigate what happens to a ray of light when it passes from air and into a glass block and then from the glass block back into air. We are going to use a glass block with parallel sides.

Before we start the investigation, we need to think about how we are going to determine if light changes direction or not. Do you remember in the investigation on reflection where we measured the angle of incidence and the angle of reflection? What did we find in this investigation?

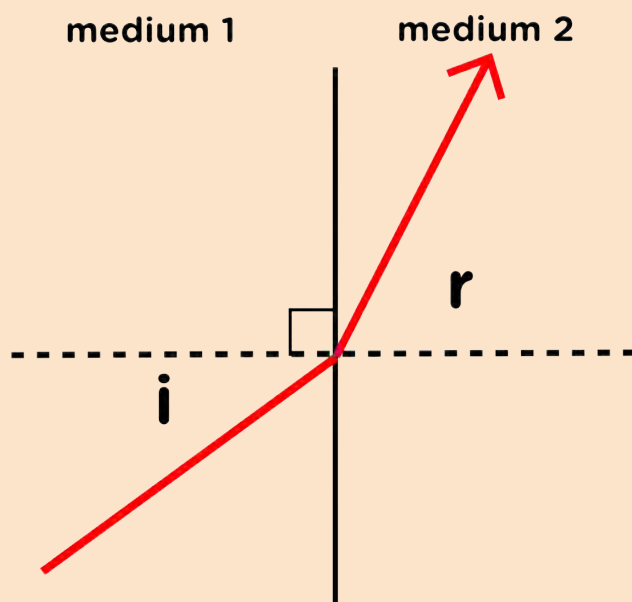
When light passes through a transparent substance, we can also measure the angles. Look at the following diagram. The angle of incidence (i) is measured between the incident light ray and the normal line. As the light passes through the transparent substance, the angle of refraction (r) is the angle between the refracted light ray and the normal.



A light ray passing from one medium to another.

In the diagram above, you can see that the angle of refraction is smaller than the angle of incidence. Therefore, the refracted light ray changed direction when it entered the transparent medium. We can also say something about which direction it bent towards. Did the light ray bend towards or away from the normal line?

The next diagram shows another outcome.



A light ray passing from one medium to another.

In the diagram above, does the refracted ray change direction when it enters the transparent medium? Give a reason for your answer.

In which direction did the refracted ray change?

We are now ready to start our investigation.

AIM: To determine whether light changes direction when it passes through a parallel-sided glass block.

HYPOTHESIS: Write a hypothesis for this investigation.

MATERIALS AND APPARATUS:

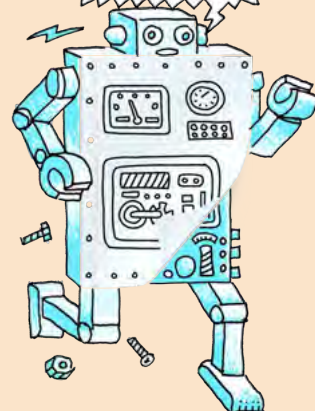
- glass block
- ray box, laser pointer or other light source
- protractor

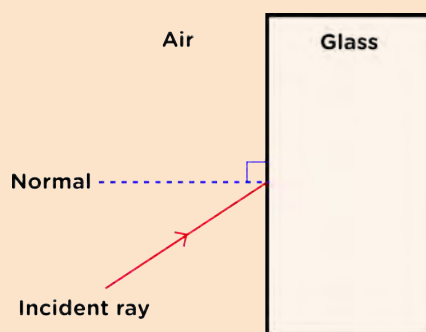
METHOD:

1. Put the glass block in the centre of a piece of white paper and trace around it.
2. Shine a ray of light into the glass block. The ray should be at an angle to the surface of the block.

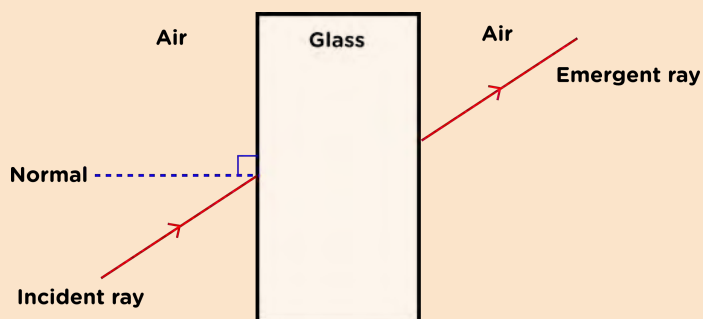
TAKE NOTE

The emergent ray from a parallel sided block is parallel to the incident ray.

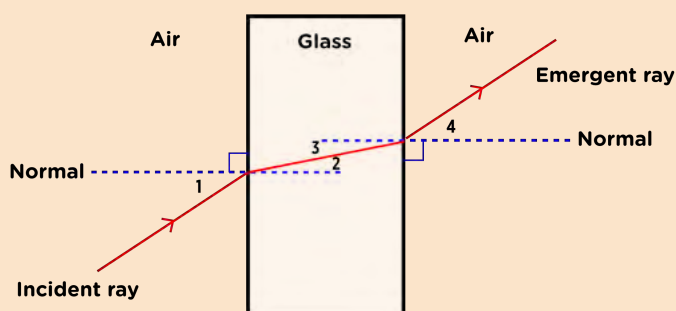




3. Trace the light ray with pencil and mark the point at which it enters the glass block.
4. The light ray emerges on the other side of the glass block. Mark the point at which it emerges with a pencil and trace the emergent ray.



5. Remove the glass block. Your diagram should look similar to the one above.
6. Draw a line joining the incident ray and emergent ray. You have traced the refracted ray through the glass block.
7. Draw the normal lines where the incident ray meets the block and where the emergent ray leaves the block.



8. Measure the angles labelled 1, 2, 3 and 4 as shown on the diagram with a protractor.
9. Fill in the measurements in the table.
10. Repeat the steps above three times using different angles of incidence (angle 1).

RESULTS AND OBSERVATIONS:

Fill your results into the following table.

Experimental repeat	Angle 1	Angle 2	Angle 3	Angle 4
1				
2				
3				
4				

1. Which pairs of angles are equal in the measurements you have taken?

2. Which of the angles you measured are the angles of incidence and which are the angles of refraction? Write this down below and mark them on the diagram above.

3. What do you notice about the angle of incidence and angle of refraction for each of your sets of measurements?

4. Did the light entering the glass block bend towards or away from the normal line?

5. Make the angle of incidence zero (make the light ray enter the block perpendicular to the surface). What is the angle of refraction?

CONCLUSION:

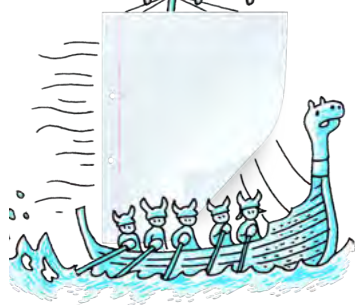
What can you conclude from your results?



The angle of incidence is not equal to the angle of refraction because the light has changed direction as it enters the glass. Therefore, when light travels from one medium to another, it bends, or changes direction. This is called **refraction**.

TAKE NOTE

Remember that although we learn about Natural Sciences in 4 strands throughout the year, there are many connections and links between the strands.



VISIT

The speed of light in glass.
bit.ly/1fcfJVZ

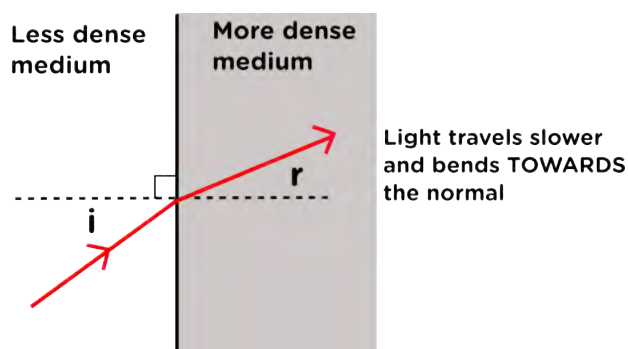


When light enters a different medium at right angles then it does not change direction.

So why does the light refract? Light behaves as a wave does and waves travel at different speeds in different media. For example, light travels faster in air than it does in water. When light enters a different medium, it changes speed, and if it entered at an angle other than 90° , then it also changes direction. The more dense the medium, the slower the light moves.

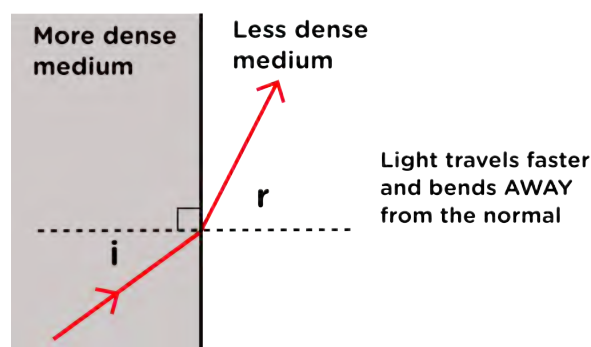
Do you remember learning about density last term in Matter and Materials? Write down your own definition for density in the space below.

If light moves from a less dense medium, like air, into a denser medium, like glass, then the light slows down. The light will bend towards the normal line.



Light travels slower and bends **TOWARDS** the normal

If light moves from a more dense medium to a less dense medium then the light speeds up and moves away from the normal.



Light travels faster and bends **AWAY** from the normal

When light refracts and changes direction as it passes through different mediums, it can distort what we see. Think back to the pencil or straw in a glass of water at the start of the section. We can now explain why a drinking straw or pencil in a glass of water looks bent. The light bends when it moves from one medium to another. Light moves from the air to glass to water, and therefore changes direction.

If you have stood in a pool of water before and looked down, have you noticed how short your legs appear to be? Let's have a look at this a bit more in the next activity.

ACTIVITY: Magic coin trick

MATERIALS:

- coin
- prestik
- opaque bowl or cup
- water

INSTRUCTIONS:

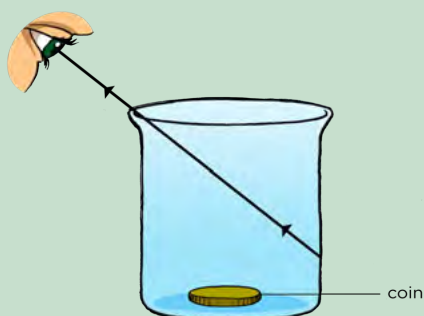
1. Work in pairs for this activity.
2. Put a small amount of prestik onto the bottom of the bowl.
3. Stick the coin to the bottom of the bowl.
4. Take small steps back from the desk/table until you cannot see the coin over the lip of the bowl.
5. Ask your partner to slowly pour water into the bowl and observe.

QUESTIONS:

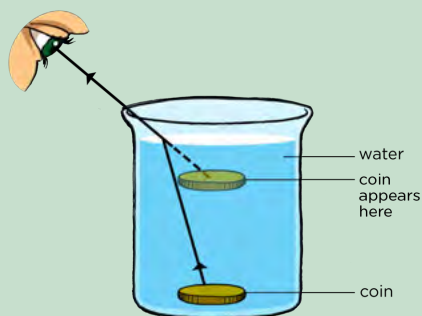
1. What happened when your partner poured the water into the bowl?

2. Where does the coin appear to be?

3. Explain why the coin can be seen when the water is added, but not before. The diagrams below will help you explain what is happening in words.



Empty container.



Container with water.



VISIT

Watch a video that shows and explains the coin activity.

bit.ly/15NmXX0



TAKE NOTE

The diagrams used here show the container as transparent so that you can see the coin inside, whereas you will actually be using an opaque container.



Refraction can be used to explain why images appear to be distorted when we view them through transparent mediums. For example, if you are looking at your legs or hands through some water, they will appear closer than they actually are as the light is refracted. Look at the photograph of the glass with water in it in front of diagonal lines. Can you see how the lines are distorted when the light travels through the water and glass compared to when it does not?



Light refraction through glass and water.

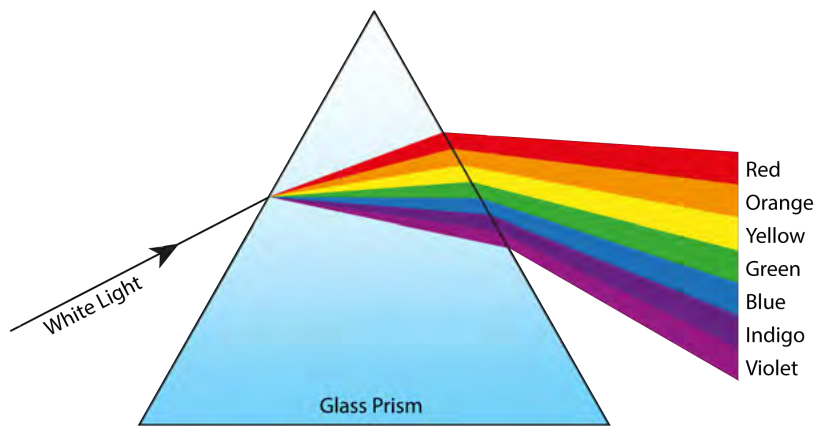
Can you remember how we split white light into the separate colours of the visible spectrum in the beginning of this chapter? What did we use to do this in the activity?

We can do this because the different colours of light bend by different amounts when the light enters a different medium. Different colours of light will slow down to different speeds, causing them to bend by different amounts.

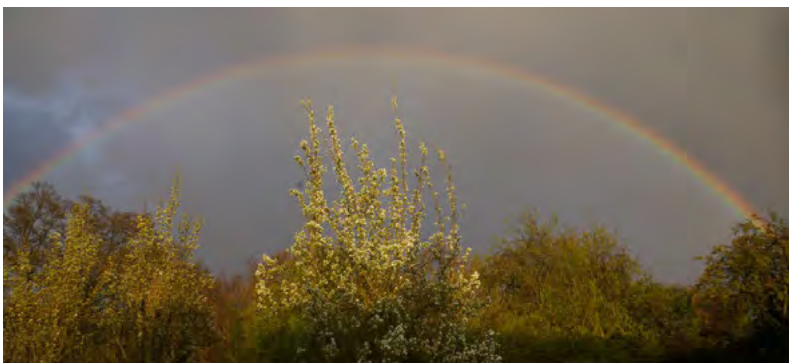


Refraction through a triangular prism.

When the white light entered the prism it refracted. The different colours of light travel at different speeds in the prism so they refracted at different angles and split up. Red light refracts the least and the violet light refracts the most as you can see in the following diagram.

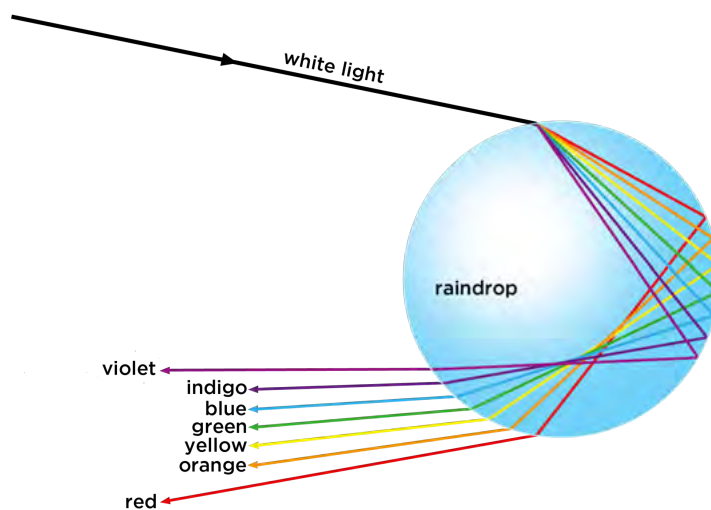


Prisms are not the only objects that can split white light into separate colours. In fact, a rainbow is a good example of white light splitting up.



A rainbow.

Light from the Sun enters the raindrops and refracts. The light is then reflected off the back of the raindrop. When the light passes out of the raindrop it is refracted again and the colours split up even more as shown in the diagram.

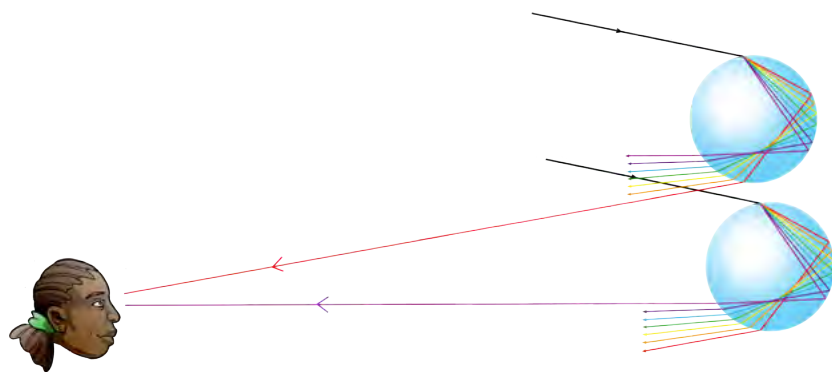
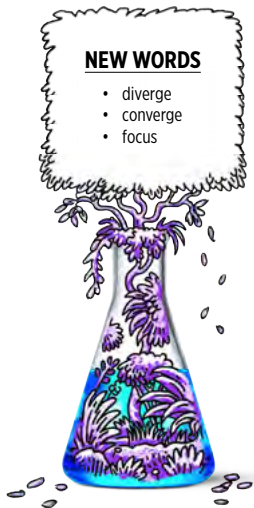


A raindrop refracts and reflects light, dispersing white light into the colours of the visible spectrum.

What colour is at the top of a rainbow and which colour is at the bottom?

Does this match the order which we see in the diagram showing how light is refracted and reflected in a raindrop?

How does this happen? When we see a rainbow, we see a combination of millions of raindrops. Although each raindrop refracts and reflects all 7 colours, we only see one colour of light reflected from each particular raindrop. This depends on the angle of the raindrop from our position. Therefore, the raindrops higher up in the sky reflect red light to us and the rain drops lower down reflect violet light to us. This is shown in the following diagram.



We see rainbows with red at the top and violet at the bottom due to the combination of millions of raindrops. We only see one colour reflected from a particular raindrop, depending on its position in the sky.

We are now going to look at an application of the refraction of light.

Lenses

Do you remember when we spoke about how we see light and the structure of the eye, we mentioned that there is a lens just behind the iris? Another place where you may have seen lenses before are in reading glasses which some people wear to correct their vision. Or, have you seen how a magnifying glass makes things appear bigger. What are lenses and how do they work?



A magnifying glass makes things look bigger.

A lens is a transparent object which focuses or refracts light. When light is spread out, we say it has **diverged**. Some lenses will diverge light while others will **converge** light, bringing the light rays together. When light rays are all brought to the same point, we say they have been **focused**. Let's have a look at this more closely.

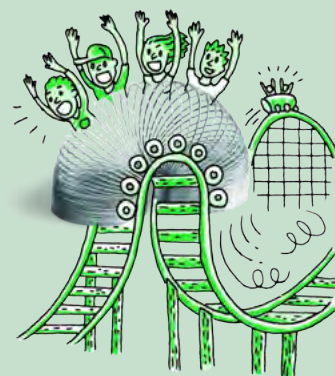
ACTIVITY: Diverging and converging light with lenses

MATERIALS:

- ray box or light source
- concave lens
- convex lens
- piece of paper
- pencil

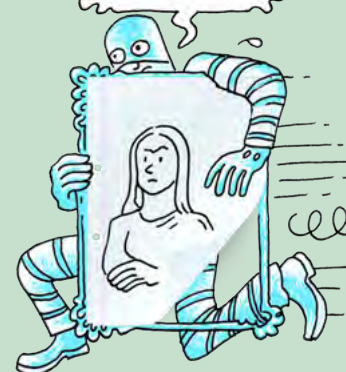
Before we start, it is important that you know the difference between a convex and a concave lens.

Convex lens	Concave lens
	
A convex lens has one side which curves or bulges outwards . A convex lens converges light.	A concave lens has one side which curves or is hollowed inwards . A concave lens diverges light.



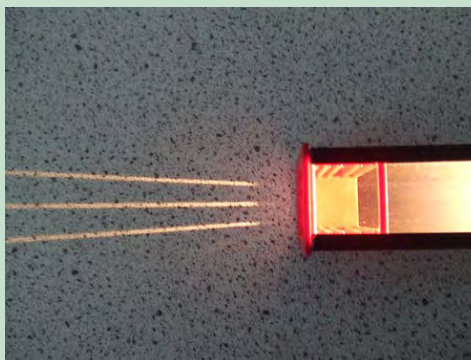
TAKE NOTE

A lens can have two sides which are concave and it is then called a biconcave lens or two sides which are convex and it is then called a biconvex lens.



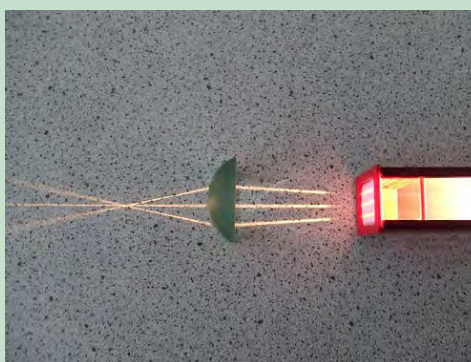
INSTRUCTIONS:

1. Place a ray box or light source on one side of a piece of paper and turn it on. Observe the light rays. You might see something as shown in the photograph here.



Three rays coming out of a ray box.

2. Turn the ray box off.
3. Place the convex lens (with the rounded surface) on the piece of paper where the light rays will pass through it. Trace around it.
4. Turn on the ray box or light source and observe what happens to the rays when they pass through the lens.

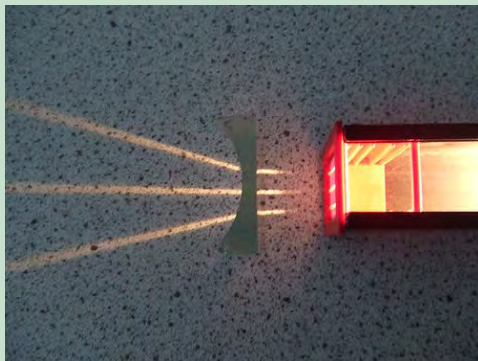


Light rays passing through a convex lens.

5. Trace the path of the light rays on your piece of paper.
6. Describe what has happened to the light rays.

7. Mark the point where the light rays cross. This is called the **focal point** of a convex lens.
8. Turn off the ray box or light source and place a new piece of paper in front of it.
9. Now place the concave lens in the path of the light rays and trace around the lens.
10. Turn on the light source and observe what happens to the rays.

11. Trace the path of the rays on the piece of paper.



A concave lens in front of the rays of light.

12. Describe what has happened to the light rays.

13. Turn off the light rays and extend the rays you have drawn until they meet at a point in front of the lens. This is the **focal point** of a concave lens.
14. If you still have your pin hole cameras, place a convex and concave lens in front of the camera and observe the image that forms.



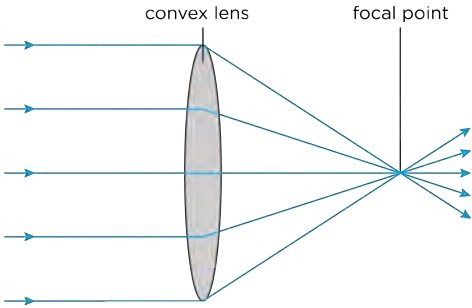
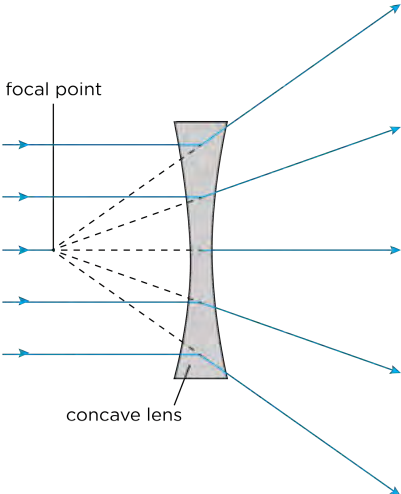
Viewing a light source through a pinhole camera with different lenses.

15. Is the image larger or smaller when you observe through a concave lens?

16. Is the image larger or smaller when you observe through a convex lens?



We have now seen how lenses can disperse or focus light. Have a look at the following diagrams which show how a biconvex lens converges light and a biconcave lens diverges light.

Converging lens	Diverging lens
	
A converging lens refracts the light entering it and bends the light rays to a focal point on the other side of the lens.	A diverging lens refracts the light entering it and bends the light rays away from each other. The light rays can be traced back to a focal point in front of the lens.

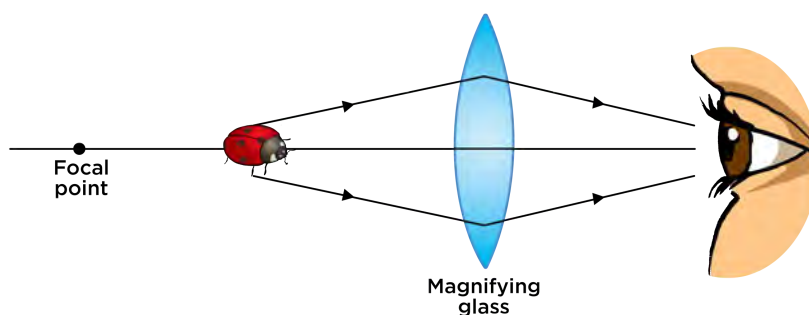
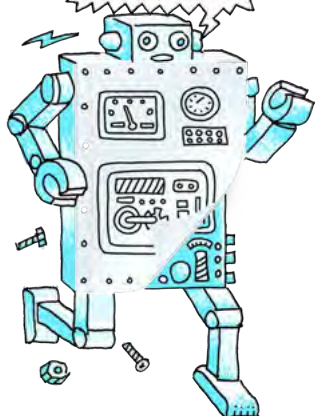
What do we use lenses for? Think of a magnifying glass. If you hold a magnifying glass over a picture or words then it enlarges the image. Is a magnifying glass an example of a diverging or converging lens?

Let's think about how this works. Imagine you are looking at the ladybird from the beginning of the chapter through a magnifying glass. The ladybird looks bigger than what it actually is. When the object you are viewing is **closer** to the lens than the focal point, you see a virtual image of the ladybird that is **larger** than the object.

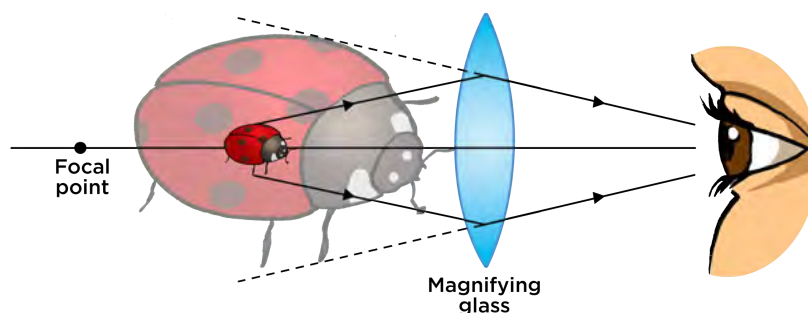
Have a look at the first diagram below. Can you see that the ladybird is between the focal point and the lens? The rays reflected from the ladybird are refracted by the magnifying glass and enter the person's eye.

TAKE NOTE

When you hold a magnifying glass up and view a distant object, the object appears smaller and upside down. Unlike when viewing the ladybird close up, the distant object is **beyond** the focal point of the lens, which results in this effect.



In the next diagram you can see how your eyes see a virtual image of the ladybird which is bigger than the object. The more curved the convex lens is in a magnifying glass, the greater its ability to magnify objects.



VISIT

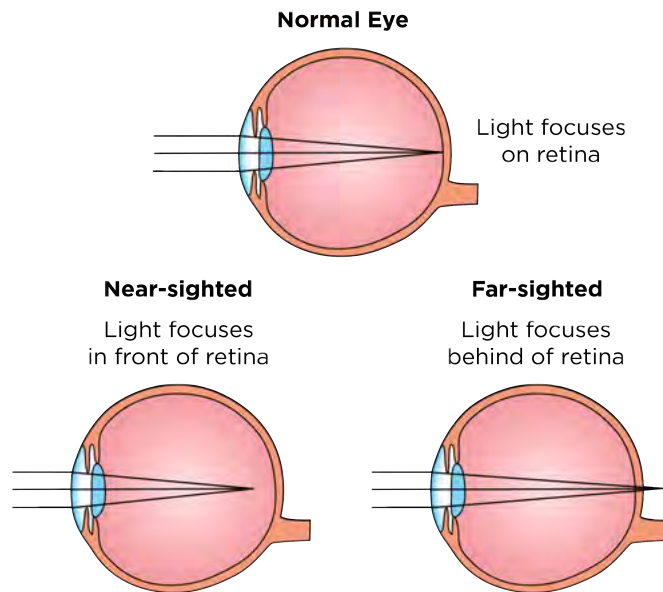
How do lenses work?

bit.ly/GABjoO

Do you remember what the human eye looks like? We have lenses in our eyes to allow us to see. The light enters the eye and passes through the lens. The lens focuses the light onto the back of our retina so that a clear image is formed. What type of lens do we have in our eyes? Give a reason for your answer.

In order for a clear image to form, the lens in our eye needs to focus the light rays coming into our eyes so that the focal point falls on the retina. This depends on the shape of the lens in our eyes. Sometimes, people have lenses in their eyes that cannot focus properly. Have a look at the following diagram which shows a normal eye and then an eye which focuses before the retina (near-sighted) and behind the retina (far-sighted).





DID YOU KNOW?

A contact lens is designed to rest on the cornea of the eye and correct vision. Leonardo da Vinci was the first to come up with the idea in the 16th century to help prevent eye infection.



Optical glasses, or spectacles, are used to correct near or far-sightedness.

If you are near-sighted you need a diverging lens. Would this be a biconcave or biconvex lens?

If you are far-sighted you need a converging lens. Would this be a biconcave or biconvex lens?

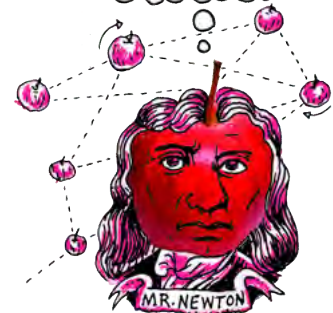
DID YOU KNOW?

A **microscope** makes a tiny, nearby object look much bigger. A **telescope** makes a large, distant object look much closer and brighter. In both, light from the object passes through two or more lenses to form an image. The lens shapes and distances between them determine how the image is produced.



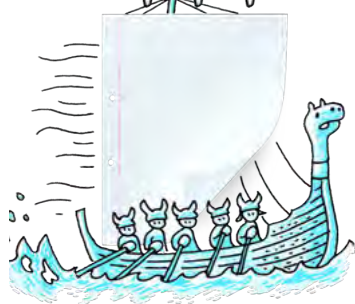
An optometrist holds a lens in front of a patient's eye to correct her vision.

The following image shows how lenses can be used to correct far and near-sightedness.



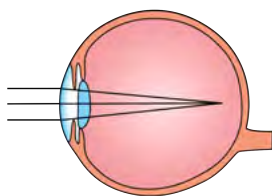
TAKE NOTE

Next term in Planet Earth and Beyond we will look at how lenses are used in optical telescopes to view objects in space.

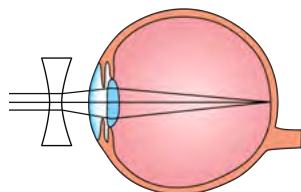


Near-sighted

Light focuses in front of retina

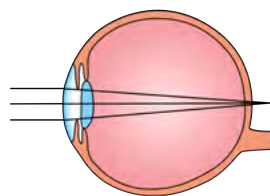


Corrected with concave lens

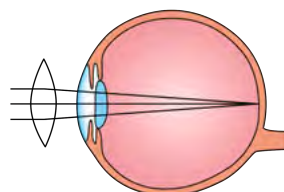


Far-sighted

Light focuses behind of retina



Corrected with convex lens



ACTIVITY: Research careers in optics

There are many different careers in the field of geometric optics.

INSTRUCTIONS:

1. Work in groups of 3.
2. Interview someone in the field of geometric optics and find out how they chose their career and what and where they studied.
3. Write a paragraph explaining the career and the study options available in order to qualify for that career.
4. Here are some examples of careers in geometric optics.
 - a) Optometry
 - b) Ophthalmology
 - c) Optoelectronics
 - d) Illumination engineering

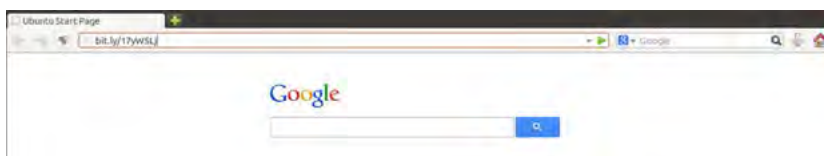
VISIT

An interview conducted with an optometrist.

bit.ly/19WxYYa



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VISIT

Want to take part in some real science research? Check out these citizen science projects to get involved easily.

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SUMMARY:

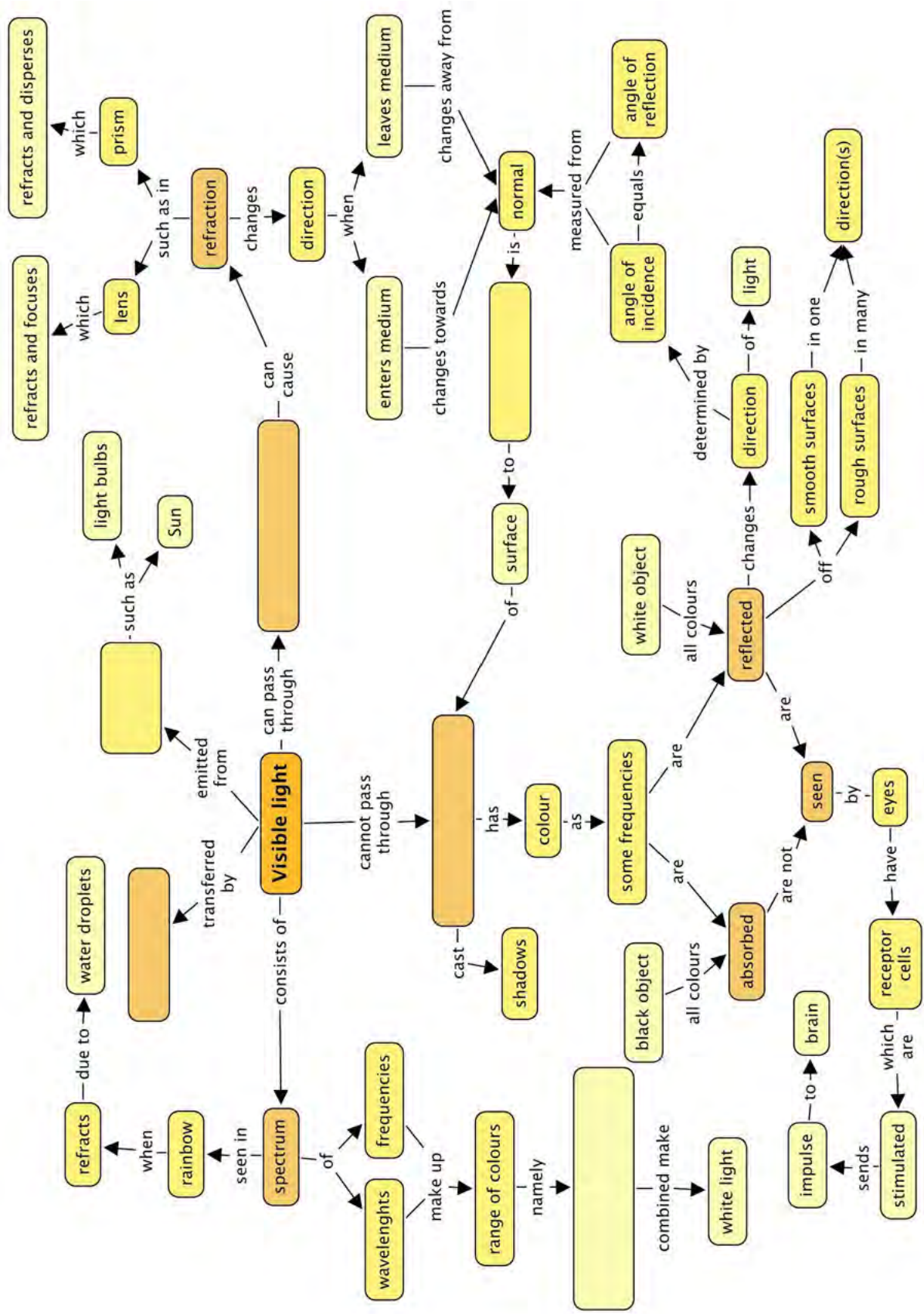
Key Concepts

- Light travels in straight lines.
- White light consists of all the colours of the visible spectrum.
- The colour spectrum can be seen when white light is dispersed by a prism or a raindrop (rainbow).
- Light cannot pass through opaque objects.
- Light can pass through transparent objects.
- Light is absorbed by some materials.
- A material appears to be a certain colour because it reflects that part of the colour spectrum. Other wavelengths of light are absorbed.
- In reflection, the angle of incidence is equal to the angle of reflection.
- On a smooth surface, parallel rays of light are all reflected at the same angle.
- On rough surfaces, the light is scattered and the image produced is not clear.
- The human eye has specialised cells in the retina which convert light into electrical nerve impulses. The nerve impulses are transmitted to the brain via the optic nerve, where they are interpreted.
- Light travels at different speeds in different media.
- When light enters a different medium at an angle, the light is refracted.
- If the light slows down, the light bends towards the normal line.
- If the light speeds up, the light bends away from the normal line.
- Converging lenses refract and focus light.
- Diverging lenses and triangular prisms refract and disperse light.
- Lenses have many applications, for example, in glasses to correct vision, microscopes, telescopes and magnifying glasses.

Concept Map

The concept map on the next page shows how all the concepts relating to visible light link together. Complete the map to reinforce what you have learned in this chapter.

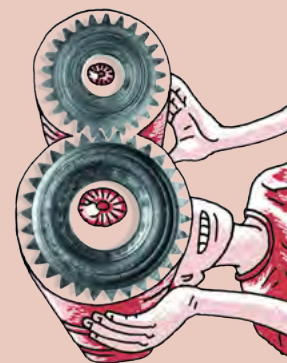




REVISION:

1. Match the correct definitions to the terms in the following table. Write the letter of the definition next to the correct number below. [12 marks]

Term	Definition
1. Radiation	A. Light cannot pass through.
2. Visible light	B. The angle of incidence equals the angle of reflection when a ray is reflected off a smooth surface.
3. Opaque	C. One of the ways in which energy is transferred, specifically through a vacuum
4. Transparent	D. When light enters a transparent medium it can change direction.
5. Absorption	E. Curved inwards.
6. Reflection	F. The spectrum of light which we are able to see.
7. Retina	G. Bulging outwards.
8. Refraction	H. A transparent object able to refract and focus light.



Term	Definition
9. Diverging	I. Light can pass through.
10. Lens	J. When light rays are spread out from a point.
11. Concave	K. A layer of tissue at the back of the eye which is sensitive to light.
12. Convex	L. When the surface of a substance absorbs certain colours of light.

Answers:

1:

2:

3:

4:

5:

6:

7:

8:

9:

10:

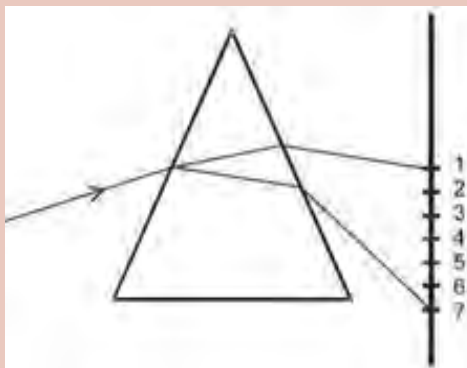
11:

12:

2. A beam of white light is shone through a glass prism. It splits up into seven colours which are shone on a screen. A learner took a photograph which is shown below and drew a ray diagram to show the prism. The colours are marked 1 to 7 in the diagram.



A photograph of the prism.



A diagram drawn by the learner.

- a) What does this tell us about white light? [1 mark]

- b) Why does the light do this when it passes through the prism? [3 marks]

- c) What colour is at label 1 and what colour is at label 7? Explain your answer. [3 marks]

- d) What label corresponds to the colour of grass? [1 mark]

- e) Can you see there are two other lighter, white rays emerging from the prism? What do you think this is the result of? [2 marks]

3. Why does an opaque object cast a shadow? [2 marks]

4. Look at the following photograph of water in a pond and answer the questions.



Water in a pond.

- a) How are we able to see the image of the wooden poles sticking up on the edge of the pond? [2 marks]

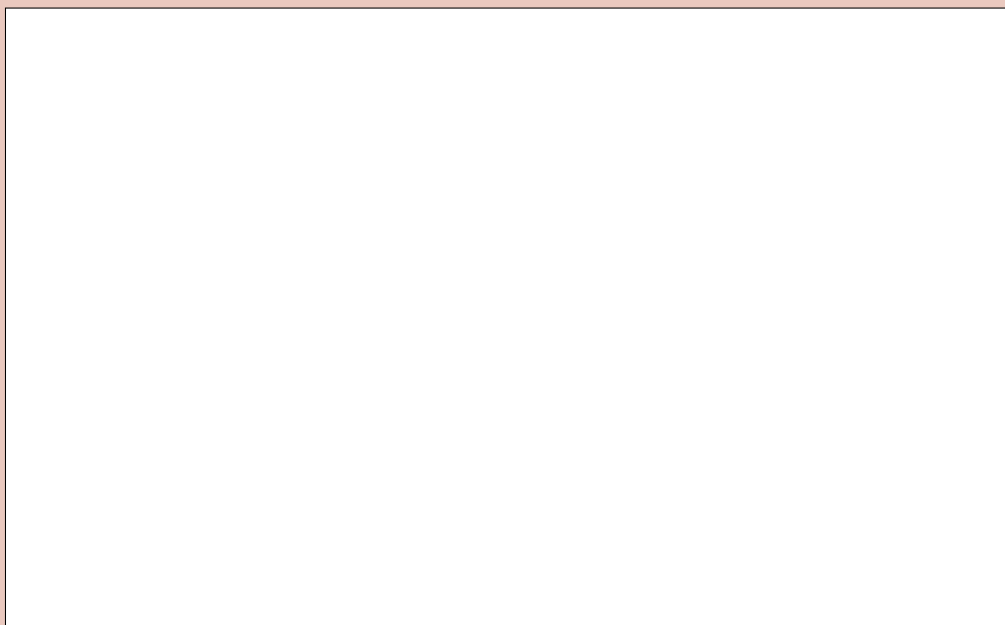
- b) Why is the image not clear, but blurred? [2 marks]

5. Two learners are discussing the colours of light. They decide that white and black are not really colours of light. If they are not colours, then how can we see them? [5 marks]

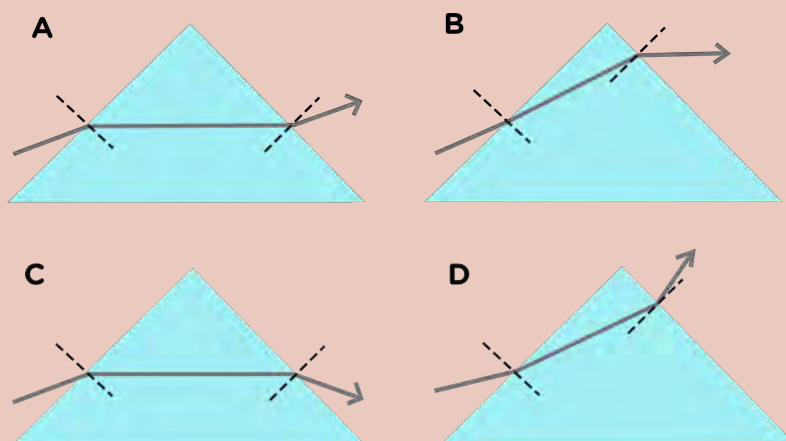
6. Explain how we are able to see the different colours on the South African flag. [6 marks]



-
-
-
-
-
-
7. Draw a ray diagram in the space provided to show how we see the green part of the flag. [5 marks]

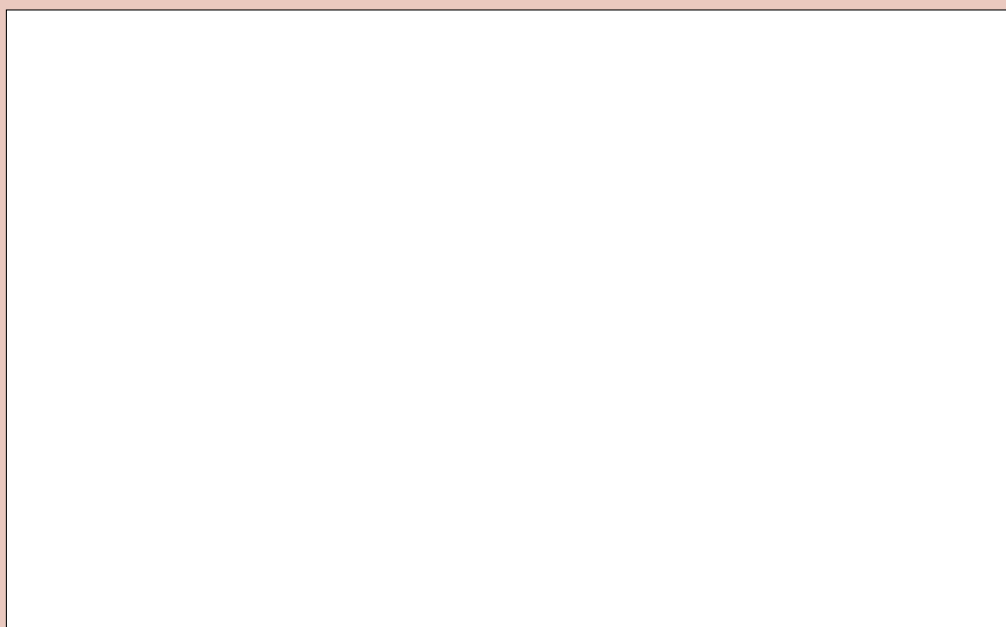


8. Which diagram shown below correctly shows the path of a ray of light through a triangular piece of glass? [2 marks]

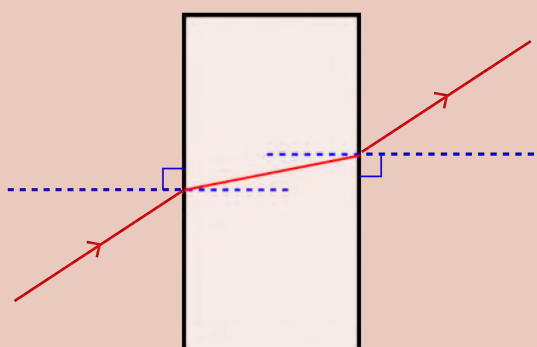


9. Complete the following sentence and write it out in full on the lines provided: When light travels from a less dense into a more dense transparent medium, it refracts and bends _____ the normal line. When light travels from more dense to a less dense medium, it refracts and bends _____ from the normal line. [2 marks]

10. Draw a diagram to show what is meant by 'when the refracted ray bends towards the normal'. Mark the angle of incidence and angle of refraction. Indicate which medium is denser [4 marks]



11. Study the following diagram and answer the questions that follow.



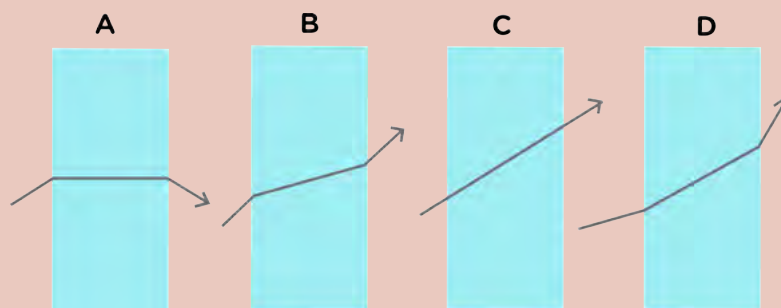
- a) This diagram is a drawing that a learner made during an investigation into the refraction of light. What does the red line represent in this diagram? [1 mark]

- b) What do the blue lines represent? Label this on the diagram. [1 mark]
c) The light passes from the air and into a block of another medium. Is this medium more or less dense than air? Give a reason for your answer. [2 marks]

- d) What type of medium could the block be made from? [1 mark]

- e) Label the incident ray and the emergent ray on the diagram. [2 marks]
f) Label the angles of incidence (i) and angles of refraction (r) on the diagram. [2 marks]

12. Which diagram shown below shows the path of a light beam passing through a rectangular glass prism correctly? [2 marks]

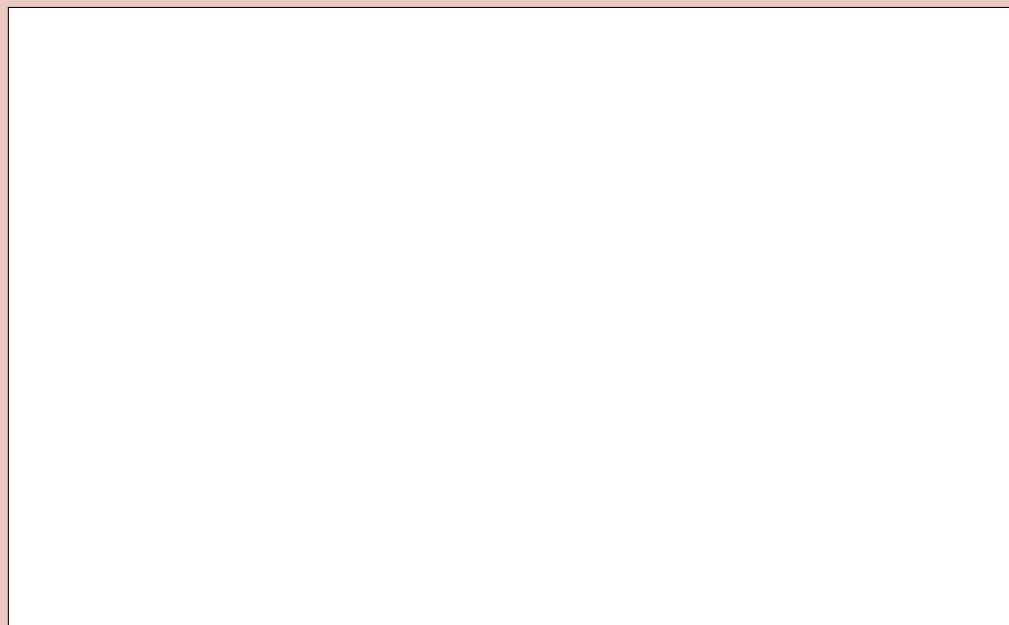


13. Why does it look like the tree trunk in the photograph is skew? [2 marks]



14. What shape does a lens have to have in order to focus the light? [1 mark]

15. Draw a ray diagram to show how a converging lens focuses light to a point. [4 marks]



16. Which eyesight defect can be fixed by using a converging lens? Explain what this defect is and why it can be corrected. [4 mark]

Total [74 marks]



GLOSSARY

ammeter:	device that measures the strength of an electric current
ampere:	the standard unit for measuring electric current
angle of incidence:	the angle between the incident ray and the normal line
angle of reflection:	the angle between the reflected ray and the normal line
attract:	to pull something closer
cell:	a source of energy for an electric circuit
component:	a part of a larger system
composition:	the parts of a mixture
conductor:	a substance which easily transmits electricity, heat, sound or light
converge:	light rays that come together and focus on a point
delocalised:	not limited to a particular place, free to move
discharge:	the sudden flow of charged particles between two electrically charged objects
dispersion:	spreading of something over an area
diverge:	light rays that spread apart as they move further and further away from a point
earth:	(or ground) to connect with a conductor to the ground, or the earth
earthing:	a way to prevent electrical charge from building up on an object, or to neutralise an electric charge, by allowing the excess charge to flow into the Earth
electric circuit:	a complete path through which electrons can move
electric current:	the movement of charge in an electric circuit
electrodes:	a conductor which allows electricity to enter a substance
electrolysis:	the use of electricity to separate chemicals in a solution
electromagnet:	a device which becomes a magnet when electric current passes through it
electroplating:	covering an object with a thin layer of metal using electrolysis
electrostatic charge:	the electric charge resulting from static electricity caused by an excess or deficiency of electrons on the surface of an object
flammable:	something is easily set on fire
focus:	bring together to the same point
friction:	the resistance that results when two surfaces are rubbed or moved against each other
fuse:	a safety device designed to melt and break the circuit if an electric current reaches too high a level

ignite:	to light something
incident ray:	the ray of light which hits a surface
luminous:	bright or shining
medium:	substance through which waves (such as light) can travel
neutral:	when the number of positive charges (from the protons) is equal to the number of negative charges (from the electrons); the (positive and negative) charges balance each other so that the object is neither positively nor negatively charged
normal line:	this is an imaginary line which is drawn at 90° to the surface
opaque:	something that you cannot see through; no light passes through the object
optical density:	a measure of how well a medium allows light to travel through it
optics:	the scientific study of sight and the behaviour of light
parallel circuit:	a circuit that provides more than one pathway for the current to pass through it
perpendicular:	at right angles
propagation:	spreading into new areas
qualitative:	describing something in terms of its properties or characteristics rather than by a number or measurement
radiation:	the emission of energy as electromagnetic waves
rectilinear:	straight lines
reflect:	throw back without absorbing
reflected ray:	the ray of light which leaves a surface
refraction:	the change in direction of a wave passing from one medium to another caused by its change in speed
repel:	to push something away
resistance:	the opposition to the movement of charge in a conductor
resistor:	a component in an electrical circuit which slows the movement of charge
retina:	a layer at the back of the eyeball which is made up of light sensitive cells
series:	components connected in series provide only one pathway for electrical current; they are connected one after another
static electricity:	the build-up of a stationary electric charge (either positive or negative) on the surface of an object
stimulate:	to cause activity
switch:	a control component in an electrical circuit which opens or closes the circuit
translucent:	semi-transparent; some light is able to pass through but not enough for details to be seen clearly
transmit:	to cause light to pass through space or medium

transparent:	something that you can see through; light passes through the object
variable:	something that can vary or change
visible spectrum:	the portion of the wave spectrum that is visible to the human eye